



**ADDIS ABABA UNIVERSITY**  
**DEPARTMENT OF CHEMISTRY**

**ORGANIC CHEMISTRY I**  
(Chem. 242)

**For Summer In-Service Program**

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|                       |                            |
|-----------------------|----------------------------|
| <b>Course Title:</b>  | <b>Organic Chemistry I</b> |
| <b>Course Number:</b> | <b>Chem. 242</b>           |
| <b>Credit Hours:</b>  | <b>3</b>                   |

## **COURSE DESCRIPTION**

This course is aimed to introduce students to fundamental principles of organic chemistry, including nomenclature, structure and bonding, and the basic principles of stereochemistry. The major reactions associated with the common functional groups will be covered with emphasis on reaction mechanisms. Although the course follows mechanistic approach to reactions of organic compounds (substitution, elimination, addition, rearrangement reactions), a chapter is devoted to brief discussion of functional groups, their typical reactions and synthesis. This will enable the students to understand the twin strategies of studying chemistry of the millions of organic compounds by either classifying them according to the reaction types they undergo (mechanistic approach) or according to their functional groups.

## **LEARNING OUTCOMES**

By the end of this course students will be able to:

- name cyclic and acyclic compounds containing alcohols, halides, alkenes and alkynes using the IUPAC system, name small compounds using common (e.g., iso, sec-, tert-) prefixes,
- identify and name compounds containing common functional groups such as halides, alkenes, alkynes, carbonyl groups, alcohols, amines, aromatic rings and ethers.
- relate properties and reactions of organic compounds to functional groups.
- know important reactions of a range of functional groups
- explain the role of hydrogen bonding in organic chemistry, including its effect on the boiling point and solubility
- know basic principles of stereochemistry
- identify conformational isomers, draw Newman projections and cyclohexane chair forms that accurately portray axial and equatorial groups, and predict the most stable conformer.
- identify stereoisomers, label a stereocenter using the Cahn-Ingold-Prelog rules, distinguish between and define the terms enantiomer, diastereomer, meso, chiral and achiral,
- draw the mechanism and predict the products of nucleophilic substitution (S<sub>N</sub>) reactions, including stereocenter inversion during an S<sub>N</sub>2 reaction, racemization of stereochemistry during an S<sub>N</sub>1 reaction, and carbocation rearrangements.
- draw the mechanism and predict the products of substitution, elimination, and addition reactions.

## Table of Contents

|                                                                                                                                   |    |
|-----------------------------------------------------------------------------------------------------------------------------------|----|
| 1. Bonding, structure, and Reactivity .....                                                                                       | 4  |
| 1.1 Historical Background of Organic Chemistry .....                                                                              | 4  |
| 1.2 Energy Levels and Atomic Orbitals .....                                                                                       | 6  |
| 1.3 Bonding in Carbon Compounds .....                                                                                             | 10 |
| 1.4 Lewis structure and Resonance .....                                                                                           | 15 |
| 1.5 Structural Formulas of Organic Molecules .....                                                                                | 19 |
| 1.6 Factors Influencing Electron Availability and Reactivity of Organic Compounds .....                                           | 22 |
| 1.6.1 Inductive Effect.....                                                                                                       | 22 |
| 1.6.2 Resonance Effect .....                                                                                                      | 23 |
| SELF -TEST EXERCISES 1 .....                                                                                                      | 26 |
| ANSWER FOR EXERCISE .....                                                                                                         | 26 |
| 2. Functional Groups in Organic Chemistry .....                                                                                   | 29 |
| 2.1 Functionality.....                                                                                                            | 29 |
| 2.2 Nomenclature of Organic Compounds .....                                                                                       | 31 |
| A. Nomenclature of ALKANE.....                                                                                                    | 31 |
| B. Nomenclature of ALKENES .....                                                                                                  | 40 |
| C. Nomenclature of ALKYNES .....                                                                                                  | 43 |
| D. Nomenclature of Aromatic Hydrocarbons .....                                                                                    | 43 |
| E. Nomenclature of ALCOHOL .....                                                                                                  | 44 |
| F. Nomenclature of Ethers and Epoxides .....                                                                                      | 45 |
| G. Nomenclature of Amines .....                                                                                                   | 46 |
| H. Nomenclature of Aldehydes and Ketones.....                                                                                     | 47 |
| I. Nomenclature of Carboxylic Acids and their Derivatives.....                                                                    | 48 |
| SELF -TEST EXERCISES 2.1.....                                                                                                     | 52 |
| 2.3 Introduction to the Common Functional Groups in Organic Chemistry: Their Typical properties, Preparations and Reactions. .... | 54 |
| 2.3.1 Alkanes, Alkenes, Alkynes,.....                                                                                             | 54 |
| A. Alkanes.....                                                                                                                   | 54 |
| B. Alkenes.....                                                                                                                   | 56 |
| C. Alkynes .....                                                                                                                  | 64 |
| SELF -TEST EXERCISES 2.2.....                                                                                                     | 69 |
| 2.3.2Alcohols.....                                                                                                                | 71 |
| 2.3.3 Ethers and Epoxides .....                                                                                                   | 77 |
| A. Ethers .....                                                                                                                   | 77 |

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| B. Epoxides .....                                                             | 78  |
| Self Test Exercise 2.2 .....                                                  | 80  |
| ANSWER FOR EXERCISES .....                                                    | 82  |
| 3. StereoChemistry .....                                                      | 84  |
| 3.0 Stereoisomerism .....                                                     | 84  |
| 3.1 Molecular Chirality: Enantiomers .....                                    | 85  |
| 3.2 The Chirality Center (*) .....                                            | 86  |
| 3.3 Configuration of Chiral Compounds (R-S Notation System) .....             | 87  |
| 3.4 Optical Activity and Plane Polarized Light.....                           | 89  |
| 3.5 Measurement of Optical Rotation.....                                      | 90  |
| 3.6 Fischer Projections.....                                                  | 91  |
| 3.7 Multiple Stereogenic Centers: Diastereomers .....                         | 93  |
| 3.8 Meso Compounds .....                                                      | 94  |
| 3.9 The Biological importance of Chirality .....                              | 95  |
| 3.10 Conformational Isomerism.....                                            | 96  |
| 3.10.1 Conformational Analysis in Alkanes: .....                              | 96  |
| 3.10.2 Conformations and Stabilities of Cycloalkanes.....                     | 98  |
| Self-Test Exercise 3.0 .....                                                  | 104 |
| ANSWER FOR EXERCISES .....                                                    | 105 |
| 4. Major organic reactions .....                                              | 106 |
| 4.1 Substitution Reaction .....                                               | 106 |
| 4.1.1 S <sub>N</sub> 1 and S <sub>N</sub> 2 Mechanism .....                   | 108 |
| 4.1.2 Factors Affecting S <sub>N</sub> 1 and S <sub>N</sub> 2 Reactions ..... | 113 |
| 4.1.3 Applications of Substitution Reactions .....                            | 118 |
| 4.2 Elimination Reactions.....                                                | 120 |
| 4.2.1 E2 and E1 Mechanism.....                                                | 121 |
| 4.2.2 Zaitsev's Rule .....                                                    | 121 |
| 4.2.3 Elimination versus Substitution .....                                   | 122 |
| 4.2.4 Applications of Elimination Reactions .....                             | 123 |
| 4.3 Addition Reactions .....                                                  | 125 |
| 4.3.1 Markovnikov's Rule .....                                                | 125 |
| 4.3.2 Application of Addition Reactions .....                                 | 126 |
| Self-Test Exercise 4.0 .....                                                  | 129 |

# UNIT 1

## 1.BONDING, STRUCTURE, AND REACTIVITY

### Learning outcomes:

At the end of this unit, you will be able to:

- define organic chemistry.
- differentiate between  $sp$ ,  $sp^2$  and  $sp^3$  hybridization.
- determine a hybridization of a carbon atom in a molecule.
- write lewis structure of a compound or an ion.
- explain the concepts of resonance contributors and draw resonance contributors of an ion or a compound.
- draw the structure of an organic compound using different types of formula.

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### 1.1 Historical Background of Organic Chemistry

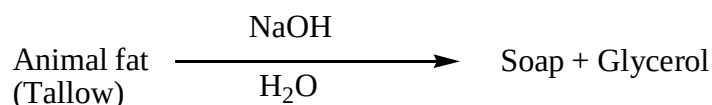
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#### What is organic chemistry?

At the beginning of the nineteenth century chemists generally thought that compounds from living organisms were too complicated in structure to be capable of artificial synthesis from non-living things, and that a 'vital force' or vitalism conferred the characteristics of living beings on this form of matter. They named these compounds 'organic', and preferred to direct their investigations toward inorganic materials that seemed more promising. Inorganic compounds: are substances derived from non-living materials (mineral constituting of the earth).

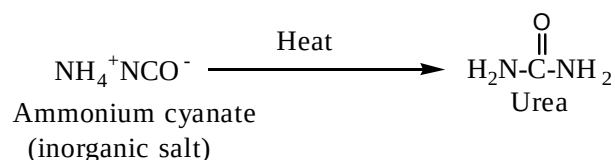
In 1780s, scientists distinguished between organic compounds and inorganic compounds (Swedish chemist Torbern Bergman). Organic compounds: as compounds derived from living organisms (vegetables and animals). A compound like urea is considered to be organic because urea is a constituent of urine. They thought that organic compounds couldn't be synthesized in the laboratory. This theory is called the **vitalism theory**. They believed that organic compounds couldn't be synthesized because of the vital force requirement, which is not present in flasks and test tubes.

In 1816 Michel Chevreull synthesized soap from animal fat.



Conversion of fat (organic) into soap and soap into fatty acid without the intervention of outside vital force was the first demonstration that organic compounds could be interconvert into each other in the laboratory. This is the first experiment that indicated the vital force theory was with drawbacks.

1828 – 1850: A number of ‘organic’ compounds were synthesized from sources that are clearly inorganic. In 1828 Friedrich Wöhler synthesized urea from inorganic compound. Urea is a substance isolated from urine. This happened by chance in his attempt of producing ammonium cyanate in the laboratory



After the synthesis of urea in laboratory, theory of Vitalism has been rejected. This clearly indicated that no special force is required to synthesis organic compounds. In science today, a different name is given to the field of study of compounds from living organisms. It is known as Natural products Chemistry.

There is no distinct boundary between organic and inorganic compounds except that almost all organic compounds contain carbon. The modern definition of **organic chemistry** is: it is the study of carbon containing compounds.

Organic chemistry, then, is the study of carbon compounds. But why is carbon special? Why, of the more than 30 million presently known chemical compounds, do more than 99% of them contain carbon? The answer to this question comes from carbon’s electron structure and its position in the periodic table. As a group 4a element, carbon can share four valance electrons and form four strong covalent bonds. Furthermore, carbon atom can bond to one another, forming long chains and rings.



How do you differentiate organic compounds from inorganic compounds?

Compounds containing carbon may or may not be organic.

For example ammonium cyanate,  $\text{NH}_4\text{NCO}$ , contains carbon, is not organic. It is an inorganic salt. Organic compounds distinguish from inorganic compounds in that they contain a characteristic functional group. You will learn about functional group in unit 2.

### Exercise 1.1

1. What is organic chemistry?
2. Classify the following compounds as organic or inorganic.
 

|                                    |                             |                                |
|------------------------------------|-----------------------------|--------------------------------|
| a. $\text{CO}_2$                   | d. $\text{CS}_2$            | g. $\text{HC}\equiv\text{CH}$  |
| b. $\text{CH}_4$                   | e. $\text{CH}_3\text{COOH}$ | h. $\text{Ln}_4(\text{C}_2)_3$ |
| c. $\text{C}_2\text{H}_5\text{OH}$ | f. $\text{CaC}_2$           | i. $\text{CO}_3^-$             |

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## 1.2 Energy Levels and Atomic Orbitals

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As you probably know, an atom consists of a dense, positively charged nucleus surrounded at a relatively large distance by negatively charged electrons. The nucleus consists of subatomic particles called neutrons, which are electrically neutral, and protons, which are positively charged.

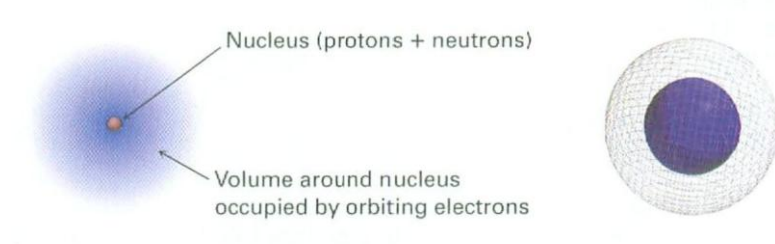


Figure 1.1 A schematic view of an atom

### Atomic Orbitals

How are electrons distributed in an atom? You might recall from your general chemistry course that, according to the quantum mechanical model, the behavior of a specific electron in an atom can be described by a mathematical expression called **wave equation**. The solution to a wave equation is called a wave function or **orbital**, and is denoted by the Greek letter psi,  $\Psi$ .

For one electron system, each solution to the equation is associated with a particular wave function, which is also called an atomic orbital. The square of the wave function gives the probability of finding the electron at any point.

The electron could be anywhere; the probability of finding it at any point never reaches zero. To visualize where the electron spends most of its time, we draw a diagram to show where the electron spent 90% of its time.

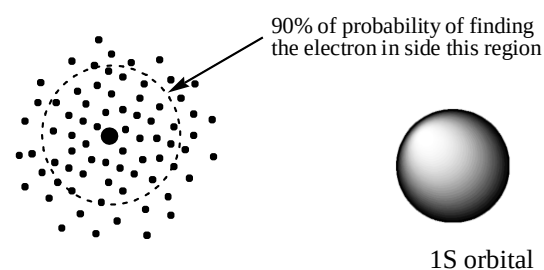


Fig. 1.2 Representation of 1S orbital

The region in which an electron is most likely found is called an **orbital**.

The electron may locate anywhere within an orbital at any instant in time, but it spends most of its time in certain high probability regions. Thus we can think of an electron as a particle that moves from place to place so rapidly that it behaves somewhat like an “electron cloud” whose density varies within the orbital.

**Atomic orbitals** are the region in which there is high probability of finding an electron in space.

## Shape of Orbitals

What do orbitals look like? There are four different kinds of orbitals, denoted s, p, d, and f, each with a different shape. Of the four, we will be concentrated primarily with s and p orbitals because these are the most common in organic chemistry. The s orbitals are spherical, with the nucleus at their center; p orbitals are dumbbell-shaped; and four of the five d orbitals are cloverleaf-shaped as shown in Figure 1.3.

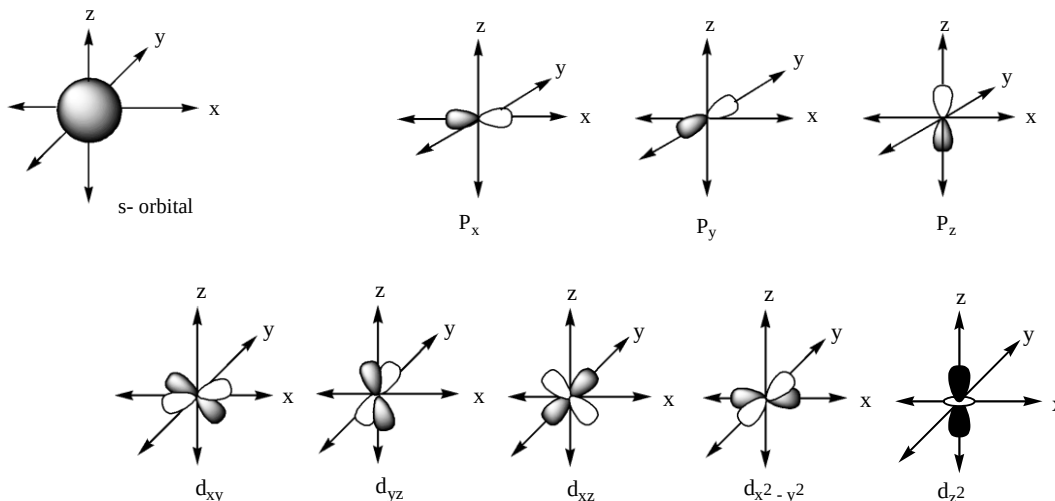


Figure 1.3 Representations of s, p, and d orbitals

## Electron Configuration

Electron configuration: is the way in which electrons are arranging around the nuclei of atoms.

It is the nature of things to seek the lowest possible energy.

The lowest energy arrangement, or **ground-state electron configuration**, of an atom is a listing of an orbital occupied by its electron. We can predict this arrangement by the following three rules.

- i. **The Aufbau principle:-** Electrons enter orbitals of lowest energy first.

In the following Augbau diagram a box ( $\square$ ) is used to represent an atomic orbital. Electrons enter the orbitals of lowest energy first, accordingly to the order  $1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s$ . Note that the filling of atomic orbitals doesn't follow a simple pattern beyond the 2<sup>nd</sup> energy level.

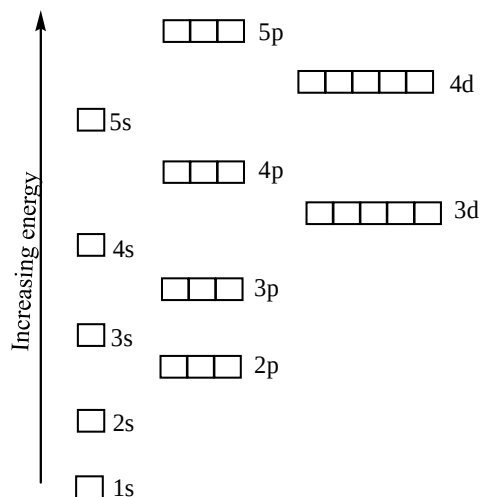
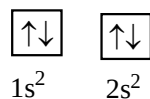


Fig. 1.4 Orbital diagram in increasing energy

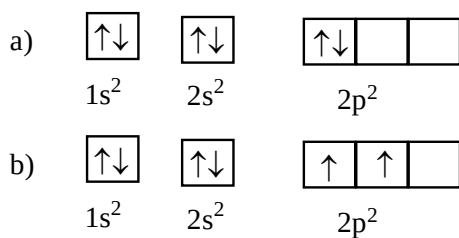
- ii. **The Pauli Exclusion Principle:** electrons act as if they were spinning around an axis, in much the same way that the earth spins. This spin can have two orientations, denoted as up  $\uparrow$  and down  $\downarrow$ . An atomic orbital may contain at most two electrons. To occupy the same orbital, two electrons must have opposite spins;

Example



- iii. **Hund's rule:** - When electrons occupy orbitals of equal energy, one electron enters each orbital unit all the orbitals contain one electron with spins parallel.

Example: The two possible orbital electronic configurations for carbon ( $1s^2, 2s^2, 2p^2$ ) are:



According to Hund's rule (b) is acceptable configuration, because parallel spins lead to the lowest electron- electron repulsion.

Table 1.1 Electron Configurations of the First Twelve Elements of the Periodic Table

| <i>Element</i> | <i>Atomic number</i> | <i>Number of electrons in indicated orbital</i> |           |            |            |            |           |
|----------------|----------------------|-------------------------------------------------|-----------|------------|------------|------------|-----------|
|                | <i>Z</i>             | <i>1s</i>                                       | <i>2s</i> | <i>2px</i> | <i>2py</i> | <i>2pz</i> | <i>3s</i> |
| Hydrogen       | 1                    | 1                                               |           |            |            |            |           |
| Helium         | 2                    | 2                                               |           |            |            |            |           |
| Lithium        | 3                    | 2                                               | 1         |            |            |            |           |
| Beryllium      | 4                    | 2                                               | 2         |            |            |            |           |
| Boron          | 5                    | 2                                               | 2         | 1          |            |            |           |
| Carbon         | 6                    | 2                                               | 2         | 1          | 1          |            |           |
| Nitrogen       | 7                    | 2                                               | 2         | 1          | 1          | 1          |           |
| Oxygen         | 8                    | 2                                               | 2         | 2          | 1          | 1          |           |
| Fluorine       | 9                    | 2                                               | 2         | 2          | 2          | 1          |           |
| Neon           | 10                   | 2                                               | 2         | 2          | 2          | 2          |           |
| Sodium         | 11                   | 2                                               | 2         | 2          | 2          | 2          | 1         |
| Magnesium      | 12                   | 2                                               | 2         | 2          | 2          | 2          | 2         |

- Fill the following orbitals of nitrogen with appropriate number of electrons using Hund's rule.

N(Z=7):

- Give the ground state electron configuration for each of the following elements:

a) Oxygen      b) silicon      c) sulfur

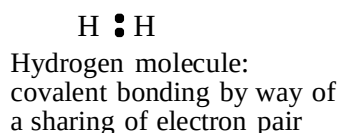
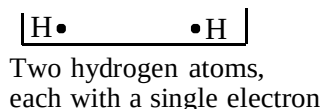
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## 1.3 Bonding in Carbon Compounds

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Carbon forms covalent bonding with other elements to produce organic molecules.

The covalent, or shared electron pair, model of chemical bonding was first suggested by G. N. Lewis of the University of California in 1916. Lewis proposed that a sharing of two electrons by two hydrogen atoms permits each one to have a stable closed-shell electron configuration analogous to helium.



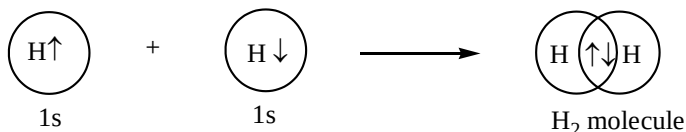
Ⓢ How does electron sharing lead to bonding between atoms?

Two models have been developed to describe covalent bonding: **valence-bond theory** and **molecular orbital theory**. Each has its strengths and weaknesses, and chemists tend to use them interchangeably depending on the circumstances. Valence-bond theory is the more easily visualized of the two, so most of the descriptions we will use derive from that approach.

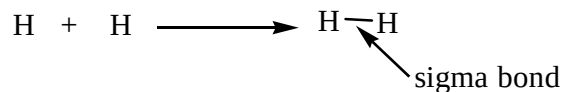
### 1.3.1 Valence Bond Theory

According to valence bond theory a covalent bond between two atoms approach each other closely and a half-filled orbital of one atom overlaps with a half-filled orbital of the other atom. The electrons are now paired in the overlapping orbitals and are attracted to the nuclei of both atoms, thus bonding the atoms together.

In the  $\text{H}_2$  molecule the H-H bond results from the overlapping of two half filled hydrogen 1s orbitals.



Bonds which are formed by end-to-end overlap of two atomic orbitals along a line drawing between the nuclei are called **sigma ( $\sigma$ ) bonds**.

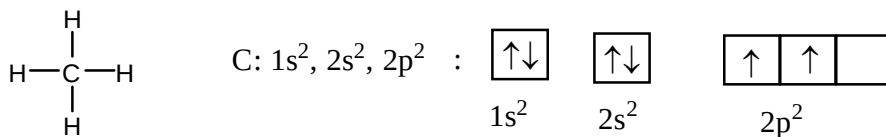


Note that: in order to overlap during covalent bond formation, the two overlapping orbitals must be half-filled.

### 1.3.2 Orbital Hybridizations

The bonding in hydrogen molecule is straightforward, but the situation is more complicated in organic molecule with tetravalent carbon atom.

Take methane,  $\text{CH}_4$ . It has 4 bonds.



This indicated only two half-filled orbitals exist in carbon for bond formation according to valance bond theory.

But in methane there are four bonds (four overlapping) exist. How can we explain this?

This can be explained by the concept of hybridization.

## ***sp*<sup>3</sup>Hybridization**

Before we describe the bonding in methane, it is worth pointing out that bonding theories attempt to describe a molecule on the basis of its component atoms; bonding theories do not attempt to explain how bonds form.

The electron configuration of a carbon bonded to other atoms need not be the same as a free carbon atom. By promoting one electron from the 2s orbital to the vacant 2p<sub>z</sub> orbital, carbon can achieve the new configuration 1s<sup>2</sup>, 2s<sup>1</sup>, 2p<sub>x</sub><sup>1</sup>, 2p<sub>y</sub><sup>1</sup>, 2p<sub>z</sub><sup>1</sup>. This new electronic configuration is called an excited state configuration. In the excited state, carbon has four unpaired electrons and can form four bonds to hydrogen. By mixing ("hybridizing") the 2s, 2p<sub>x</sub>, 2p<sub>y</sub>, and 2p<sub>z</sub> orbitals, four new orbitals are obtained (Figure 1.5). These four new orbitals are called **sp<sup>3</sup>** hybrid orbitals because they come from one s orbital and three p orbitals. Each sp<sup>3</sup> hybrid orbital has 25% s character and 75% p character.

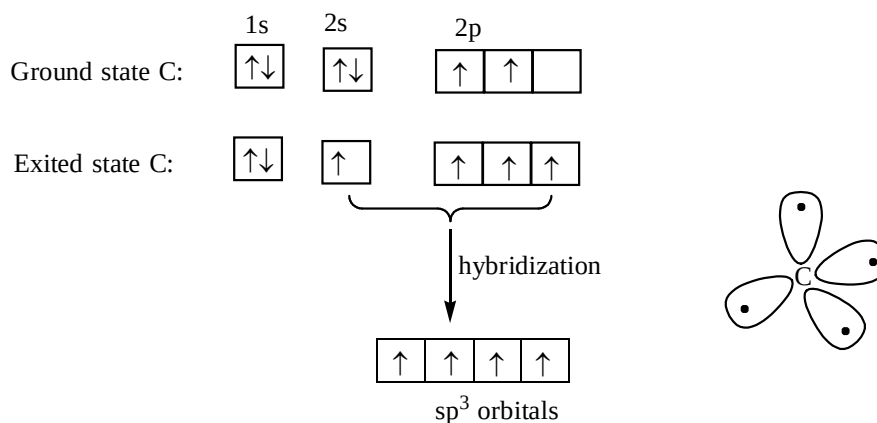


Figure 1.5 Mixing the s orbital with the three p orbitals generates four  $sp^3$  hybrid orbitals. The four  $sp^3$  hybrid orbitals are of equal energy.

Now carbon can overlap with hydrogen atoms to form four bonds.

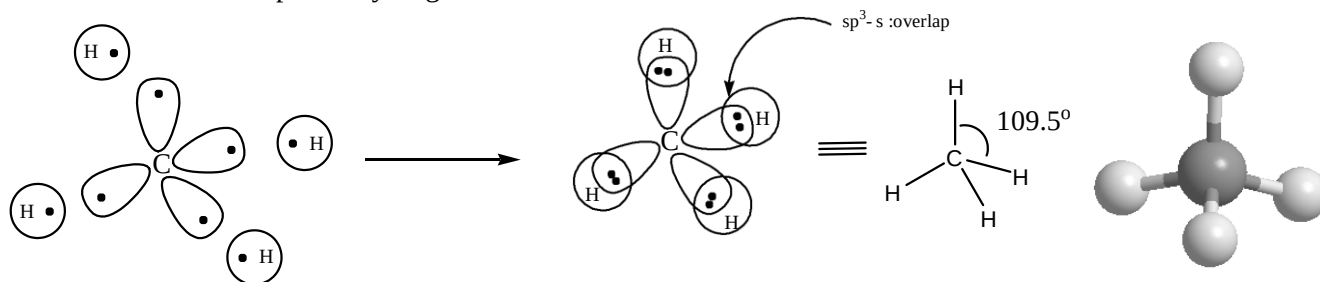


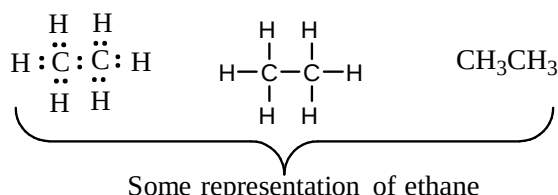
Fig 1.6 Each half-filled  $sp^3$  orbital overlaps with a half-filled hydrogen  $1s$  orbital along a line between them giving a tetrahedral arrangement of four sigma bonds.

Such end-to-end overlapping of orbitals is called sigma overlapping and the resulting bond called **sigma ( $\sigma$ ) bond**

$sp^3$  hybrid orbital has a tetrahedral shape with all bond angles equal  **$109.5^\circ$**

## Ethane

The same kind of orbital hybridization that accounts for methane structure also accounts for the bonding of carbon atoms in to chains and rings to make possible many millions of organic compounds. Ethane is simple compound containing carbon-carbon bond.



In ethane the two carbon atoms bond to each other by  $\sigma$  overlap of  $sp^3$  hybrid orbital from each (Fig 1.7). The remaining three  $sp^3$  hybrid orbitals of each carbon overlap with  $1s$  orbitals of three hydrogens.

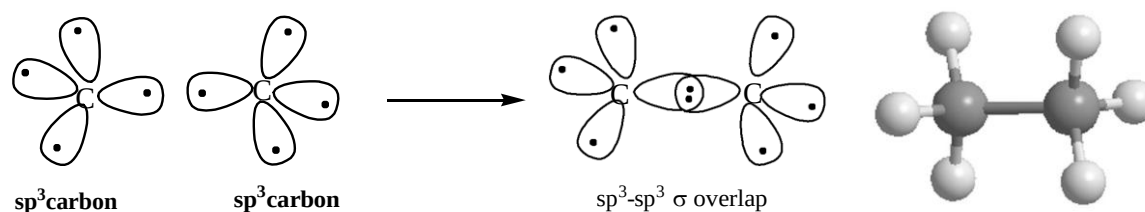
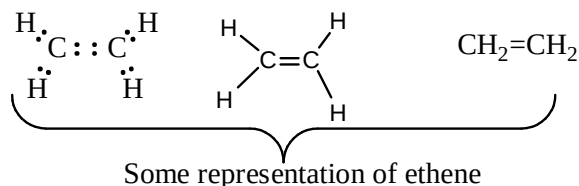


Fig 1.7 The structure of ethane. The carbon-carbon bond is formed by  $\sigma$  overlap of two carbon  $sp^3$  hybrid orbitals.

## $sp^2$ Hybridization

Ethene (ethylene) is planar with bond angles close to  $120^\circ$ ; therefore, some hybridization state other than  $sp^3$  is required. The hybridization scheme is determined by the number of atoms to which carbon is

directly attached. In  $sp^3$  hybridization, four atoms are attached to carbon by  $\sigma$  bonds, and so four equivalent  $sp^3$  hybrid orbitals are required. In ethylene, three atoms are attached to each carbon, so three equivalent hybrid orbitals are needed.



In ethane the 2s orbital combines with only two of three available 2p orbitals. One of the 2p orbitals is left unhybridized. Three  $sp^2$  hybrid orbitals result. The three  $sp^2$  orbitals lie in a plane at angle of  $120^\circ$  to one another, with the remaining p orbital perpendicular to the  $sp^2$  plane.

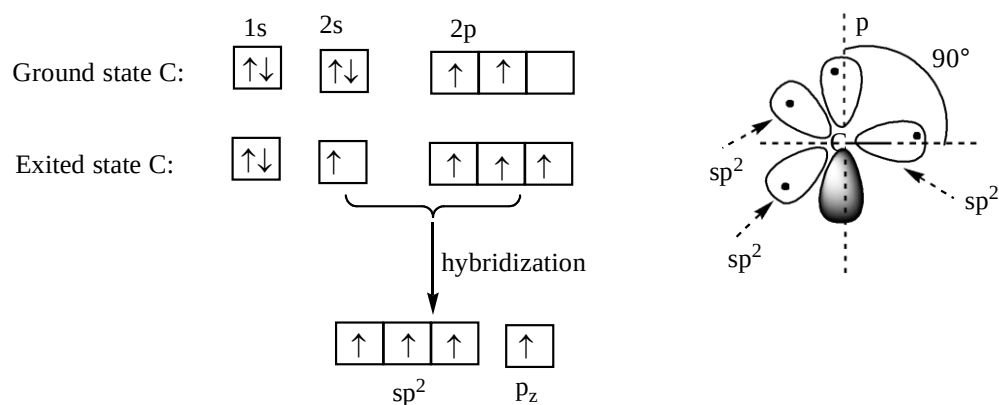


Figure 1.8 Mixing the s orbital with the two p orbitals generates three  $sp^2$  hybrid orbitals.

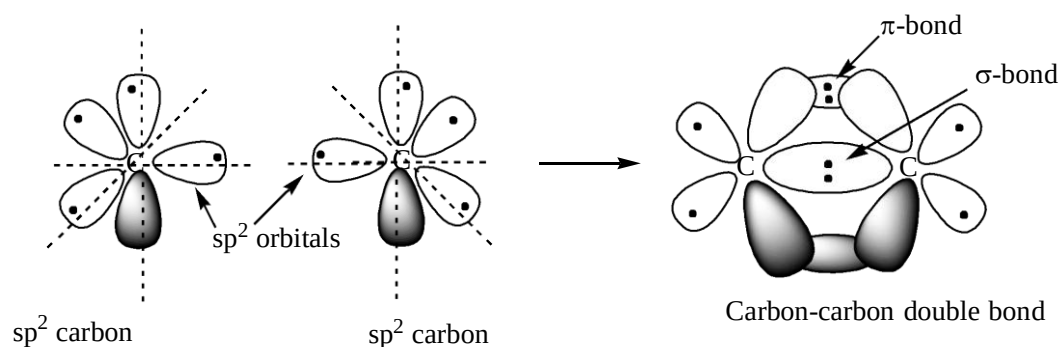
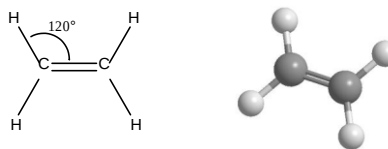


Figure 1.9 The structure of Ethene. orbital overlap of two  $sp^2$  hybrid carbon forms double bond

Sigma bond ( $\sigma$ ) resulted due to end- to-end overlap of orbitals

Pi bond ( $\pi$ ) resulted due to side-to-side overlap of p orbitals



## sp Hybridization

Consider bonding in acetylene,  $\text{H}-\text{C}\equiv\text{C}-\text{H}$ . To explain the formation of triple bond imagine that, instead of combining with two or three  $p$  orbitals, a carbon  $2s$  orbital hybridizes with only a single  $p$  orbital. Two  $sp$  hybrid orbitals result and two  $p$  orbitals left unhybridized. The two  $sp$  orbitals are oriented  $180^\circ$  apart on the  $x$ -axis, while the remaining two  $p$  orbitals are perpendicular on the  $y$ -axis (Figure 1.10).

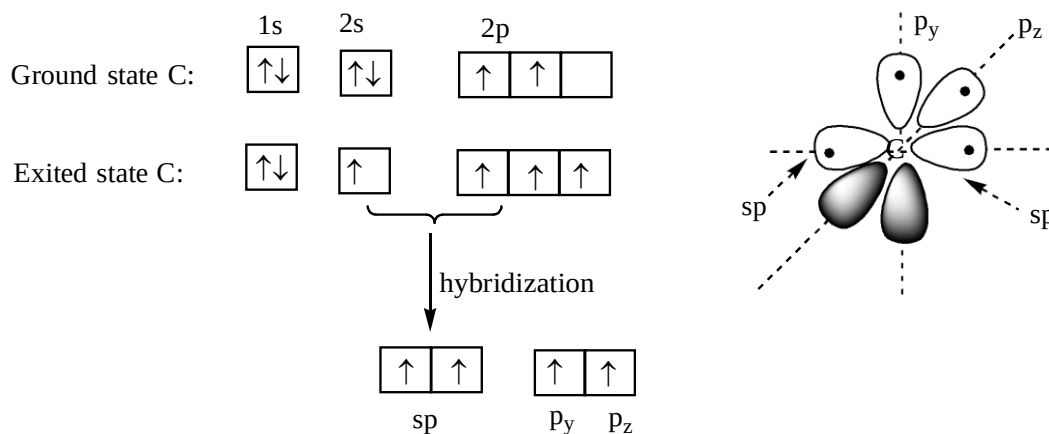
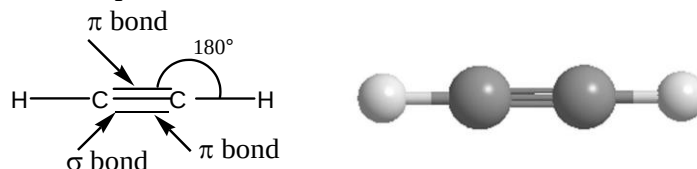


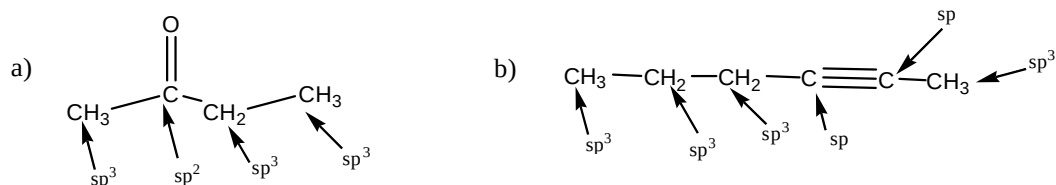
Figure 1.10 Mixing the  $s$  orbital with the one  $p$  orbitals generates two  $sp$  hybrid orbitals. The hybrid orbital indicates the presence of two unhybridized  $p$  orbitals that can overlap side-to-side. As a result two  $\pi$  bonds produced.



Generally **single** bond is has sigma bond only, **double** bond has one sigma and one pi, and **triple** bond has one sigma and two pi bonds.

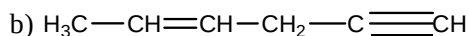
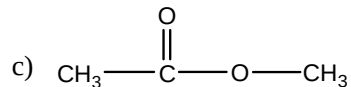
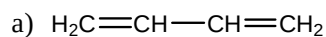
A doubly bonded carbon has  $sp^2$  hybridization and a triple bonded carbon has  $sp$  hybridization. Now you can extend the concept of hybridization to other organic compounds and be able to state the hybridization of each carbon.

Examples: the hybridization of each carbon in the following compounds is:



### Exercise 1.3

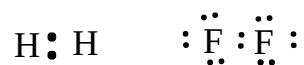
1. Indicate the hybridization of each carbons in the following compounds



## 1.4 Lewis structure and Resonance

### LEWIS STRUCTURE

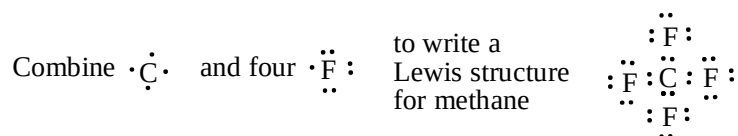
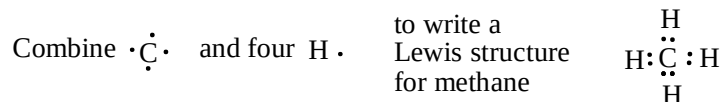
Structural formulas in which electrons are represented as dots are called **Lewis structures**.



Covalent bonding in  $\text{F}_2$  gives each fluorine eight electrons in its valence shell and a stable electron configuration equivalent to that of the noble gas neon:

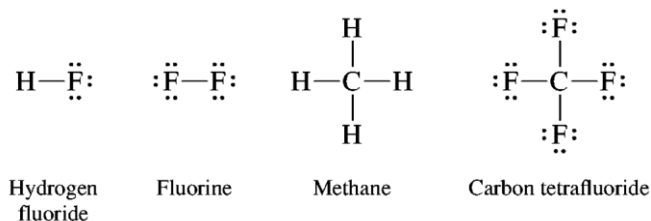
The Lewis model limits second-row elements (Li, Be, B, C, N, O, F, Ne) to a total of eight electrons (shared plus unshared) in their valence shells. Hydrogen is limited to two.

Now let's apply the Lewis model to the organic compounds methane and carbon tetrafluoride.

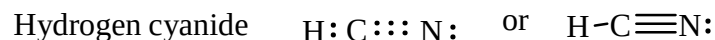
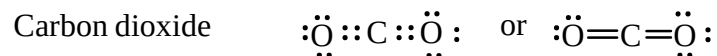


Carbon has eight electrons in its valence shell in both methane and carbon tetrafluoride.

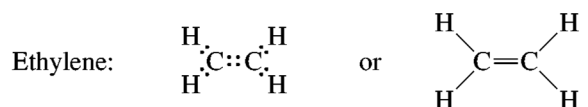
By forming covalent bonds to four other atoms, carbon achieves a stable electron configuration analogous to neon. Representing a two-electron covalent bond by a dash (-), the Lewis structures for hydrogen fluoride, fluorine, methane, and carbon tetrafluoride become:



Lewis's concept of shared electron pair bonds allows for four-electron double bonds and six-electron triple bonds. Carbon dioxide ( $\text{CO}_2$ ) has two carbon-oxygen double bonds, and the octet rule is satisfied for both carbon and oxygen. Similarly, the most stable Lewis structure for hydrogen cyanide ( $\text{HCN}$ ) has a carbon-nitrogen triple bond.



Multiple bonds are very common in organic chemistry. Ethylene ( $\text{C}_2\text{H}_4$ ) contains a carbon-carbon double bond in its most stable Lewis structure. The most stable Lewis structure for acetylene ( $\text{C}_2\text{H}_2$ ) contains a carbon-carbon triple bond. Here again, the octet rule is satisfied.



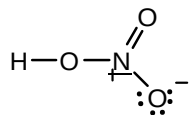
### Exercise 1.4

Write the most stable Lewis structure for

- Formaldehyde,  $\text{CH}_2\text{O}$
- Tetrafluoroethylene,  $\text{C}_2\text{F}_4$ .
- Acrylonitrile,  $\text{C}_3\text{H}_3\text{N}$ .

## FORMAL CHARGE

Lewis structures frequently contain atoms that bear a positive or negative charge. If the molecule as a whole is neutral, the sum of its positive charges must equal the sum of its negative charges. An example is nitric acid,  $\text{HNO}_3$ :



As written, the structural formula for nitric acid depicts different bonding patterns for its three oxygens. One oxygen is doubly bonded to nitrogen, another is singly bonded to both nitrogen and hydrogen, and the third has a single bond to nitrogen and a negative charge. Nitrogen is positively charged. The positive and negative charges are called **formal charges**.

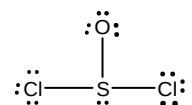
The positive and negative charges on each atom in a compound or an ion are called formal charges.

We calculate formal charges by counting the number of electrons "owned" by each atom in a Lewis structure and comparing this electron count with that of the neutral atom.

The method described for calculating formal charge has been one of reasoning through a series of logical steps. It can be reduced to the following equation:

|                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------|
| $\text{Formal charge} = \text{Group number in periodic Tabel} - \text{Number of bonds} - \text{Number of unshared electrons}$ |
|-------------------------------------------------------------------------------------------------------------------------------|

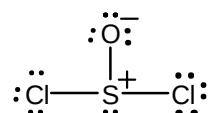
**Example:** Calculate the formal charge on each of the atoms in the Lewis structure of thionyl chloride given below.



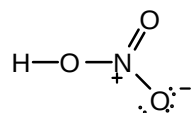
**Solution:** The formal charge is the difference between the number of valence electrons in the neutral atom and the electron count in the Lewis structure. (The number of valence electrons is the same as the group number in the periodic table for the main-group elements.)

| Valence electrons of neutral atom |   | Formal charge |
|-----------------------------------|---|---------------|
| Sulfur                            | 6 | $6-3-2 = +1$  |
| Oxygen                            | 6 | $7-1-6 = -1$  |
| Chlorine                          | 7 | $7-7-0 = 0$   |

The formal charges are shown in the Lewis structure of thionyl chloride as:



**Example:** Calculate the formal charge on each of the atoms in  $\text{HNO}_3$



Formal charge of N = +1

Formal charge of double bonded O = 0

Formal charge of single bonded O = -1

Formal charge of single bonded O (bonded to H) = 0

The formal charges are shown in the Lewis structure of nitric acid as:

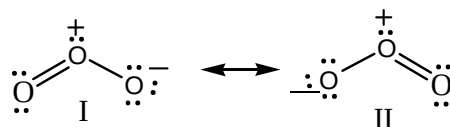
### Exercise 1.5

Calculate the formal charge on atoms in a)  $\text{POCl}_3$  b)  $\text{H}_2\text{SO}_4$

### RESONANCE

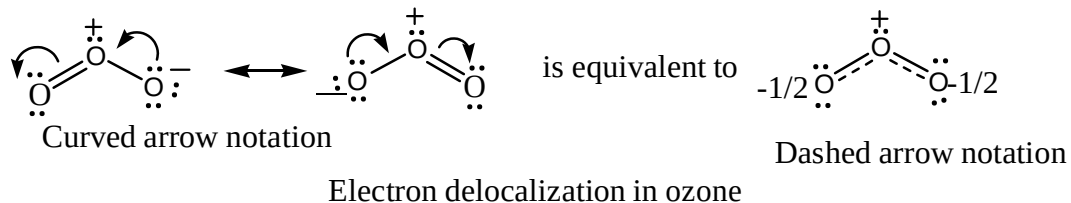
According to the resonance concept, when more than one Lewis structure may be written for a molecule, a single structure is not sufficient to describe it. Rather, the true structure has an electron distribution that is a “hybrid” of all the possible Lewis structures.

In the case of ozone, two equivalent Lewis structures may be written.

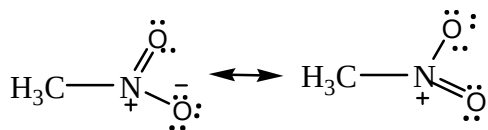


Structure I and II are resonance structures of ozone. We use a double-headed arrow ( $\leftrightarrow$ ) to represent resonance between these two Lewis structures.

Organic chemists often use curved arrows to show this electron delocalization. Alternatively, an average of two Lewis structures is sometimes drawn using a dashed line to represent a “partial” bond. In the dashed-line notation the central oxygen is linked to the other two by bonds that are halfway between a single bond and a double bond, and the terminal oxygens each bear one half of a unit negative charge.



Example: The two possible resonance structures of  $\text{CH}_3\text{NO}_2$  are



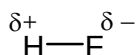
### Exercise 1.6

Write all resonance structures that can be generated for  $\text{SO}_3$ ,  $\text{NO}_3^-$

## POLAR COVALENT BONDS AND ELECTRONEGATIVITY

Electrons in covalent bonds are not necessarily shared equally by the two atoms that they connect. If one atom has a greater tendency to attract electrons toward itself than the other, we say the electron distribution is polarized, and the bond is referred to as a **polar covalent bond**.

Hydrogen fluoride, for example, has a polar covalent bond. Because fluorine attracts electrons more strongly than hydrogen, the electrons in the H-F bond are pulled toward fluorine, giving it a partial negative charge, and away from hydrogen giving it a partial positive charge. This polarization of electron density is represented in various ways.



The symbols  $\delta^+$  and  $\delta^-$  indicate partial positive and partial negative charge, respectively



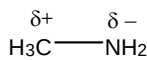
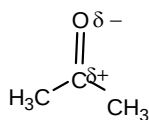
The symbol  $\text{+} \longrightarrow$  represents the direction of polarization of electrons in the H-F bond

The tendency of an atom to draw the electrons in a covalent bond toward itself is referred to as its electronegativity. An electronegative element attracts electrons; an electropositive one donates them. Electronegativity increases across a row in the periodic table. The most electronegative of the second-row elements is fluorine. Others include O, N, Cl.

| Element                       | F   | O   | N   | Cl  | Br  | C   | H   |
|-------------------------------|-----|-----|-----|-----|-----|-----|-----|
| Pauling electronegative scale | 4.0 | 3.5 | 3.0 | 3.0 | 2.8 | 2.5 | 2.1 |

When carbon forms a bond with those electronegative elements, the resulting bond is polar covalent bond.

Polar covalent bonds determine different properties of organic molecules.



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### 1.5 Structural Formulas of Organic Molecules

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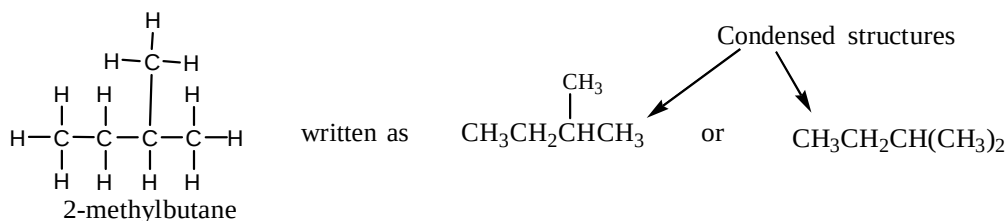
#### A) Molecular Formula

This indicates the total number of atoms present a molecule.

Example:  $\text{C}_6\text{H}_6$ ,  $\text{C}_2\text{H}_5\text{OH}$ ,  $\text{C}_{20}\text{H}_{42}$

#### B) Condensed structural formula

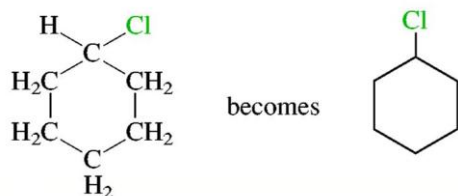
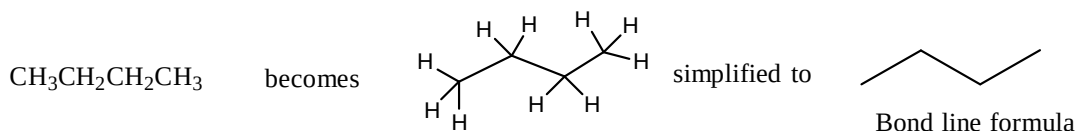
We leave out some, many, or all of the covalent bonds and use sub-scripts to indicate the number of identical groups attached to a particular atom.



In condensed structures, Carbon-hydrogen and carbon-carbon single bonds aren't shown; instead, they are understood.

### C) Bond-line formulas or carbon skeletal structure

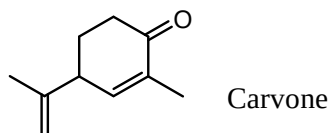
- Carbon atoms aren't usually shown. Instead, a carbon atom is assumed to be at each intersection of two lines (bonds) and at the end of each line.
- The hydrogen atoms bonded to carbon are not shown. Since carbon always has a valance of 4, we mentally supply the correct number of hydrogen atoms for each carbon.
- Atoms other than carbon and hydrogen are shown.



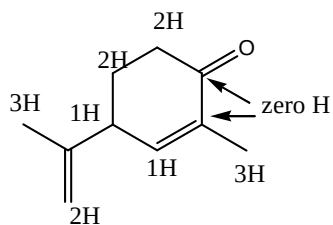
### Interpreting bond-line formula

We can find number of hydrogens bonded to each carbon and the molecular formula of a molecule from its bond-line formula.

Carvone is a substance responsible for the odor of spearmint, has the following structure. Find its molecular formula



The end of a line represents a carbon atom with 3 hydrogens,  $\text{CH}_3$ ; a two-way intersection is a carbon with 2 hydrogens,  $\text{CH}_2$ ; a three-way intersection is a carbon with 1 hydrogen,  $\text{CH}$ ; and a four intersection is a carbon atom with no attached hydrogen.

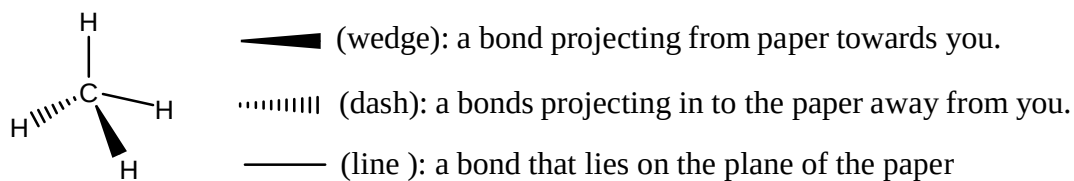


Carvone  
molecular formula,  $C_{10}H_{14}O$

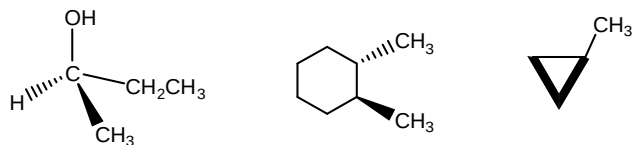
#### D) Wedge- line formula (three dimensional formula)

Organic molecules are three dimensional. We use this formula to draw three dimensional structure on a paper.

Methane is a tetrahedral molecule; its four hydrogens occupy the corners of a tetrahedron with carbon at its center has the following structure.

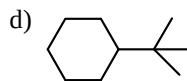
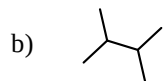
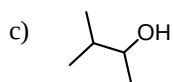
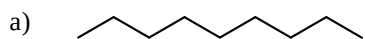


This type of formula is useful to show the stereochemistry of a molecule.

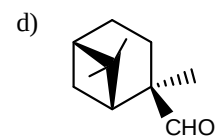
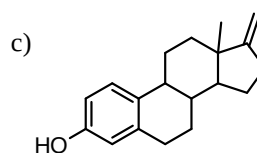
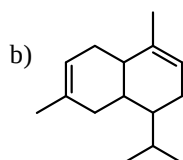
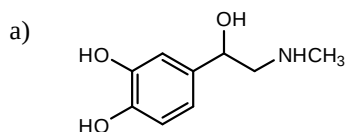


#### Exercise 1.7

- Expand the following bond-line representations to show all the atoms including carbon and hydrogen.



- Find the molecular formula for the following molecules



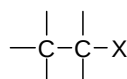
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## 1.6 Factors Influencing Electron Availability and Reactivity of Organic Compounds

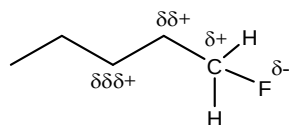
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### 1.6.1 Inductive Effect

Inductive effect depends up on the “intrinsic” tendency of a substituent to release or withdraw through the molecular chain or through space (through sigma bonds). Hydrogen of organic molecules can be replaced by various substituents. Substituents like F, Cl, Br, I, OH, NH<sub>2</sub>, NO<sub>2</sub> are more electronegative than hydrogen, and thus exert electron withdrawing inductive effects. Suppose X is one the groups listed above, withdraws or attracts electrons towards it.



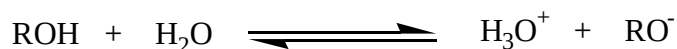
Thus, the carbon where X is attached feels positive. Moreover, the electro withdrawing effect of X is also transmitted through the sigma bond and carbon-2 will also be positive even though it is less positive than carbon-1. This phenomenon of transmission of charge through of a chain of atoms is called an **inductive effect**. The classic inductive effect is through bonds:



Inductive effects rationalize reactivities of organic compounds. Let's see some examples

### ACIDITY OF ALCOHOLS

The acidity of alcohols in water is expressed by the equilibrium constant K.



i.e when an alcohol is dissolved in water it will dissociate in to proton and alkoxide where the proton is taken by water to produce hydronium ion. The presence of electron withdrawing groups in the proximity of the oxygen-hydrogen bond increases the acidity of the alcohol. For example CF<sub>3</sub>CH<sub>2</sub>OH is more acidic than CH<sub>3</sub>CH<sub>2</sub>OH.

### INDUCTIVE EFFECT AND RELATIVE STRENGTH OF CARBOXYLICACIDS

In the examples below you can observe the effect of the presence of chlorine groups on the acidity of the carboxylic acid. As more chlorine groups which exert electron withdrawing inductive effect exist, the acidity strength increases (bigger K<sub>a</sub> Value)

|                                                                          |                                |
|--------------------------------------------------------------------------|--------------------------------|
| $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$          | $K_a$<br>$1.75 \times 10^{-5}$ |
| $\text{ClCH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$        | $1.86 \times 10^{-5}$          |
| $\text{Cl}_2\text{HC}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ | $5530 \times 10^{-5}$          |
| $\text{Cl}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$  | $23,200 \times 10^{-5}$        |

The electron withdrawing inductive effect of groups reduces as the group exists far from the reaction center. In the following example, the electron withdrawing inductive effect of chlorine is having little influence on the acidity strength as it is attached far from the carboxylic functional group.

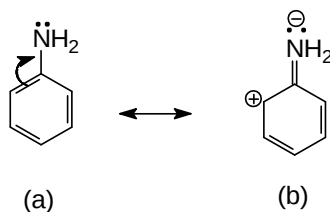
|                                                                                                                       |               |
|-----------------------------------------------------------------------------------------------------------------------|---------------|
|                                                                                                                       | $\text{PK}_a$ |
| $\text{CH}_3\text{CH}_2\text{CH}_2\overset{\text{O}}{\parallel}{\text{C}}\text{OH}$                                   | 4.82          |
| $\text{ClCH}_2\text{CH}_2\text{CH}_2\overset{\text{O}}{\parallel}{\text{C}}\text{OH}$                                 | 4.52          |
| $\text{CH}_3\underset{\text{Cl}}{\underset{ }{\text{CH}}}\text{CH}_2\overset{\text{O}}{\parallel}{\text{C}}\text{OH}$ | 4.05          |
| $\text{CH}_3\text{CH}_2\underset{\text{Cl}}{\underset{ }{\text{CH}}}\overset{\text{O}}{\parallel}{\text{C}}\text{OH}$ | 2.86          |

As you can see from the above data, the last compound is strong acid than the rest of the compounds because chlorine is closest to the reaction center i.e the carboxylic acid functional group. Its effect will be so great that the compound is strong acid.

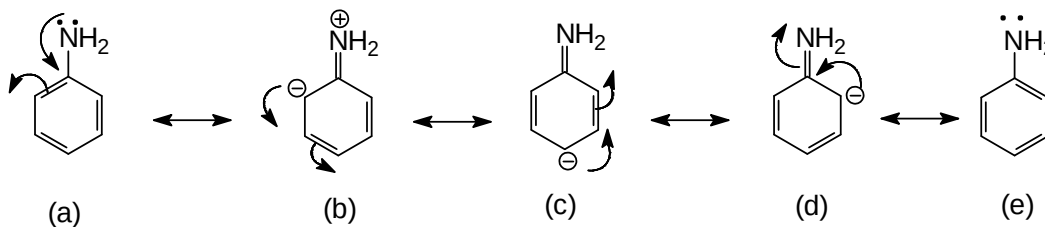
### 1.6.2 Resonance Effect

This involves delocalization  $\pi$ (pi) electrons and non-bonded electrons. It depends upon the overlap of certain orbitals. It also has effects on the reactivity of compounds.

The following example illustrates the resonance contributors of aniline.



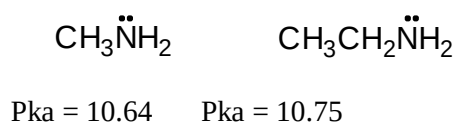
In (b) nitrogen consists of two more electrons than the maximum limit of eight electrons. Thus, (b) is an impossible resonance contributor of aniline. Here are the resonance contributors of aniline.



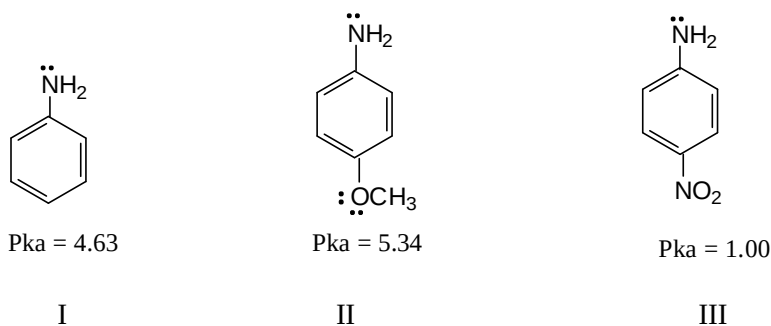
## APPLICATIONS OF RESONANCE EFFECTS

### BASICITY OF AMINES

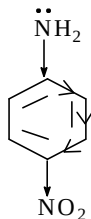
The strength of basicity depends on the degree of availability of lone pair electrons.



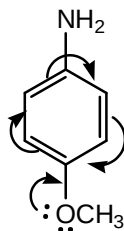
Obviously, ethylamine is more basic than methylamine because of the better electron releasing capacity of the ethyl group than methyl group.



The methoxy-substituted aniline is more basic than the two other compounds. Of course, p-nitro aniline is less basic because the nitro group withdraws electrons inductively. This reduces the charge density of the amino group making the lone pair electrons less available for an electron deficient species.

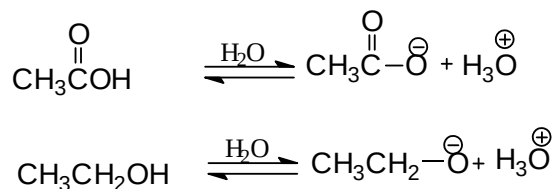
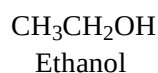
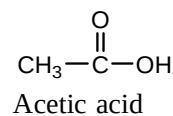


Thus, methoxy group increases the electron density of the amino group and makes the electron more available to be used by an electron-seeking reagent. *P*-methoxy aniline (III) is therefore more basic than the other two compounds (I and II).

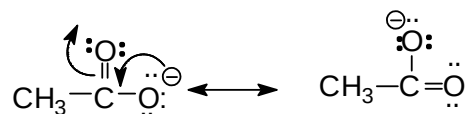


## ACIDITY

The degree of strength of acidity can be compared in a similar approach by using the concept of inductive and resonance effects. Let's look at acidity of acetic acid and ethanol.



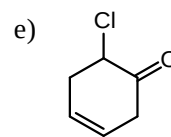
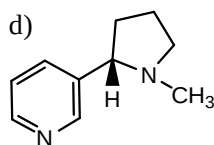
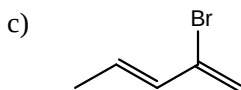
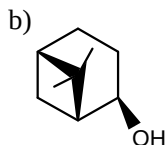
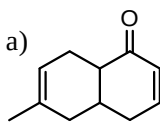
The conjugate base of acetic acid (acetate ion) is resonance stabilized while the conjugate base of ethanol (ethoxide) is not. Thus, acetate ion is formed readily from acetic than the formation of ethoxide from ethanol. We can say that acetic acid is stronger acid than ethanol.



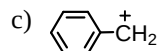
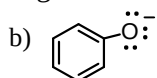
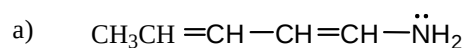
One more reason that supports the reason why acetic acid is stronger acid than ethanol lays on the fact that carbonyl group in acetic acid is an inductive electron withdrawing group that makes the oxygen-hydrogen bond more polarized. The more polar the oxygen-hydrogen bond is, the more ready to release proton and hence stronger acid. This is not observed in ethanol

## SELF -TEST EXERCISES 1

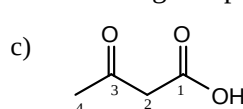
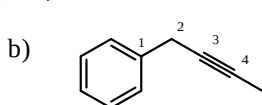
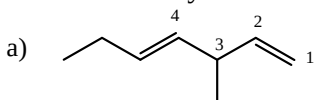
1 Find the molecular formula.



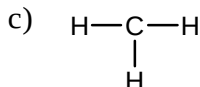
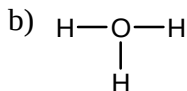
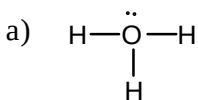
2 Write all resonance contributors for the following



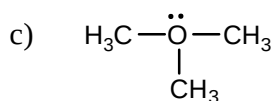
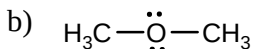
3 Determine the hybridization of C1, C2, C3 and C4 for each of the following compounds



4 Determine the formal charge at all the atoms in each of the following species and the net charge on the species as a whole



5 What is the formal charge of oxygen in each of the following Lewis structure?



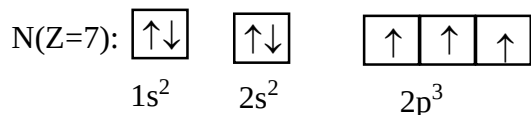
## ANSWER FOR EXERCISE

### Exercise 1.1

- Organic chemistry is the study of carbon containing compounds.
- Organic:  $\text{CH}_4$ ,  $\text{C}_2\text{H}_5\text{OH}$ ,  $\text{CH}_3\text{COOH}$ ,  $\text{HC}\equiv\text{CH}$   
Inorganic:  $\text{CO}_2$ ,  $\text{CS}_2$ ,  $\text{CaC}_2$ ,  $\text{CO}_3^-$ ,  $\text{Ln}_4(\text{C}_2)_3$

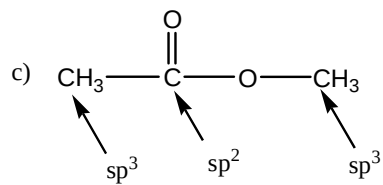
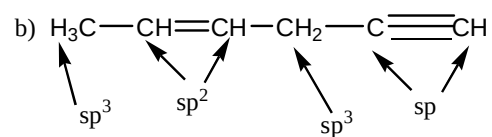
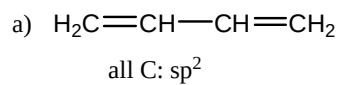
### Exercise 1.2

1.

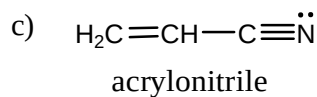
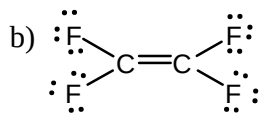
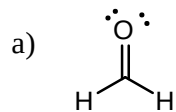


- Oxygen (Z=6):  $1s^2, 2s^2, 2p^4$   
Silicon (Z=14):  $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$   
Sulfur (Z=16):  $1s^2, 2s^2, 2p^6, 3s^2, 3p^4$

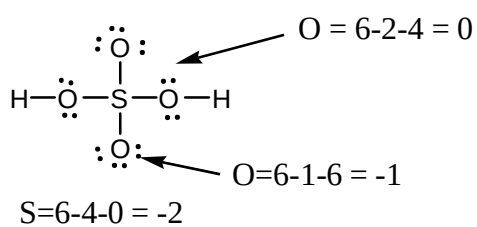
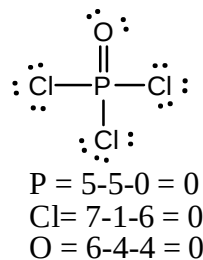
### Exercise 1.3



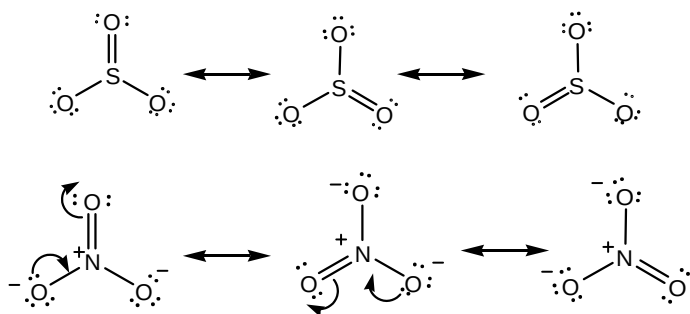
#### Exercise 1.4



#### Exercise 1.5

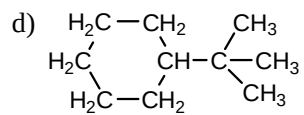
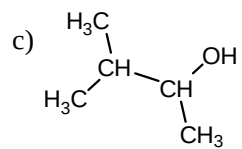
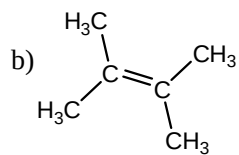
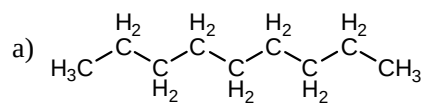


#### Exercise 1.6

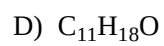
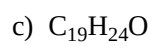
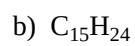
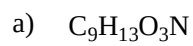


#### Exercise 1.7

1.



2.



# UNIT 2

## 2. FUNCTIONAL GROUPS IN ORGANIC CHEMISTRY

### Learning outcomes:

At the end of this unit, you will be able to:

- identify common functional groups.
- write common name and iupac name of organic compounds.
- relate properties and reactions of organic compounds to functional groups.
- know important reactions of a range of functional groups.
- predict reaction products of range of functional groups.
- explain the role of hydrogen bonding in organic chemistry, including its effect on the boiling point and solubility

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### 2.1 Functionality

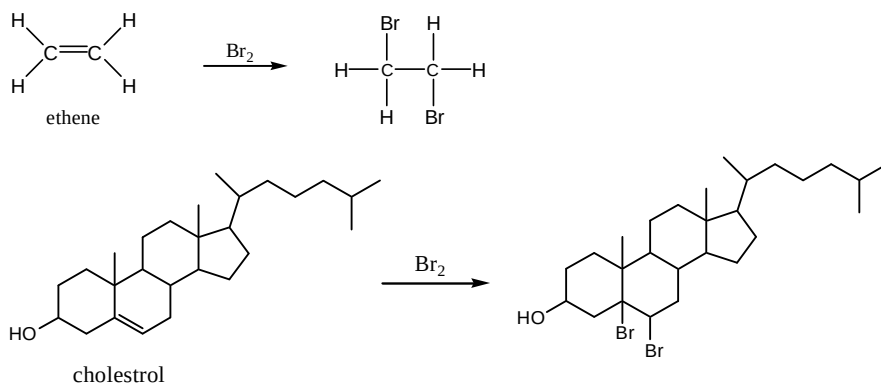
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❓ What is functional group?

A **functional group** is a particular combination of atoms in a molecule that is particularly responsible for characteristic set of reactions that the molecule exhibit.

The chemistry of every organic molecule, regardless of size and complexity, is determined by the functional group it contains.

For example, the functional group of an alkene is carbon-carbon double bond. The following two compounds which have carbon-carbon double bond undergo remarkably similar reaction, addition of bromine on double bonds.



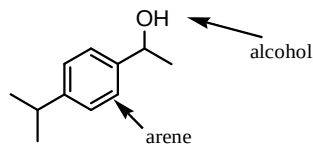
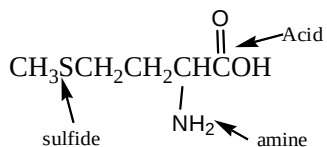
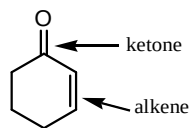
Nowaday, there are more than 10 million known compounds. Attempting to deal with each and individual compounds may make learning organic chemistry bulky. Instead of dealing with millions of compounds, we deal with dozens of functional groups which logically categorizes compounds according to their structural features.

### Structure of Some Functional Groups

| Name    | Functional Group Structure | Example                                       | Name            | Functional Group Structure | Example |
|---------|----------------------------|-----------------------------------------------|-----------------|----------------------------|---------|
| Alkene  |                            | $\text{H}_2\text{C}=\text{CH}-\text{CH}_3$    | Carbonyl        |                            |         |
| Alkyne  | $-\text{C}\equiv\text{C}-$ | $\text{H}-\text{C}\equiv\text{C}-\text{CH}_3$ | Aldehyde        |                            |         |
| Arene   |                            | Benzene                                       | Ketone          |                            |         |
| Halide  | <br>(X=F, Cl, Br, I)       | $\text{CH}_3\text{CH}_2\text{Cl}$             | Carboxylic acid |                            |         |
| Alcohol |                            | $\text{CH}_3\text{CH}_2\text{OH}$             | Ester           |                            |         |
| Ether   |                            | $\text{H}_3\text{C}-\text{O}-\text{CH}_3$     | Amide           |                            |         |
| Amine   |                            | $\text{CH}_3\text{CH}_2\text{NH}_2$           | Acid anhydride  |                            |         |
| Nitrile | $-\text{C}\equiv\text{N}$  | $\text{CH}_3-\text{C}\equiv\text{N}$          | Acid chloride   |                            |         |
| Nitro   |                            | $\text{CH}_3\text{CH}_2\text{NO}_2$           |                 |                            |         |
| Thiol   |                            | $\text{CH}_3\text{CH}_2\text{SH}$             |                 |                            |         |
| Sulfide |                            | $\text{H}_3\text{C}-\text{S}-\text{CH}_3$     |                 |                            |         |

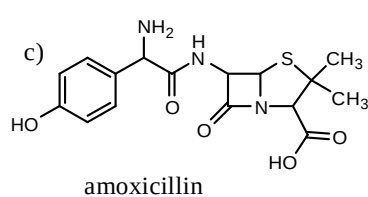
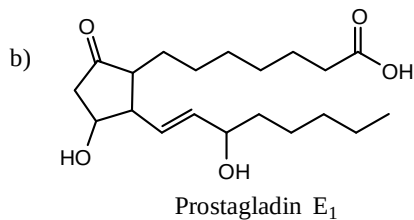
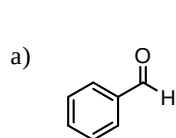
Many compounds contain more than one functional group.

Example: functional group identification



#### Exercise 2.1

Identify the functional groups in each of the following compounds.



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## 2.2 Nomenclature of Organic Compounds

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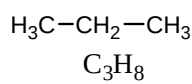
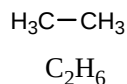
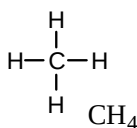
Nomenclature in organic chemistry is of two types: **common** (or "trivial") and **systematic**. Some common names existed long before organic chemistry became an organized branch of chemical science. Methane, ethane, propane, n-butane, isobutane, n-pentane, isopentane, and neopentane are common names. So long as there are only a few names and a few compounds, the task is manageable.

But there are millions of organic compounds already known. A system built on common names is not adequate to the task of communicating structural information. Beginning in 1892, chemists developed a set of rules for naming organic compounds based on their structures. We call these the **IUPAC** rules; IUPAC stands for the "International Union of Pure and Applied Chemistry."

Under this section we will deal with nomenclature of different organic molecules.

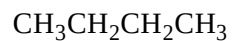
### A. Nomenclature of ALKANE

- are saturated hydrocarbons with C-C single bond. They are Saturated, because contain maximum possible number of H per C.

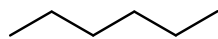


- General formula:  $\text{C}_n\text{H}_{2n+2}$
- Occur as straight chain and branched chain.
- Unbranched alkanes are named according to the number of C-atoms in their chain.

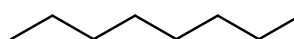
| Molecular Formula            | Structural Formula                        | IUPAC Name | Molecular Formula              | Structural Formula                           | IUPAC Name   |
|------------------------------|-------------------------------------------|------------|--------------------------------|----------------------------------------------|--------------|
| $\text{CH}_4$                | $\text{CH}_4$                             | Methane    | $\text{C}_{11}\text{H}_{24}$   | $\text{CH}_3-(\text{CH}_2)_9-\text{CH}_3$    | Undecane     |
| $\text{C}_2\text{H}_6$       | $\text{CH}_3-\text{CH}_3$                 | Ethane     | $\text{C}_{12}\text{H}_{26}$   | $\text{CH}_3-(\text{CH}_2)_{10}-\text{CH}_3$ | Dodecane     |
| $\text{C}_3\text{H}_8$       | $\text{CH}_3-\text{CH}_2-\text{CH}_3$     | Propane    | $\text{C}_{13}\text{H}_{28}$   | $\text{CH}_3-(\text{CH}_2)_{11}-\text{CH}_3$ | Tridecane    |
| $\text{C}_4\text{H}_{10}$    | $\text{CH}_3-(\text{CH}_2)_2-\text{CH}_3$ | Butane     | $\text{C}_{14}\text{H}_{30}$   | $\text{CH}_3-(\text{CH}_2)_{12}-\text{CH}_3$ | Tetradecane  |
| $\text{C}_5\text{H}_{12}$    | $\text{CH}_3-(\text{CH}_2)_3-\text{CH}_3$ | Pentane    | $\text{C}_{15}\text{H}_{32}$   | $\text{CH}_3-(\text{CH}_2)_{13}-\text{CH}_3$ | Pentadecane  |
| $\text{C}_6\text{H}_{14}$    | $\text{CH}_3-(\text{CH}_2)_4-\text{CH}_3$ | Hexane     | $\text{C}_{20}\text{H}_{42}$   | $\text{CH}_3-(\text{CH}_2)_{18}-\text{CH}_3$ | Icosane      |
| $\text{C}_7\text{H}_{16}$    | $\text{CH}_3-(\text{CH}_2)_5-\text{CH}_3$ | Heptane    | $\text{C}_{30}\text{H}_{62}$   |                                              | triacontane  |
| $\text{C}_8\text{H}_{18}$    | $\text{CH}_3-(\text{CH}_2)_6-\text{CH}_3$ | Octane     | $\text{C}_{40}\text{H}_{82}$   |                                              | tetracontane |
| $\text{C}_9\text{H}_{20}$    | $\text{CH}_3-(\text{CH}_2)_7-\text{CH}_3$ | Nonane     | $\text{C}_{50}\text{H}_{102}$  |                                              | tentacontane |
| $\text{C}_{10}\text{H}_{22}$ | $\text{CH}_3-(\text{CH}_2)_8-\text{CH}_3$ | Decane     | $\text{C}_{100}\text{H}_{202}$ |                                              | hectane      |



Butane



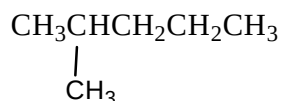
Hexane



Octane

## IUPAC NOMENCLATURE OF BRANCHED ALKANES

The IUPAC rules name branched alkanes as substituted derivatives of the unbranched alkanes listed in  
Consider the  $\text{C}_6\text{H}_{14}$  isomer represented by the structure



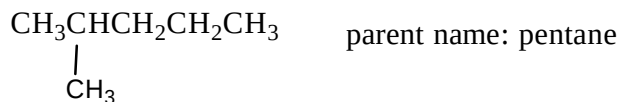
IUPAC name typically has four parts: prefix, locant, parent, and suffix.



### Step 1: Find the parent hydrocarbon

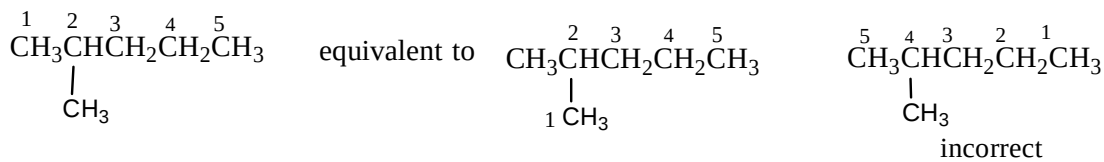
Pick out the longest continuous carbon chain, and find the IUPAC name to the unbranched alkane having that number of carbons. This is the parent alkane from which the IUPAC name is to be derived.

In this case, the longest continuous chain has five carbon atoms; the compound is named as a derivative of pentane. The key word here is continuous. It does not matter whether the carbon skeleton is drawn in an extended straight-chain form or in one with many bends and turns. All that matters is the number of carbons linked together in an uninterrupted sequence.



### Step 2: Number the longest continuous chain

Number the longest continuous chain in the direction that gives the lowest number to the substituent group at the first point of branching.

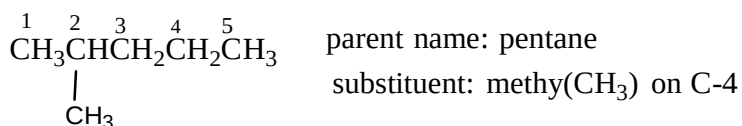


Both schemes count five carbon atoms in their longest continuous chain and bear a methyl group as a substituent at the second carbon. An alternative numbering sequence that begins at the other end of the chain is incorrect:

**Step 3:** Identify and number the substituents.

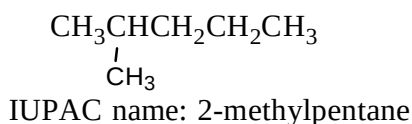
Identify the substituent groups attached to the parent chain.

The parent pentane chain bears a methyl (CH<sub>3</sub>) group as a substituent on carbon 4.



**Step 4 :** write the name as a single word.

Write the name of the compound. The parent alkane is the last part of the name and is preceded by the names of the substituent groups and their numerical locations (locants). Hyphens separate the locants from the words.



The same sequence of four steps gives the IUPAC name for the isomer that has its methyl group attached to the middle carbon of the five-carbon chain.



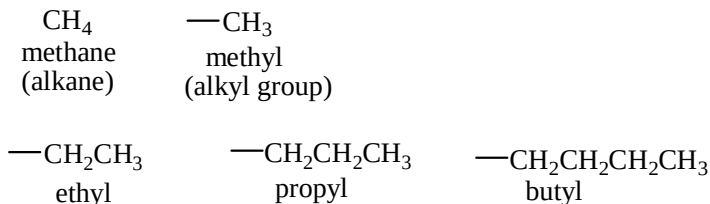
Both remaining C<sub>6</sub>H<sub>14</sub> isomers have two methyl groups as substituents on a four carbon chain. Thus the parent chain is butane. When the same substituent appears more than once, use the multiplying prefixes **di-**, **tri-**, **tetra-**, and so on. A separate locant is used for each substituent, and the locants are separated from each other by commas and from the words by hyphens.



This basic IUPAC rule is applicable to other organic functional group with some modification at each level. So far, the only branched alkanes that we've named have methyl groups attached to the main chain. What about groups other than CH<sub>3</sub>? What do we call these groups, and how do we name alkanes that contain them?

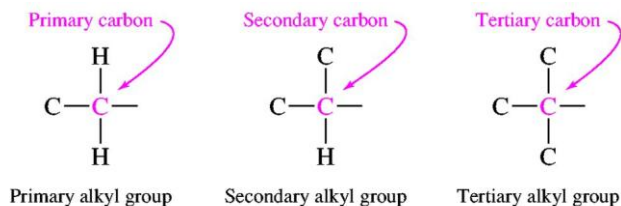
## ALKYL GROUPS

An alkyl group lacks one of the hydrogens of an alkane. A methyl group ( $\text{CH}_3\text{-}$ ) is an alkyl group derived from methane ( $\text{CH}_4$ ). Unbranched alkyl groups in which the point of attachment is at the end of the chain are named in IUPAC nomenclature by replacing the -ane endings -yl.



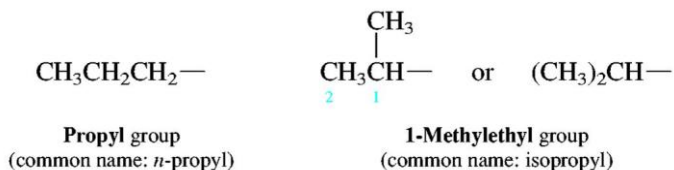
The dash at the end of the chain represents a potential point of attachment for some other atom or group.

Carbon atoms are classified according to their degree of substitution by other carbons. A **primary** carbon ( $1^\circ$ ) is directly attached to one other carbon. Similarly, a **secondary** carbon ( $2^\circ$ ) is directly attached to two other carbons, a **tertiary** carbon ( $3^\circ$ ) to three, and a **quaternary** carbon ( $4^\circ$ ) to four. Alkyl groups are designated as primary, secondary, or tertiary according to the degree of substitution of the carbon at the potential point of attachment.

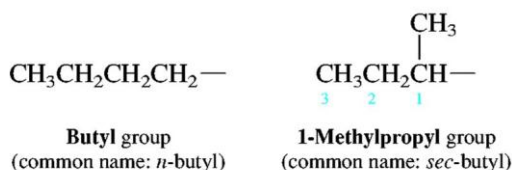


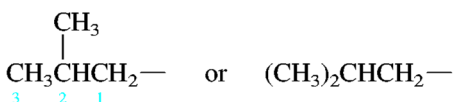
Ethyl ( $\text{CH}_3\text{CH}_2\text{-}$ ), propyl ( $\text{CH}_3\text{CH}_2\text{CH}_2\text{-}$ ) and butyl are examples of primary alkyl groups.

Branched alkyl groups are named by using the longest continuous chain that begins at the point of attachment as the base name. Thus, the systematic names of the two  $\text{C}_3\text{H}_7$  alkyl groups are propyl and 1-methylethyl. Both are better known by their common names, n-propyl and isopropyl, respectively.

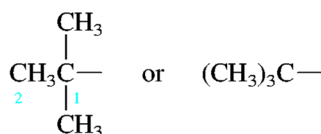


The  $\text{C}_4\text{H}_9$  alkyl groups may be derived either from the unbranched carbon skeleton of butane or from the branched carbon skeleton of isobutane. Those derived from butane are the butyl (n-butyl) group and the 1-methylpropyl (sec-butyl) group.

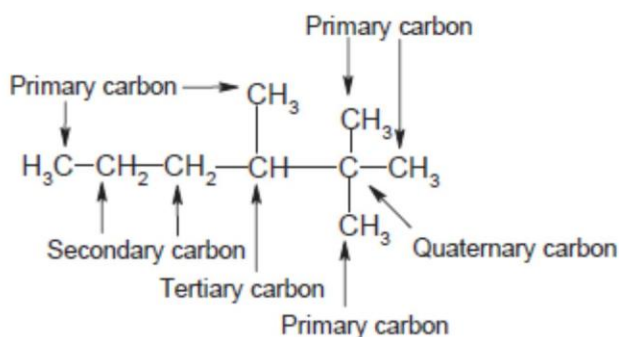




**2-Methylpropyl group**  
(common name: isobutyl)

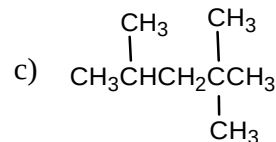
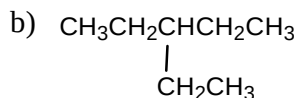
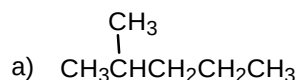


**1,1-Dimethylethyl group**  
(common name: *tert*-butyl)

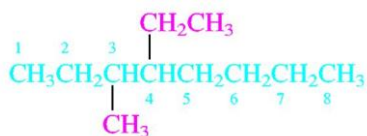


### Exercise 2.2

Identify the carbon atoms in the following molecules as primary, secondary and tertiary, or quaternary.

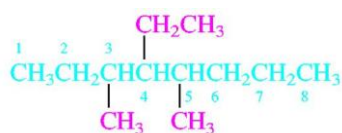


### IUPAC NAMES OF HIGHLY BRANCHED ALKANES



The compound becomes an octane derivative that bears a C-3 methyl group and a C-4 ethyl group. When two or more different substituents are present, they are listed in alphabetical order in the name. The IUPAC name for this compound is 4-ethyl-3-methyloctane.

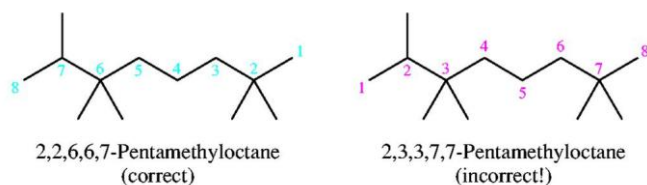
Replicating prefixes such as di-, tri-, and tetra- are used as needed but are ignored when alphabetizing. Adding a second methyl group to the original structure, at C-5, for example, converts it to 4-ethyl-3,5-dimethyloctane.



4-ethyl-3,5-dimethyloctane.

Italicized prefixes such as *sec*- and *tert*- are ignored when alphabetizing except when they are compared with each other. *tert*-Butyl precedes *isobutyl*, and *sec*-butyl precedes *tert*-butyl.

An additional feature of IUPAC nomenclature that concerns the direction of numbering is the "first point of difference" rule. Consider the two directions in which the following alkane may be numbered:



When deciding on the proper direction, a point of difference occurs when one order gives a lower locant than another. Thus, although 2 is the first locant in both numbering schemes, the tie is broken at the second locant, and the rule favors 2,2,6,6,7, which has 2 as its second locant, whereas 3 is the second locant in 2,3,3,7,7. Notice that locants are not added together, but examined one by one.

Finally, when equal locants are generated from two different numbering directions, choose the direction that gives the lower number to the substituent that appears first in the name. (Remember, substituents are listed alphabetically.)

N.B: Iso-is considered for alphabet order

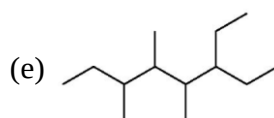
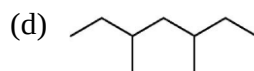
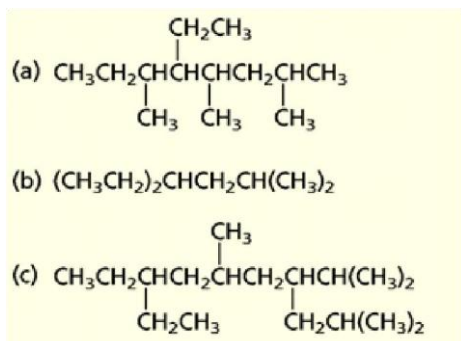
Isobutyl, isopropyl....

Sec-, tert-(Hyphend prefixes) not considered for alphabet order.

Sec-btyl, tert-butl....

### Exercise 2.3

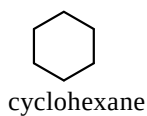
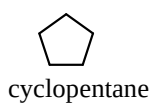
Give an acceptable IUPAC name for the following alkanes.



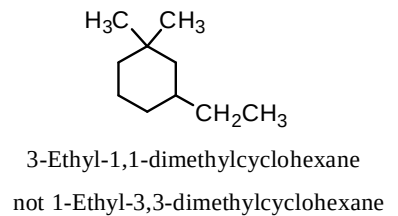
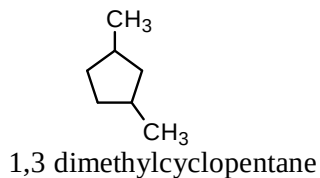
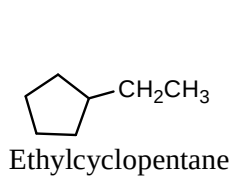
## CYCLOALKANE NOMENCLATURE

- Cycloalkanes are alkanes that contain a ring of three or more carbons.
- Are characterized by the molecular formula  $\text{C}_n\text{H}_{2n}$ .

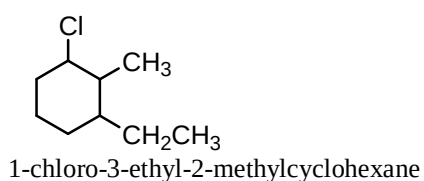
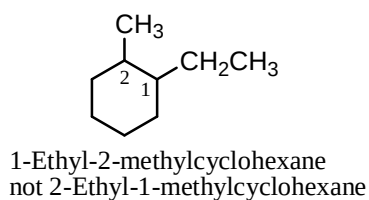
- Named by adding the prefix cyclo- to the name of the unbranched alkane with the same number of carbons as the ring.



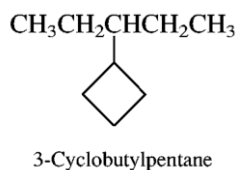
Substituent groups are identified in the usual way. Choose a point of attachment at C-1 and number the ring in the direction that gives the lowest number to the substituents at the first point of difference.



For different alkyl groups, number them by alphabetical priority.

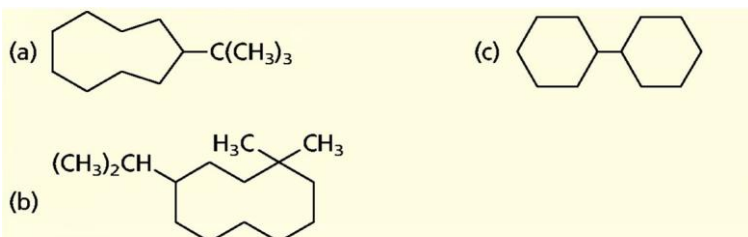


When the ring contains fewer carbon atoms than an alkyl group attached to it, the compound is named as an alkane, and the ring is treated as a cycloalkyl substituent:



### Exercise 2.4

Name for the following Compounds



## POLYCYCLIC ALKANES

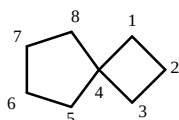
- are alkanes containing more than one ring system

a) **spirocyclic compounds:** two rings share one common atom.

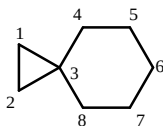
Named as spiro[n,m]alkane

n and m are number of atoms in each bridge (excluding the spiro atom)

Parent name is total number of carbon in the ring. The C-atom in the smaller ring containing the common atom (spiro atom) is assigned no. 1



Spiro[3,4]octane



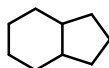
Spiro[2,5]octane



Spiro[3,3]heptane

b) **Fused ring compounds:** two rings share in common side

i.e. there are two common atoms which are adjacent to each other.



c) **Bridge- ring compounds:** two rings share more than one common side

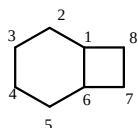
There are more than two common atoms which are adjacent to each other.



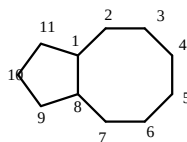
Both Fused ring Bridge- ring compounds named using the same set of rules.

Bicyclic compounds are named in the IUPAC system by counting the number of carbons in the ring system, assigning to the structure the base name of the unbranched alkane having the same number of carbon atoms, and attaching the prefix bicyclo-. The number of atoms in each of the bridges connecting the common atoms is then placed, in descending order, within brackets.

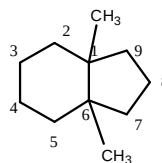
Number the compound by beginning at one bridgehead, then proceeding around the 2<sup>nd</sup> longest and then finally to the shortest bridge.



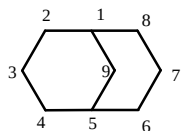
Bicyclo[4,2,0]octane



Bicyclo[6,3,0]undecane



1,6-dimethylbicyclo[4,3,0]nonane



Bicyclo[3,3,1]nonane



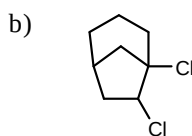
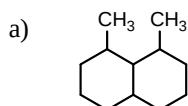
Bicyclo[2,2,1]heptane



Bicyclo[3.2.1]octane

### Exercise 2.5

1. Name for the following Compounds



2. Write the structural formula for each of the following compounds.

a) Bicyclo[2,2,2] octane

b) Bicyclo[5,2,0]nonane

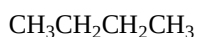
## Isomerism in Alkanes

Different compounds that have the same molecular formula are called **Isomers**

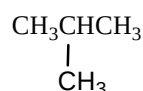
**Constitutional isomers:** isomers that have the same molecular formula but differ in the order in which their atoms are connected.

Alkanes show constitutional isomerism.

$C_4H_{10}$  isomers

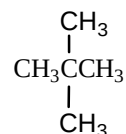
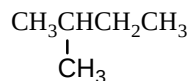


n-butane



isobutane

$C_5H_{12}$  isomers : there are 3 constitutional isomers of  $C_5H_{12}$



How many isomers possible for  $C_6H_{14}$ ?

You cannot easily calculate the number of isomers. The best way to ensure that you have written all the isomers of a particular molecular formula is to work systematically, beginning with the unbranched chain and then shortening it while adding branches one by one.

The data in Table 2.1 were determined by a mathematician who concluded that no simple expression can calculate the number of isomers.

Table 2.1 number of constitutional isomers

| Molecular formula              | No. of constitutional isomers | Molecular formula               | No. of constitutional isomers |
|--------------------------------|-------------------------------|---------------------------------|-------------------------------|
| CH <sub>4</sub>                | 1                             | C <sub>7</sub> H <sub>16</sub>  | 9                             |
| C <sub>2</sub> H <sub>6</sub>  | 1                             | C <sub>8</sub> H <sub>18</sub>  | 18                            |
| C <sub>3</sub> H <sub>8</sub>  | 1                             | C <sub>9</sub> H <sub>20</sub>  | 35                            |
| C <sub>4</sub> H <sub>10</sub> | 2                             | C <sub>10</sub> H <sub>22</sub> | 75                            |
| C <sub>5</sub> H <sub>12</sub> | 3                             | C <sub>15</sub> H <sub>32</sub> | 4347                          |
| C <sub>6</sub> H <sub>14</sub> | 5                             |                                 |                               |

### Exercise 2.6

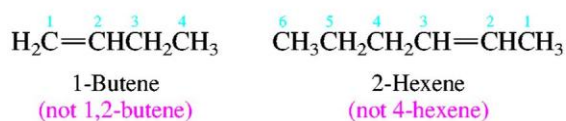
Write the structure of the 5 isomers of C<sub>6</sub>H<sub>12</sub>

## B. Nomenclature of ALKENES

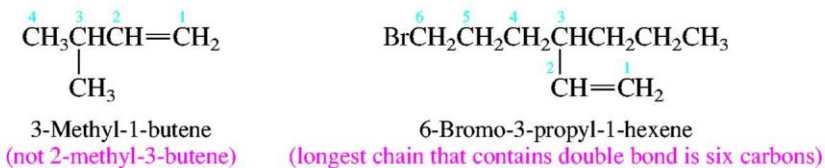
Alkenes are unsaturated hydrocarbons with carbon-carbon double bond.

Rules similar to alkanes. We give alkenes IUPAC names by replacing the -ane ending of the corresponding alkane with -ene.

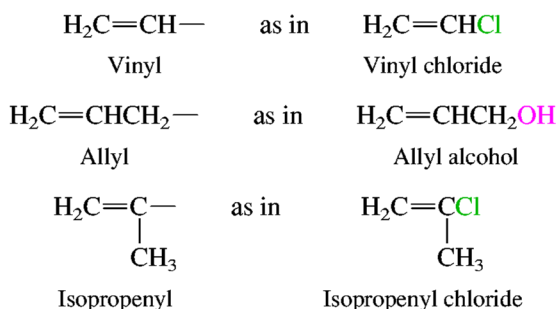
The longest continuous chain that includes the double bond forms the base name of the alkene, and the chain is numbered in the direction that gives the doubly bonded carbons their lower numbers. The locant (or numerical position) of only one of the doubly bonded carbons is specified in the name; it is understood that the other doubly bonded carbon must follow in sequence.



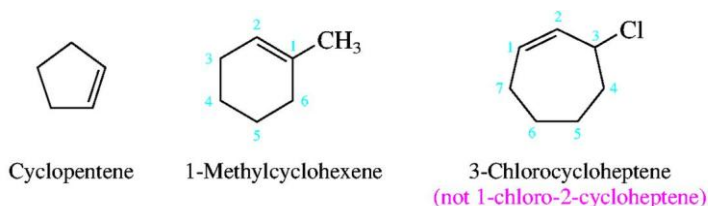
Carbon-carbon double bonds take precedence over alkyl groups and halogens in determining the main carbon chain and the direction in which it is numbered.



There are three alkenyl groups—vinyl, allyl, and isopropenyl—are frequently encountered groups, are acceptable in the IUPAC system like other alkyl groups.

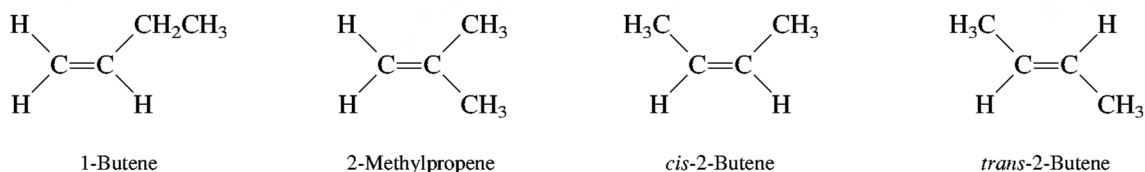


**Cycloalkenes** and their derivatives are named by adapting cycloalkane terminology to the principles of alkene nomenclature. Number the cycloalkane so that the double bond is between C-1 and C-2.



## ISOMERISM IN ALKENES

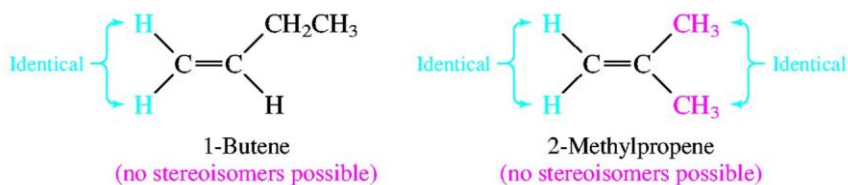
Alkenes show cis-trans isomerism. There are four isomeric alkenes of molecular formula  $\text{C}_4\text{H}_8$ :



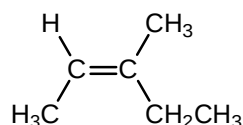
Cis isomer has identical groups on the same side of the double bond, but in the trans isomer they are on opposite sides of the double bond.

Isomers that have the same constitution but differ in the arrangement of their atoms in space are classified as **stereoisomers**. cis-2-Butene and trans-2-butene are stereoisomers, and the terms cis and trans specify the configuration of the double bond.

Cis-trans stereoisomerism in alkenes is not possible when one of the doubly bonded carbons bears two identical substituents. Thus, neither 1-butene nor 2-methylpropene can have cis-trans isomers.



The terms cis and trans are ambiguous, however, when it is not obvious which substituent on one carbon is "similar" or "analogous" to a reference substituent on the other.



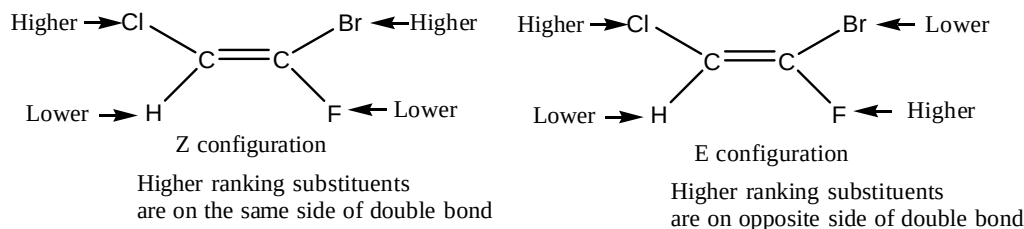
Cis or trans ?

Fortunately, a completely unambiguous system for specifying double bond stereochemistry has been developed based on an atomic number criterion for ranking substituents on the doubly bonded carbons. This system is called the E/Z system.

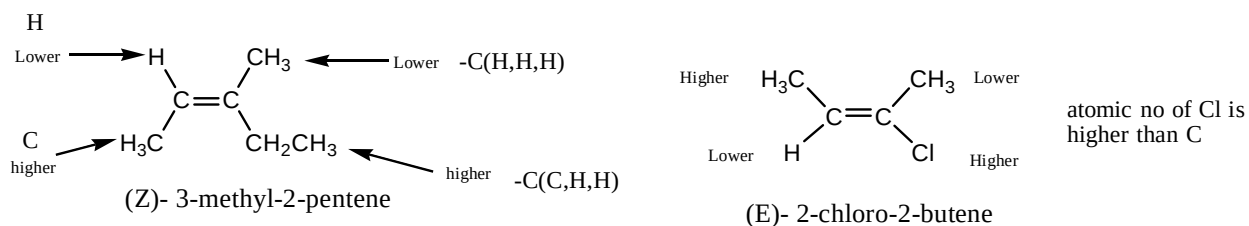
### THE E-Z NOTATIONAL SYSTEM

Rank the atoms directly attached to the double bond. When atoms of higher atomic number are on the same side of the double bond, we say that the double bond has the Z configuration, where Z stands for the German word *zusammen*, meaning "together."

When atoms of higher atomic number are on opposite sides of the double bond, we say that the configuration is E. The symbol E stands for the German word *entgegen*, meaning "opposite."

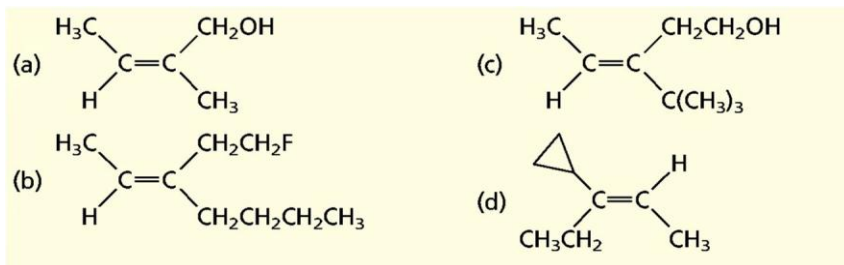


### Example



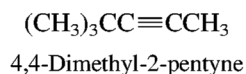
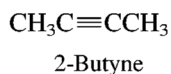
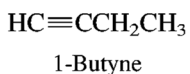
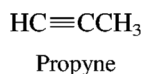
### Exercise 2.7

Determine the configuration of each of the following alkenes as Z or E as appropriate:



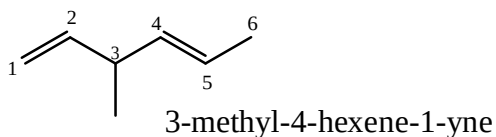
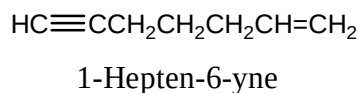
### C. Nomenclature of ALKYNES

- carbon-carbon triple bond
- -yne suffix (with number)
- rules similar to alkenes



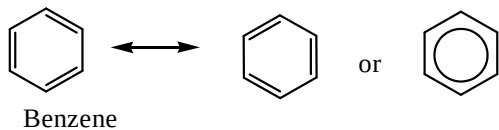
**Enynes:** containing both double and triple bond.

- with both -enes and -ynes, suffix is -en-yne and numbering is from the end closer to a multiple bond.
- When there is equal chance in numbering, double bond receives lower no. than triple bond.

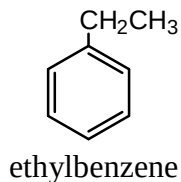
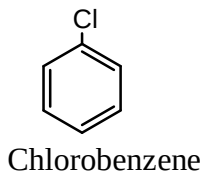


### D. Nomenclature of Aromatic Hydrocarbons

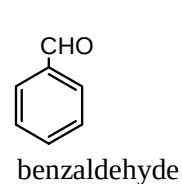
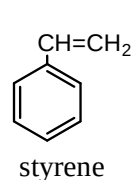
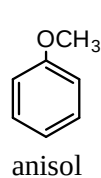
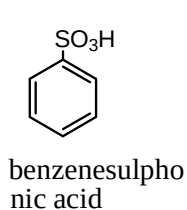
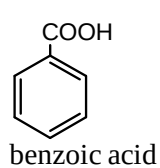
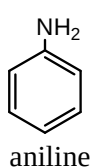
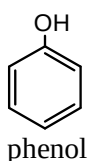
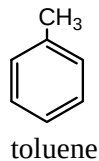
Aromatic hydrocarbon includes benzene and its derivative.



**Monosubstituted** benzenes are named as “substituent benzene”

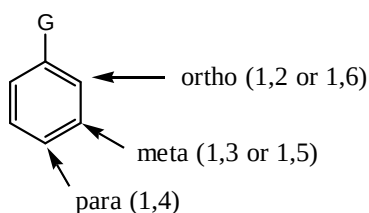


Common names of special monosubstituted benzenes that are also used in IUPAC system

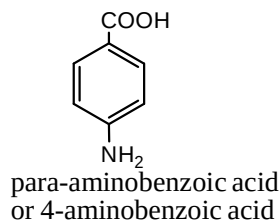
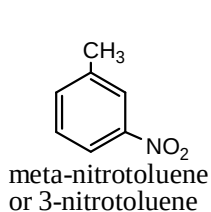
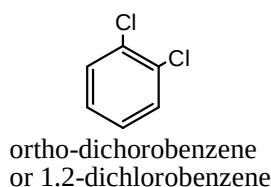


## Disubstituted benzene

There are two ways that disubstituted benzene derivatives can be named: Ortho-Meta-Para Designation and Numbering approach.

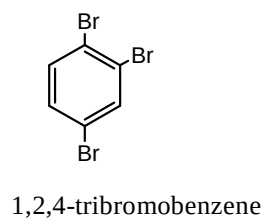
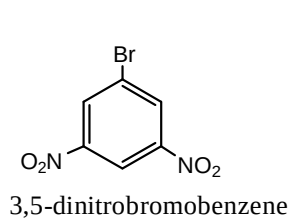
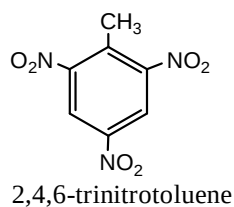


For groups that are part of special monosubstituted benzenes; the special group is at position **1**.



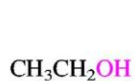
## Polysubstituted benzene

If more than two substituents are present their position is indicated by the use of number, the benzene ring is numbered so as to give the lowest possible number to the substituent.

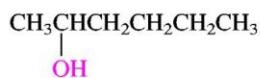


## E. Nomenclature of ALCOHOL

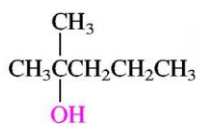
- OH group takes priority (even over -ene or -yne)
- it must be in the parent chain
- the direction of numbering gives it the lowest possible number
- ol suffix with number designation
- name other substituents and multiple bonds as usual



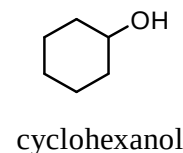
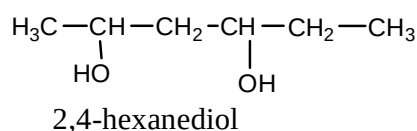
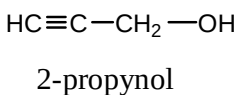
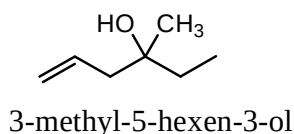
Ethyl alcohol  
Ethanol



1-Methylpentyl alcohol  
2-Hexanol

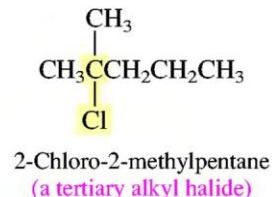
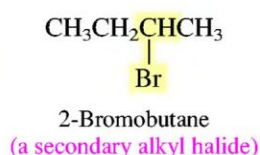
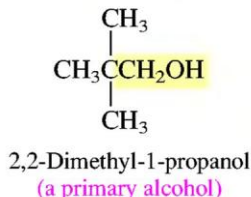
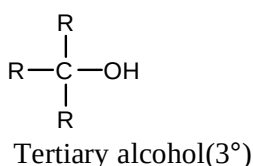
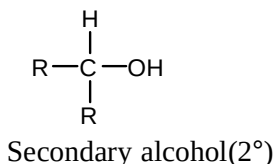
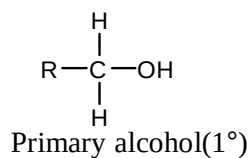


1,1-Dimethylbutyl alcohol  
2-Methyl-2-pentanol



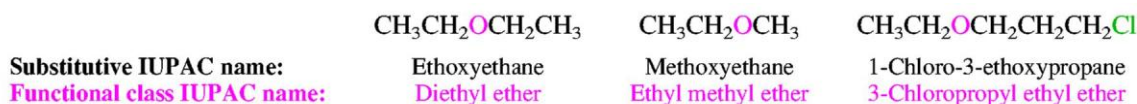
## CLASSES OF ALCOHOLS

Alcohols are classified as primary, secondary, or tertiary according to the degree of substitution of the carbon that bears the functional group.

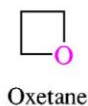


## F. Nomenclature of Ethers and Epoxides

Ethers are named, in substitutive IUPAC nomenclature, as alkoxy derivatives of alkanes. Functional class IUPAC names of ethers are derived by listing the two alkyl groups in the general structure ROR' in alphabetical order as separate words, and then adding the word ether at the end. When both alkyl groups are the same, the prefix di- precedes the name of the alkyl group.

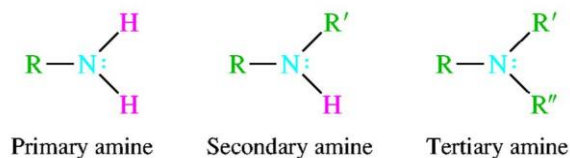


Cyclic ethers have their oxygen as part of a ring--they are heterocyclic compounds. Several have specific IUPAC names.

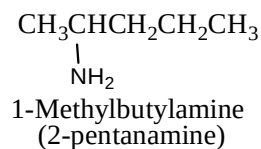
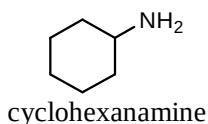
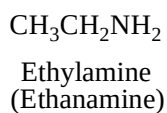


## G. Nomenclature of Amines

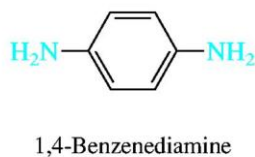
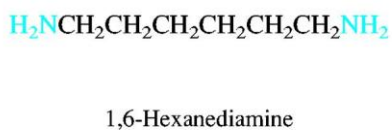
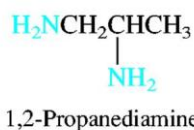
Unlike alcohols and alkyl halides, which are classified as primary, secondary or tertiary according to the degree of substitution at the carbon that bears the functional group, amines are classified according to their degree of substitution at nitrogen. An amine with one carbon attached to nitrogen is a primary amine; an amine with two is a secondary amine, and an amine with three is a tertiary amine.



Amines are named in two main ways in the IUPAC system, either as *alkylamines* or as *alkanamines*. When primary amines are named as alkylamines, the ending -amine is added to the name of the alkyl group that bears the nitrogen. When named as alkanamines, the alkyl group is named as an alkane and the -e ending replaced by -amine.



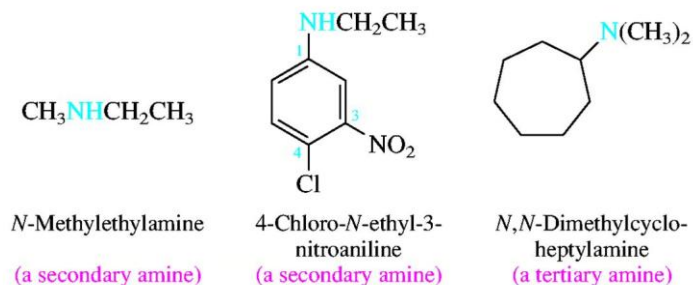
Compounds with two amine groups are named by adding the suffix -diamine to the name of the corresponding alkane or arene. The final -e of the parent hydrocarbon is retained.



Amino groups rank rather low in seniority when the parent compound is identified for naming purposes. Hydroxyl groups and carbonyl groups outrank amine groups. In these cases, the amine group is named as a substituent. (see priority order below)



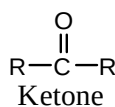
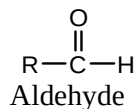
Secondary and tertiary amines are named as N-substituted derivatives of primary amines. The parent primary amine is taken to be the one with the longest carbon chain. The prefix N- is added as a locant to identify substituents on the amine nitrogen as needed.



When two or more functional groups are found in a molecule the parent hydrocarbon is determined by the following Priority order:

**Priority order from highest to lowest:** Carboxylic acid > ester > amide > nitrile > aldehyde > ketone > alcohol > amine > alkene > alkyne

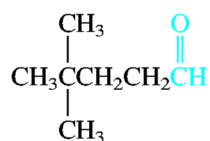
## H. Nomenclature of Aldehydes and Ketones



### Aldehyde

The longest continuous chain that contains the  $-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$  group provides the base name for aldehydes. The *-e* ending of the corresponding alkane name is replaced by *-al*, and substituents are specified in the usual way.

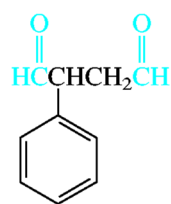
It is not necessary to specify the location of the  $-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$  group in the name, because the chain must be numbered by starting with this group as C-1. The suffix *-dial* is added to the appropriate alkane name when the compound contains two aldehyde functions.



4,4-Dimethylpentanal

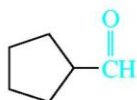


5-Hexenal

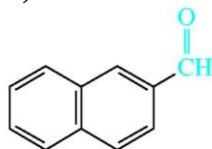


2-Phenylbutanedial

When a formyl group ( $-\text{CH}=\text{O}$ ) is attached to a ring, the ring name is followed by the suffix *-carbaldehyde*.

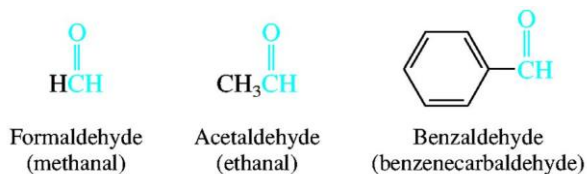


Cyclopentanecarbaldehyde



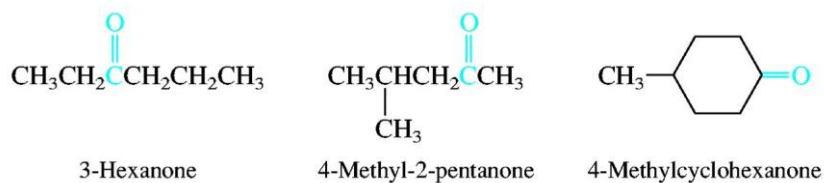
2-Naphthalenecarbaldehyde

Certain common names of familiar aldehydes are acceptable as IUPAC names. A few examples include

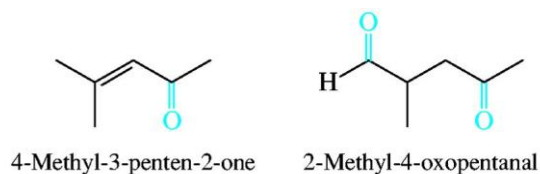


## Ketones

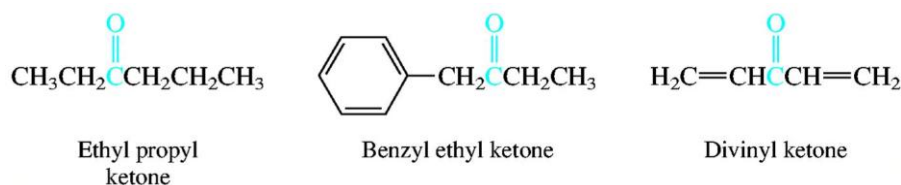
With ketones, the *-e* ending of an alkane is replaced by *-one* in the longest continuous chain containing the carbonyl group. The chain is numbered in the direction that provides the lower number for this group. The carbonyl carbon of a cyclic ketone is C-1 and the number does not appear in the name.



Aldehydes outrank ketones, however, and a compound that contains both an aldehyde and a ketone carbonyl group is named as an aldehyde. In such cases, the carbonyl oxygen of the ketone is considered an oxo-substituent on the main chain.

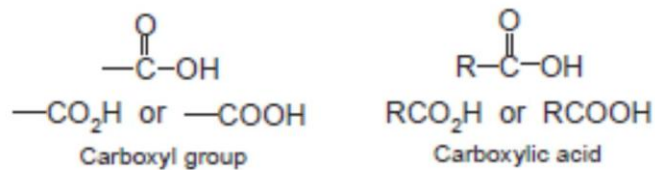


Ketones also named by functional class nomenclature. The groups attached to the carbonyl group are named as separate words followed by the word ketone. The groups are listed alphabetically.

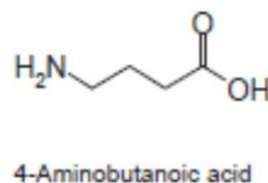
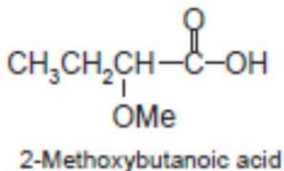
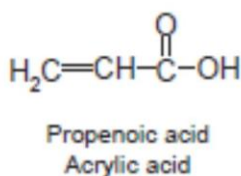
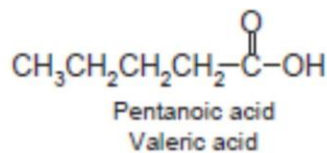
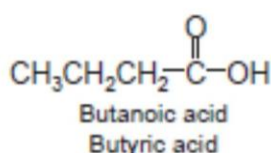
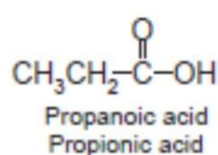


## I. Nomenclature of Carboxylic Acids and their Derivatives

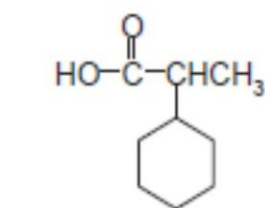
Carboxylic acid is an organic acid that has an acyl group ( $R-C=O$ ) linked to a hydroxyl group ( $-OH$ )



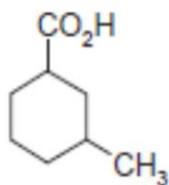
The root name is based on the longest continuous chain of carbon atoms bearing the carboxyl group. The -e is replaced by -oic acid. The chain is numbered starting with the carboxyl carbon atom. The carboxyl group takes priority over any other functional groups



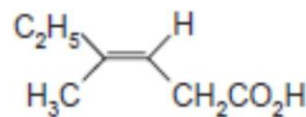
Cycloalkanes with carboxyl substituents are named as cycloalkanecarboxylic acids. Unsaturated acids are named using the name of the alkene with -e replaced with -oic acid. The chain is numbered starting with the carboxyl group, a number designates the location of the double bond and Z or E is used.



2-Cyclohexylpropanoic acid



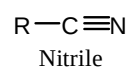
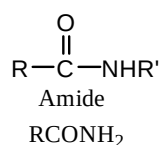
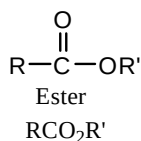
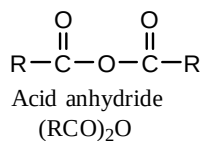
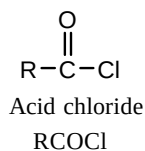
3-Methylcyclohexanecarboxylic acid



(E)-4-Methyl-3-hexenoic acid

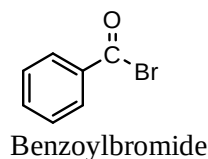
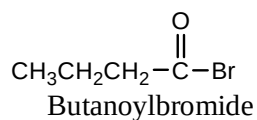
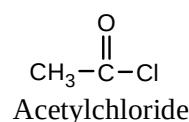
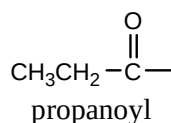
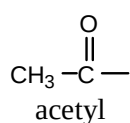
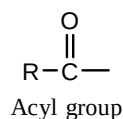
## Carboxylic Acids Derivatives

Carboxylic acid derivatives are compounds that possess an acyl group ( $\text{R}-\text{C}(=\text{O})$ ) linked to an electronegative atom, e.g. -Cl,  $-\text{CO}_2\text{R}$ , -OR or  $-\text{NH}_2$ . They can be converted to carboxylic acids via simple acidic or basic hydrolysis. The important acid derivatives are acid chlorides, acid anhydrides, esters and amides. Usually nitriles are also considered as carboxylic acid derivatives. Although nitriles are not directly carboxylic acid derivatives, they are conveniently hydrolysed to carboxylic acids by acid or base catalysts. Moreover, nitriles can be easily prepared through dehydration of amides, which are carboxylic acid derivatives.

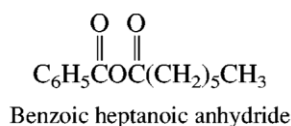
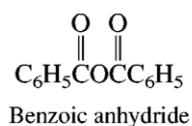
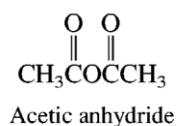


Acyl groups are named by replacing the -ic acid ending of the corresponding carboxylic acid by -yl.

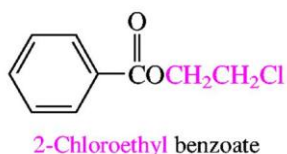
Acyl halides are named by placing the name of the appropriate halide after that of the acyl group.



In naming carboxylic **acid anhydrides** in which both acyl groups are the same, we simply specify the acid and replace acid by anhydride. When the acyl groups are different, they are cited in alphabetical order.

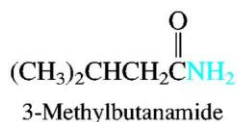
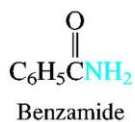
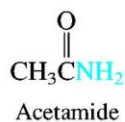


The alkyl group and the acyl group of an *ester* are specified independently. Esters are named as *alkyl alkanoates*. The alkyl group R' of  $\text{RCOR}'$  is cited first, followed by the acyl portion  $\overset{\text{O}}{\parallel}{\text{RC}}-$ . The acyl portion is named by substituting the suffix *-ate* for the *-ic acid* ending of the corresponding acid.

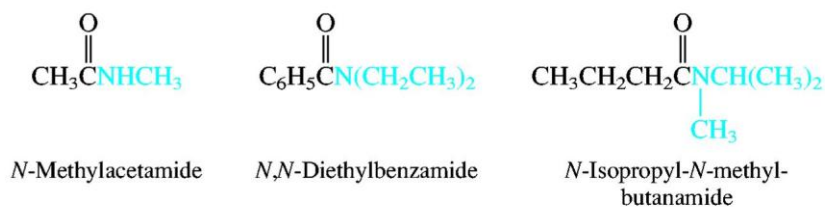


Aryl esters, that is, compounds of the type  $\overset{\text{O}}{\parallel}{\text{RCOAr}}$ , are named in an analogous way.

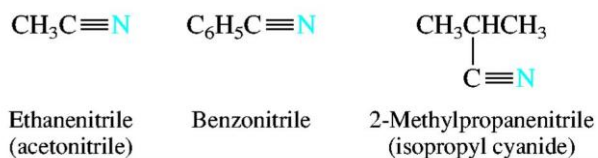
The names of *amides* of the type  $\overset{\text{O}}{\parallel}{\text{RCNH}_2}$  are derived from carboxylic acids by replacing the suffix *-oic acid* or *-ic acid* by *-amide*.



We name compounds of the type  $\text{RCNHR}'$  and  $\text{RCNR}'_2$  as *N*-alkyl- and *N,N*-dialkyl-substituted derivatives of a parent amide.

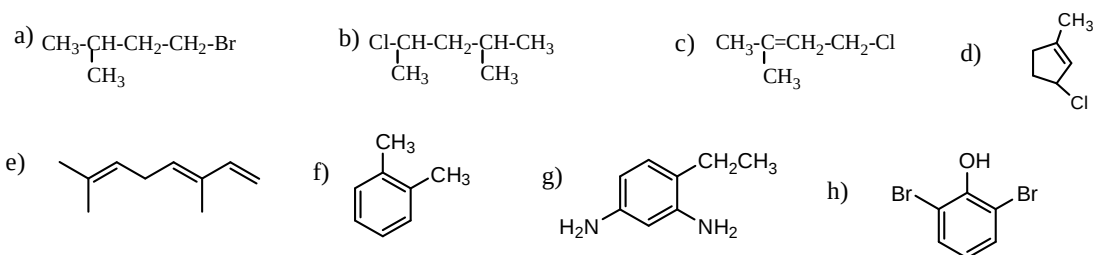


Substitutive IUPAC names for nitriles add the suffix -nitrile to the name of the parent hydrocarbon chain that includes the carbon of the cyano group. Nitriles may also be named by replacing the -ic acid or -oic acid ending of the corresponding carboxylic acid with *-onitrile*. Alternatively, they are sometimes given functional class IUPAC names as alkyl cyanides.

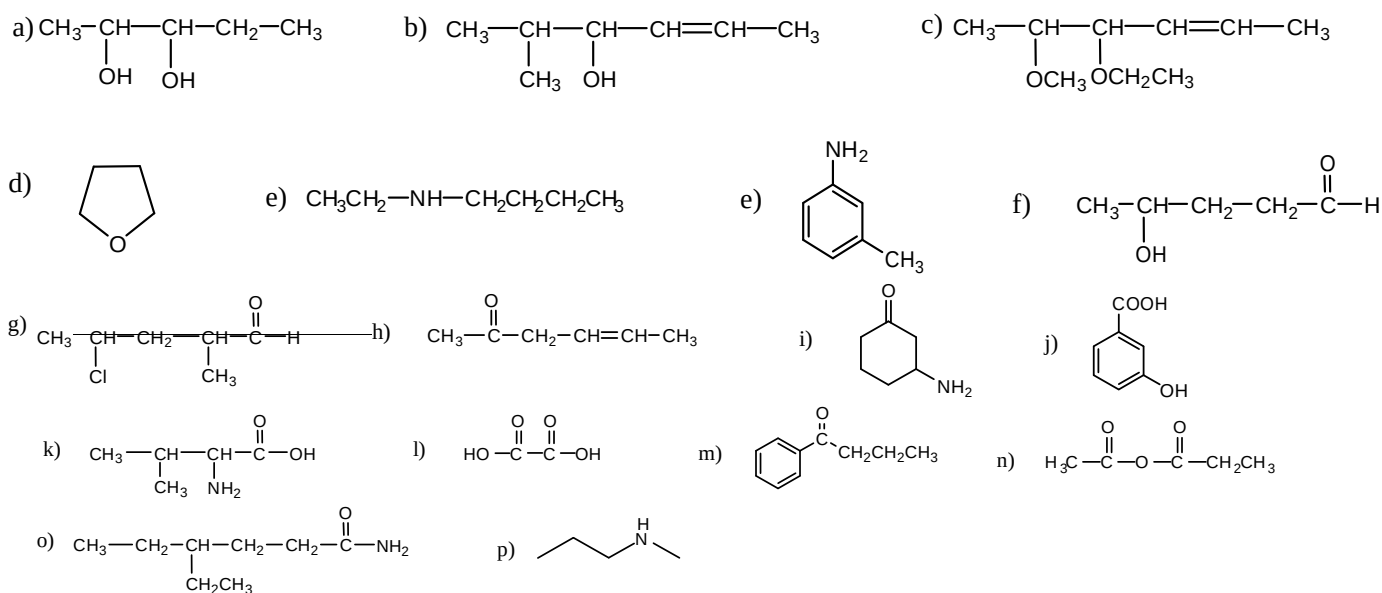


## SELF -TEST EXERCISES 2.1

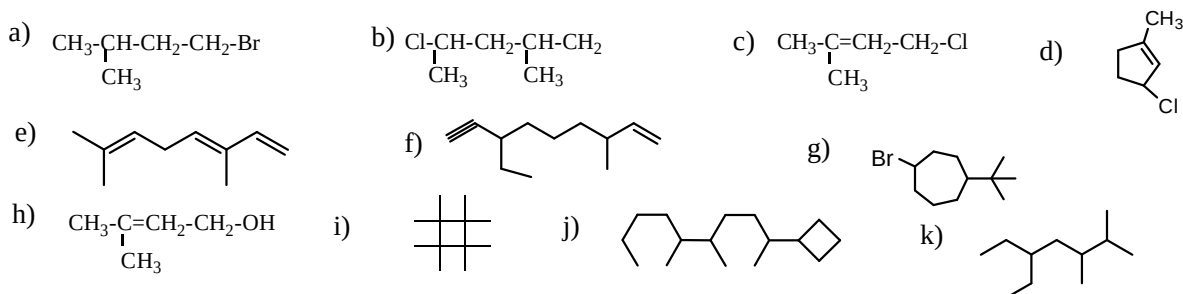
1. Write the **IUPAC** names of the following hydrocarbons



2. Write the **IUPAC** names of the following compounds



3. Name the following compounds according to IUPAC rule



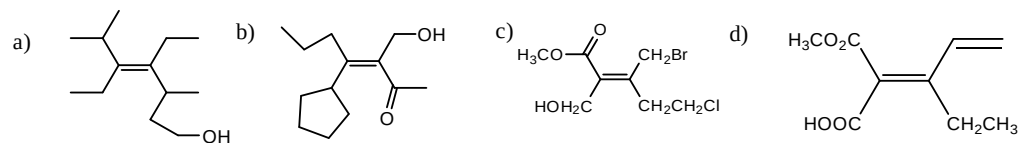
4. Draw the structure of the following

- a) 5-Chloro-4-iodo-6-methyl-1-heptyne      e) Diethyl ether  
 b) 6-tert-butyl-5-isopropyl-2,7-dimethylnonane      f) (Z)-3-Fluoro-2-methyl-3-hexene.  
 c) (E)-1-bromo-2-chloro-1-pentene      g) 1-ethoxy-1-methylcyclohexane

5. The name given below is **incorrect** IUPAC name for the hydrocarbon. Find the correct IUPAC name.

- a) 1,2,4,4-tetramethylcyclohexane      d) 3-methylbicyclo[2,2,1]heptane  
 b) 3-sec-butylpentane      e) 2,2-diethylheptane  
 c) 1-cyclopropyl-3,3-dimethylcycloheptane.      f) 3,6-dipropylspiro[5,5]undecane

6. Assign E or Z configuration to the following alkenes:



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## 2.3 Introduction to the Common Functional Groups in Organic Chemistry: Their Typical properties, Preparations and Reactions.

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### 2.3.1 Alkanes, Alkenes, Alkynes,

#### A. Alkanes

#### Physical Properties of Alkanes

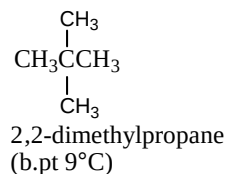
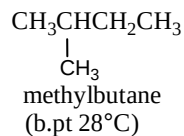
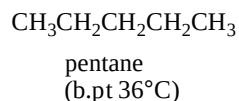
##### Boiling Points

Molecular size and degree of intermolecular forces that exists between the molecules will affect the boiling points of organic compounds. Generally speaking, the boiling points of alkanes are the lowest of any organic family if you compare roughly the same size molecules. This is because the intermolecular forces holding the alkane molecules together are very weak London Dispersion forces. These are the only intermolecular forces that would be present. As a result, the temperature required to separate the molecules into the vapor state is relatively low.

Methane, ethane, propane, and butane are gases at room temperature. The unbranched alkanes pentane ( $C_5H_{12}$ ) through heptadecane ( $C_{17}H_{36}$ ) are liquids, whereas higher homologs are solids. The boiling points of unbranched alkanes increase with the number of carbon

| Name     | Number of carbons | Molecular formula | Condensed structure   | bp ( $^{\circ}C$ ) | mp ( $^{\circ}C$ ) |
|----------|-------------------|-------------------|-----------------------|--------------------|--------------------|
| Methane  | 1                 | $CH_4$            | $CH_4$                | -164               | -182.5             |
| Ethane   | 2                 | $C_2H_6$          | $CH_3CH_3$            | -88.6              | -183.3             |
| Propane  | 3                 | $C_3H_8$          | $CH_3CH_2CH_3$        | -42.1              | -189.7             |
| Butane   | 4                 | $C_4H_{10}$       | $CH_3(CH_2)_2CH_3$    | -0.60              | -138.4             |
| Pentane  | 5                 | $C_5H_{12}$       | $CH_3(CH_2)_3CH_3$    | 36.1               | -129.7             |
| Hexane   | 6                 | $C_6H_{14}$       | $CH_3(CH_2)_4CH_3$    | 68.9               | -93.5              |
| Heptane  | 7                 | $C_7H_{16}$       | $CH_3(CH_2)_5CH_3$    | 98.4               | -90.6              |
| Octane   | 8                 | $C_8H_{18}$       | $CH_3(CH_2)_6CH_3$    | 125.7              | -56.8              |
| Nonane   | 9                 | $C_9H_{20}$       | $CH_3(CH_2)_7CH_3$    | 150.8              | -51.0              |
| Decane   | 10                | $C_{10}H_{22}$    | $CH_3(CH_2)_8CH_3$    | 174.1              | -29.7              |
| Undecane | 11                | $C_{11}H_{24}$    | $CH_3(CH_2)_9CH_3$    | 196                | -26                |
| Dodecane | 12                | $C_{12}H_{26}$    | $CH_3(CH_2)_{10}CH_3$ | 216                | -10                |

Branched alkanes have lower boiling points than their unbranched isomers. Isomers have, of course, the same number of atoms and electrons, but a molecule of a branched alkane has a smaller surface area than an unbranched one. The extended shape of an unbranched alkane permits more points of contact for intermolecular associations. Compare the boiling points of pentane and its isomers:

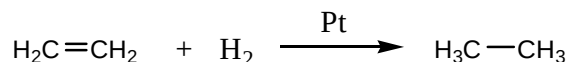


## Solubility

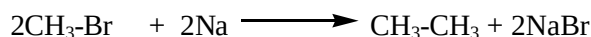
Alkanes are insoluble in water because the polar water molecules are not attracted to the non-polar alkane molecules. However alkanes are soluble in non-polar solvents such as Carbon Tetrachloride,  $\text{CCl}_4$ . Generally speaking like dissolves in like. So non-polar substances will be insoluble in polar solvents such as water, but they will be soluble in non-polar solvents.

## Preparations of Alkanes

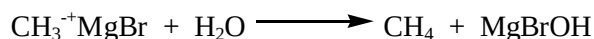
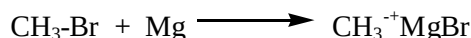
1. Alkanes can be prepared by catalytic hydrogenation of alkenes or alkynes. For example:



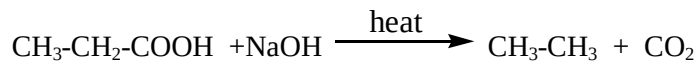
2. Alkanes can be prepared by a coupling reaction known as the Wurtz Reaction:



3. Alkanes can be prepared by forming an organo-metallic reagent known as a Grignard reagent from an alkyl halide and magnesium metal, then hydrolyzing of it (react it with water) gives alkane:



4. Thermal Decarboxylation of a Carboxylate Salt

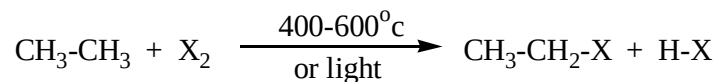


## Reactions of Alkanes

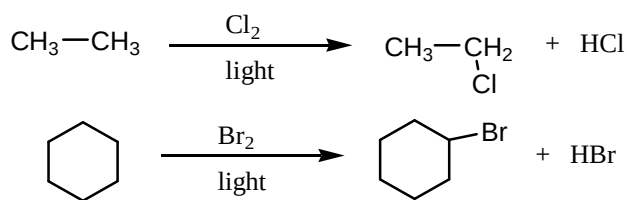
Alkanes are sometimes referred to as paraffins, meaning “little affinity”. They show little chemical affinity for other substances and are chemically inert to most laboratory reagent. Alkanes do, however, react with oxygen, halogens, and a few other substances under appropriate conditions.

### Free Radical Halogenation

Alkanes react with halogens in the presence of heat or light



X is a symbol used by Organic Chemists to mean any halogen atom: F, Cl, Br, or I.

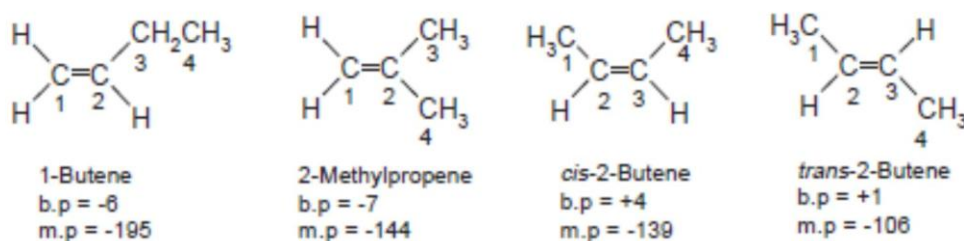


## B. Alkenes

### Physical properties of alkenes

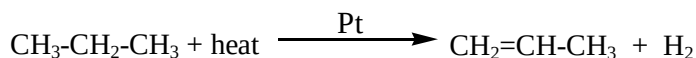
Alkenes are also referred to as "olefins". The word means fat dissolving since fats are relatively non-polar as alkenes are. The boiling points and solubilities of alkenes are very similar to the alkanes.

As with alkanes, the boiling points and melting points of alkenes decrease with increasing molecular weight, but show some variations that depend on the shape of the molecule. Alkenes with the same molecular formula are isomers of one another if the position and the stereochemistry of the double bond differ. For example, there are four different acyclic structures that can be drawn for butene ( $\text{C}_4\text{H}_8$ ). They have different b.p. and m.p. as follows.



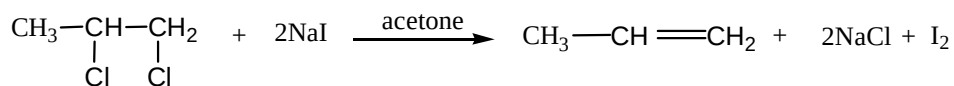
### Preparation of Alkenes

#### 1. Dehydrogenation of Alkanes

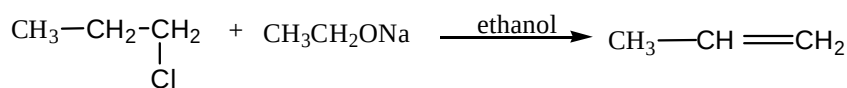


#### 2. Dehalogenation of Vicinyl Di-halides

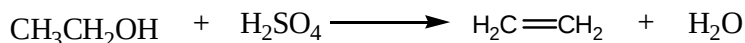
A vicinylDihalide is an alkyl halide that has two atoms of halogen attached to adjacent carbons in the molecule.



#### 3. Dehydrohalogenation of an alkyl halide

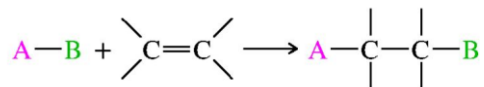


#### 4. Dehydration of an Alcohol



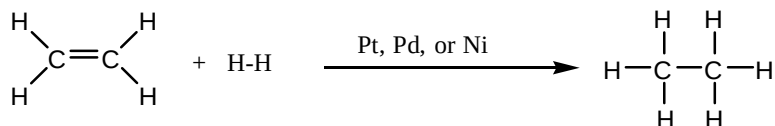
## Reactions of Alkenes

Alkenes are electron-rich species. The characteristic reaction of alkenes is addition to the double bond according to the general equation.

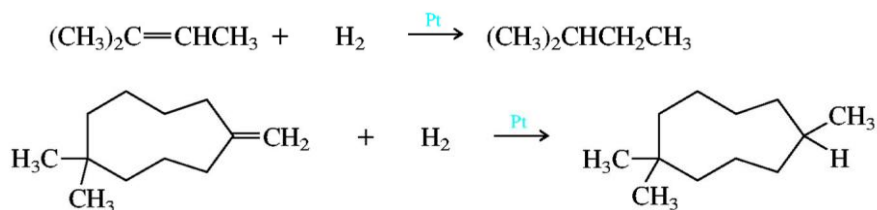


### 1. Hydrogenation of alkenes

This is addition of hydrogen to a double bond to produce saturated hydrocarbons.

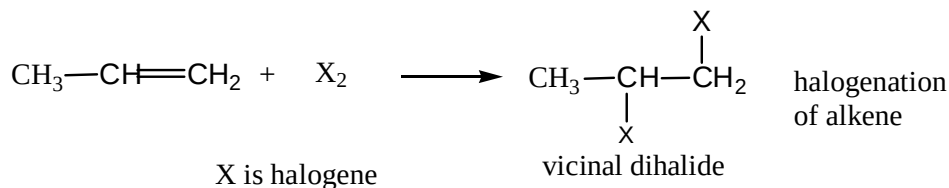


The uncatalyzed addition of hydrogen to an alkene, although exothermic, is very slow. The rate of hydrogenation increases dramatically, however, in the presence of certain finely divided metal catalysts. Platinum is the hydrogenation catalyst most often used, although *palladium*, *nickel*, and *rhodium* are also effective.

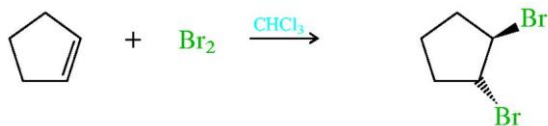


### 2. Halogenation of an alkene

Alkenes react with halogens to produce a *vicinal dihalide*.

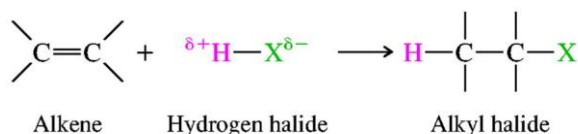


Two substituents, in this case the halogens, are vicinal if they are attached to adjacent carbons. The word is derived from the Latin *vicinalis*, which means "neighboring."



### 3. Addition of Hydrogen Halide (HX)

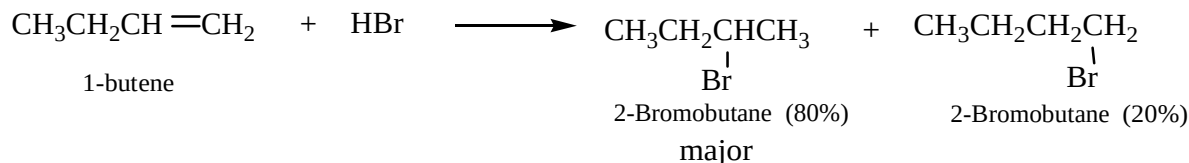
In many addition reactions the attacking reagent is a polar molecule. Hydrogen halides are among the simplest examples of polar substances that add to alkenes.



Addition occurs rapidly in a variety of solvents, including pentane, benzene, dichloromethane, chloroform, and acetic acid.

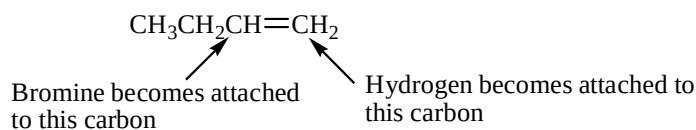


In principle a hydrogen halide can add to an unsymmetrical alkene (an alkene in which the two carbons of the double bond are not equivalently substituted) in either of two directions. In practice, addition is so highly *regioselective* as to be considered *regiospecific*.

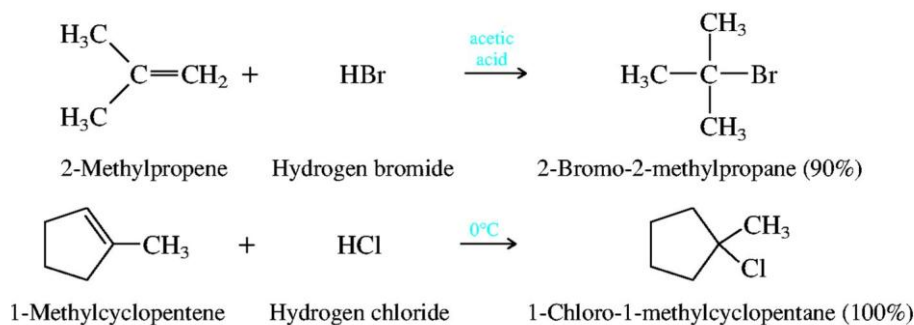


The major product is determined by Markovnikov's rule

**Markovnikov's rule** states that when an unsymmetrically substituted alkene reacts with a hydrogen halide, the hydrogen adds to the carbon that has the greater number of hydrogens, and the halogen adds to the carbon having fewer hydrogens.



The following are some examples.



### Exercise 2.8

Write the structure of the major organic product formed in the reaction of HCl with the following

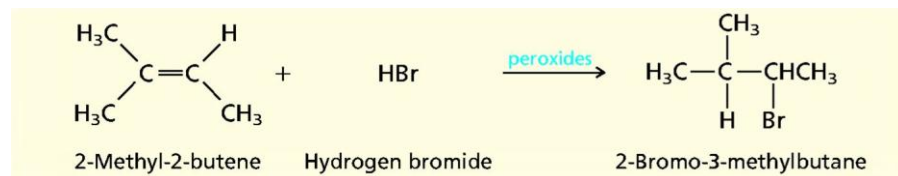
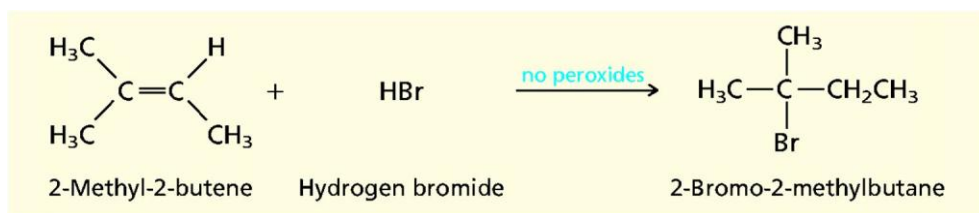
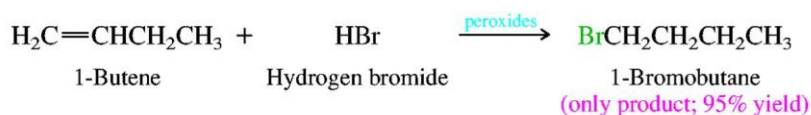
a) 2-methyl-butene

c) cis-2-Butene

b) 2-methyl-2-butene

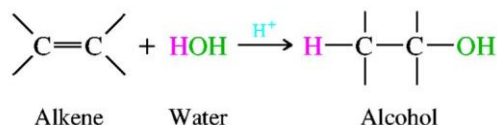
d)  $\text{CH}_3\text{-CH=}$  

In the presence of peroxide (ROOR) the hydrobromination of an alkene takes a different pathway leading to a different product. Only HBr addition occurs by this pathway. It is referred to as an **Anti Markovnikov addition** because the opposite constitutional isomer is the predominant product. This is because the mechanistic pathway is a free radical instead of an ionic pathway.

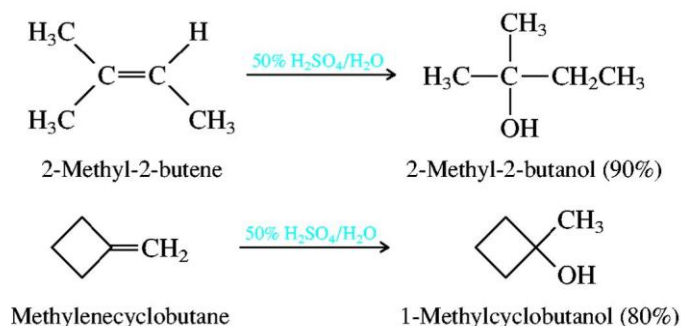


## 4. Addition of water (Hydration)

Acid-Catalyzed Hydration of Alkenes

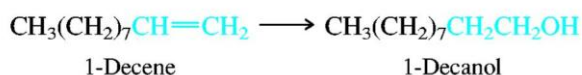


This reaction is carried out in a dilute acid medium. A 50% water/sulfuric acid solution is often used, yielding the alcohol directly without the necessity of a separate hydrolysis step. Markovnikov's rule is followed:



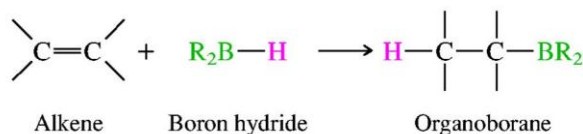
## 5. Hydroboration-Oxidation of Alkenes

Acid-catalyzed hydration converts alkenes to alcohols with regioselectivity according to Markovnikov's rule. Frequently, however, one needs an alcohol having a structure that corresponds to hydration of an alkene with a regioselectivity opposite to that of Markovnikov's rule. The conversion of 1-decene to 1-decanol is an example of such a transformation.

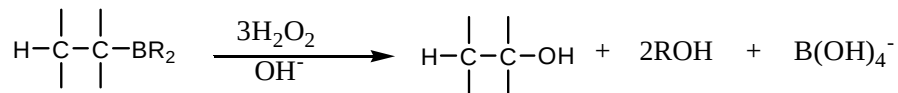


The synthetic method used to accomplish this is an indirect one known as hydroboration-oxidation.

**Hydroboration** is a reaction in which a boron hydride, a compound of the type  $\text{R}_2\text{BH}$ , adds to a carbon-carbon bond. A carbon-hydrogen bond and a carbon-boron bond result.

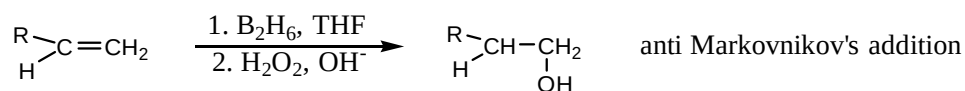


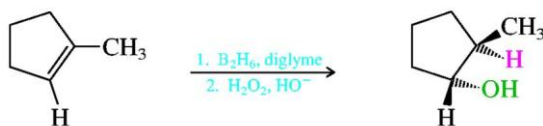
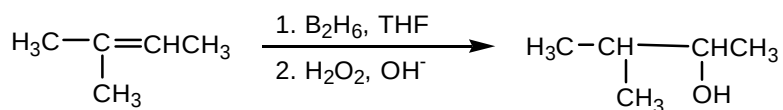
Following hydroboration, the organoborane is oxidized by treatment with hydrogen peroxide in aqueous base. This is the oxidation stage of the sequence; hydrogen peroxide is the oxidizing agent, and the organoborane is converted to an alcohol.



A more convenient hydroborating agent is the borane-tetrahydrofuran complex ( $\text{H}_3\text{B}-\text{THF}$ ). It is very reactive, adding to alkenes within minutes at  $0^\circ\text{C}$ , and is used in tetrahydrofuran as the solvent.

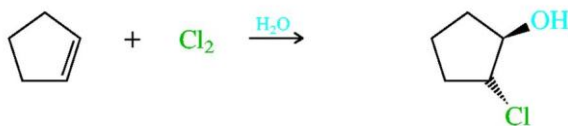
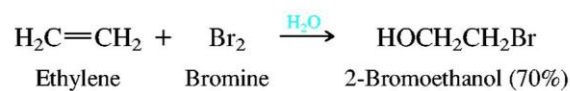
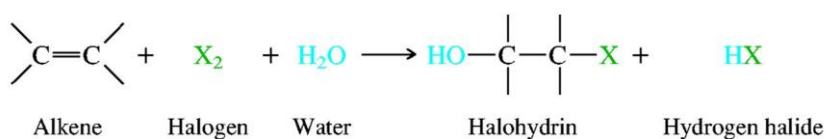
The overall general equation for hydroboration is



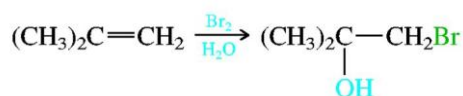


## 6. Conversion of Alkenes to Vicinal Halohydrins

In aqueous solution chlorine and bromine react with alkenes to form vicinal halohydrins, compounds that have a halogen and a hydroxyl group on adjacent carbons.

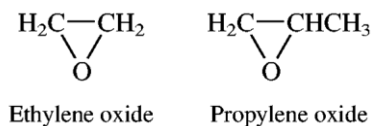


The regioselectivity of addition is established when water attacks one of the carbons of the halonium ion. In the following example, the structure of the product tells us that water attacks the more highly substituted carbon.

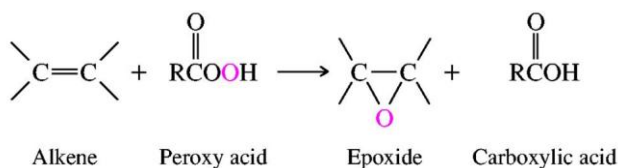


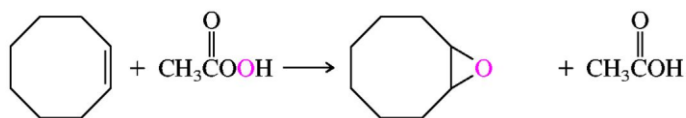
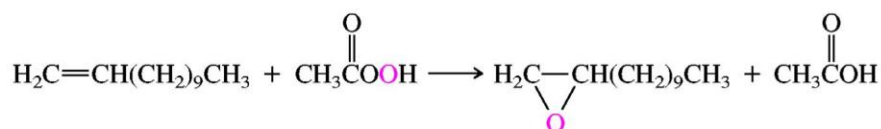
## 7. Epoxidation of Alkenes

Three-membered rings that contain oxygen are called epoxides.



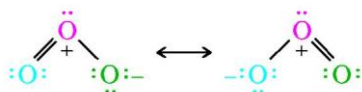
Epoxides are very easy to prepare via the reaction of an alkene with a peroxy acid. This process is known as **epoxidation**.



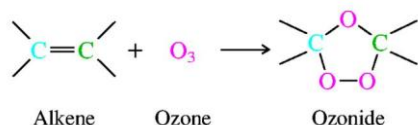


## 8. Ozonolysis of Alkenes

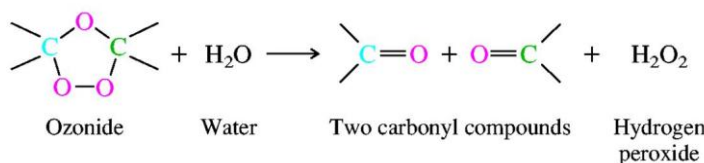
Ozone ( $\text{O}_3$ ) is the triatomic form of oxygen. It is a neutral but polar molecule that can be represented as a hybrid of its two most stable Lewis structures.



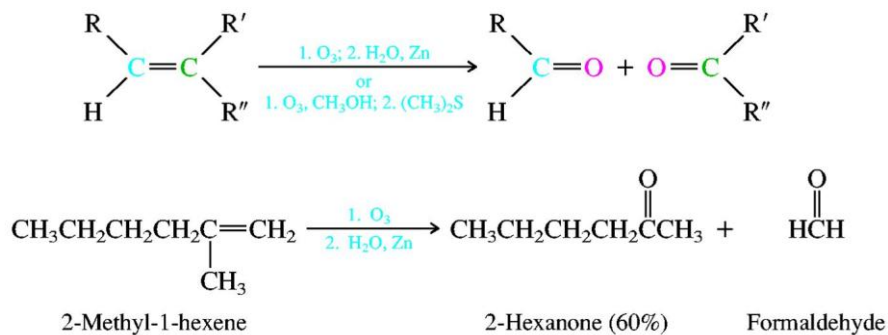
Ozone is a powerful electrophile and undergoes a remarkable reaction with alkenes in which both the  $\pi$  and  $\sigma$  components of the carbon-carbon double bond are cleaved to give a product referred to as an **ozonide**.



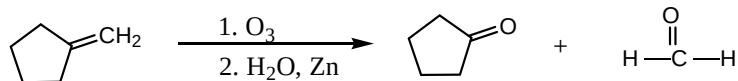
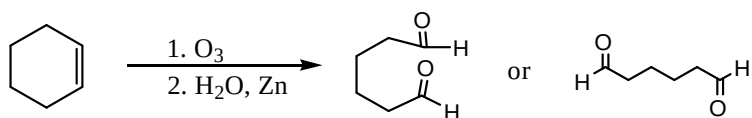
Ozonides undergo hydrolysis in water, giving carbonyl compounds.



The two-stage reaction sequence is called **ozonolysis** and is represented by the general equation



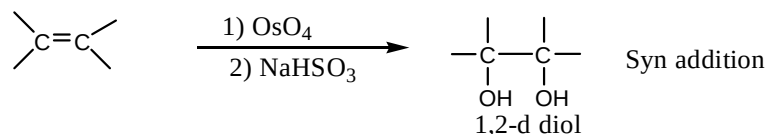
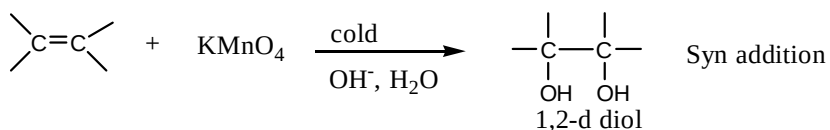
Ozonolysis break a double bond and open ring alkenes also.



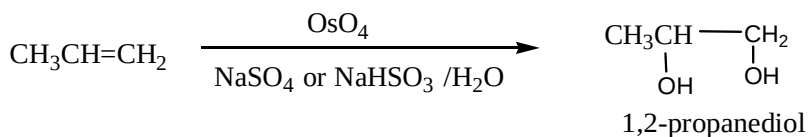
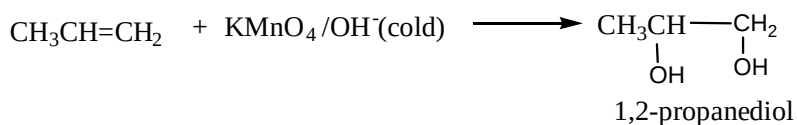
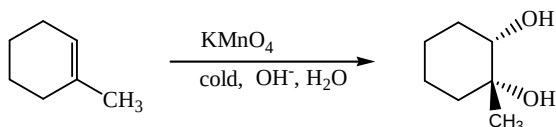
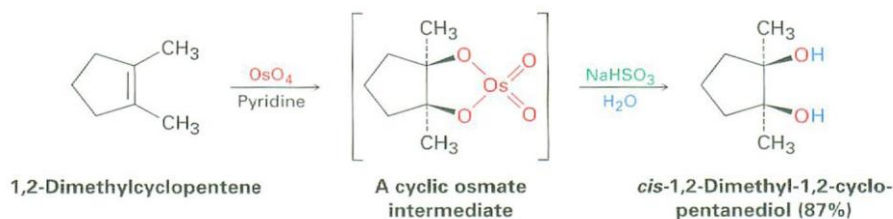
## 9. Hydroxylation

This is oxidation of alkene using oxidizing reagents.

- Hydroxylation is addition of two OH groups to each of the two double bonded carbon



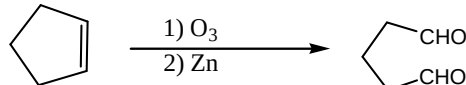
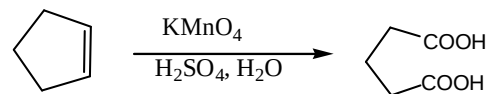
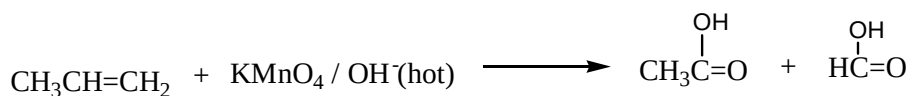
- The two OH adds to the same side of the double bond (syn)



## 10. Oxidative cleavage of an alkene using hot alkaline KMnO<sub>4</sub>

Acidic KMnO<sub>4</sub> also causes double bond cleavage like ozone.

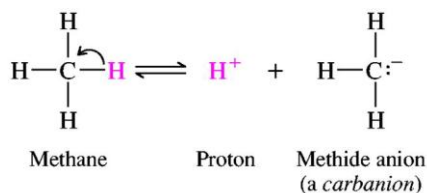
Alkenes are oxidatively cleaved by hot alkaline potassium permanganate solution to give carboxylic acid.



## C. Alkynes

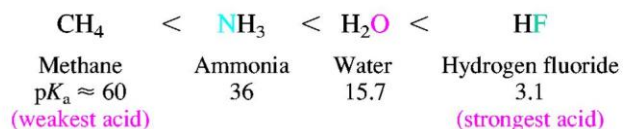
Alkynes resemble alkanes and alkenes in their physical properties. They share with these other hydrocarbons the properties of low density and low water-solubility. They are slightly more polar and generally have slightly higher boiling points than the corresponding alkanes and alkenes.

The C-H bonds of hydrocarbons show little tendency to ionize, and alkanes, alkenes, and alkynes are all very weak acids. The acid-dissociation constant  $K_a$  for methane, for example, is too small to be measured directly but is estimated to be about  $10^{-60}$  ( $\text{p}K_a=60$ ).

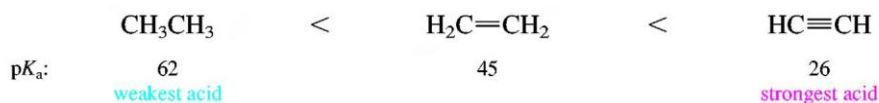


The conjugate base of a hydrocarbon is called a **carbanion**. It is an anion in which the negative charge is borne by carbon. Because it is derived from a very weak acid, a carbanion such as  $:\text{CH}_3^-$  is an exceptionally strong base.

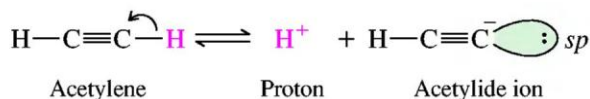
The ability of an atom to bear a negative charge is related to its electronegativity. Both the electronegativity of an atom X and the acidity of H-X increase across a row in the periodic table.



Using the relationship from the preceding section that the effective electronegativity of carbon in a C-H bond increases with its s character ( $\text{sp}^3 < \text{sp}^2 < \text{sp}$ ), the order of hydrocarbon acidity behaves much like the preceding methane, ammonia, water, hydrogen fluoride series.



The acidity increases as carbon becomes more electronegative. Ionization of acetylene gives an anion in which the unshared electron pair occupies an orbital with 50% s character.



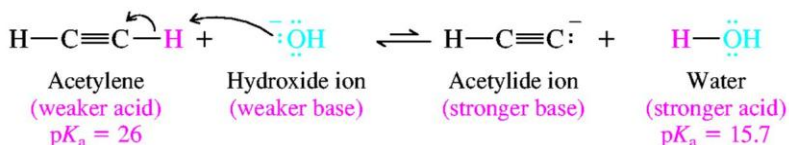
In the corresponding ionizations of ethylene and ethane, the unshared pair occupies an orbital with 33% ( $sp^2$ ) and 25% ( $sp^3$ ) s character, respectively.

Terminal alkynes ( $\text{RC}\equiv\text{CH}$ ) resemble acetylene in acidity.

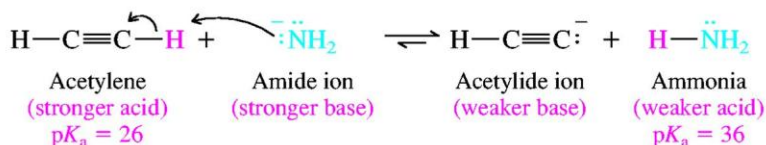


3,3-Dimethyl-1-butyne

Although acetylene and terminal alkynes are far stronger acids than other hydrocarbons, we must remember that they are, nevertheless, very weak acids-much weaker than water and alcohols, for example. Hydroxide ion is too weak a base to convert acetylene to its anion in meaningful amounts.



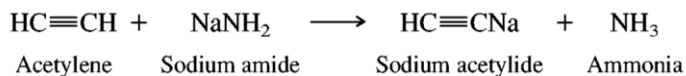
*Amide* ion is a much stronger base than acetylide ion and converts acetylene to its conjugate base quantitatively.



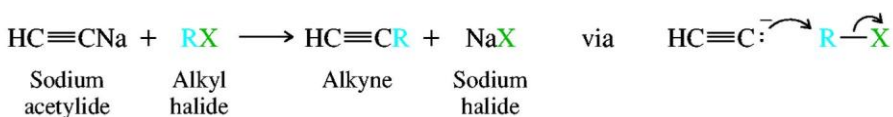
## Preparation of Alkynes

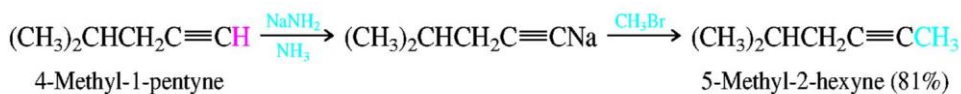
### A. By Alkylation of Acetylene

Alkylation of acetylene involves a sequence of two separate operations. In the first one, acetylene is converted to its conjugate base by treatment with sodium amide.



Next, an alkyl halide (the alkylating agent) is added to the solution of sodium acetylide. Acetylide ion acts as a nucleophile, displacing halide from carbon and forming a new carbon-carbon bond. Substitution occurs.



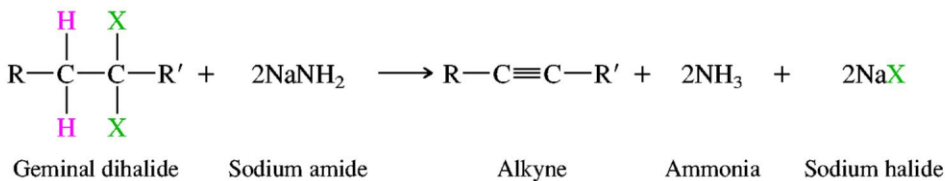


Dialkylation of acetylene can be achieved by carrying out the sequence twice.

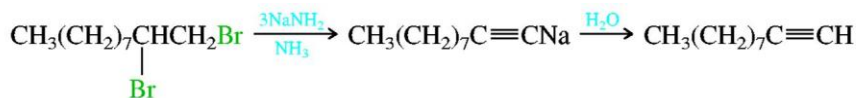
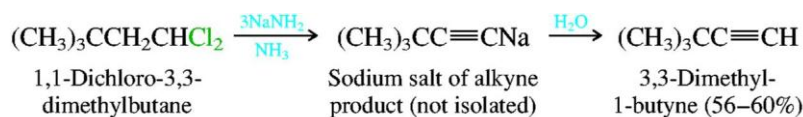
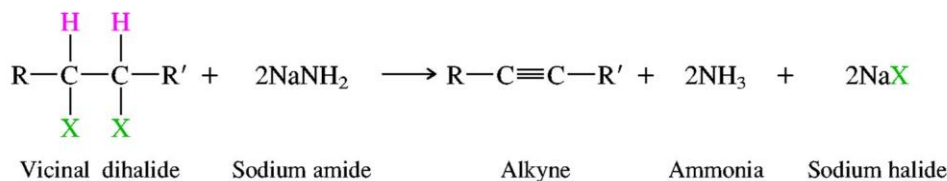


## B. By Elimination Reactions

*Double dehydrohalogenation of a geminal dihalide*

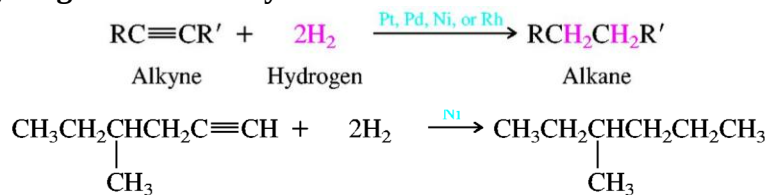


*Double dehydrohalogenation of a vicinal dihalide*



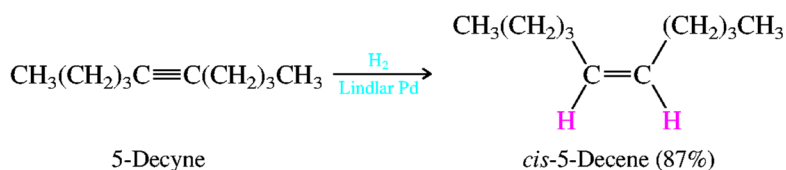
## REACTIONS OF ALKYNES

### A. Hydrogenation of Alkynes

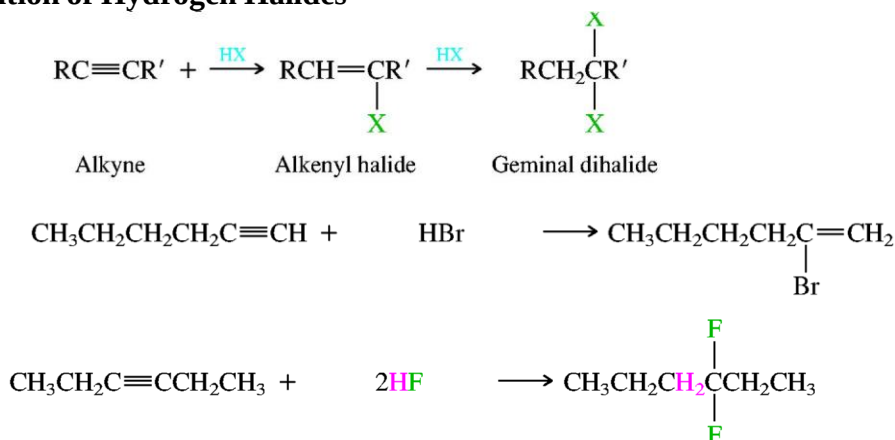


Partial hydrogenation of an alkyne to produce alkene could be achieved using Lindlar catalyst.

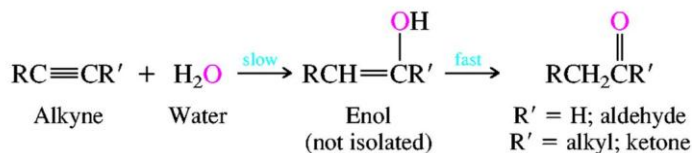
**Lindlar catalyst** is a palladium on calcium carbonate combination to which lead acetate and quinoline have been added. Hydrogenation of alkynes with internal triple bonds gives cis alkenes.



## B. Addition of Hydrogen Halides

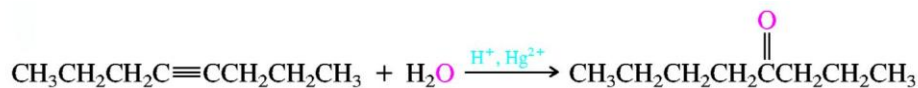


## C. Hydration of Alkynes

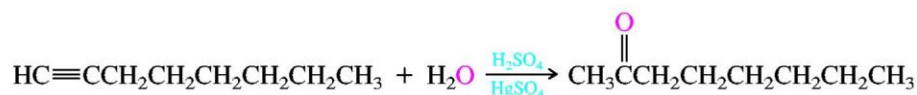


The aldehyde or ketone is called the keto form, and the keto $\leftrightarrow$ enol equilibration referred to as keto-enol isomerism or keto-enol tautomerism. Tautomers are constitutional isomers that equilibrate by migration of an atom or group, and their equilibration is called tautomerism.

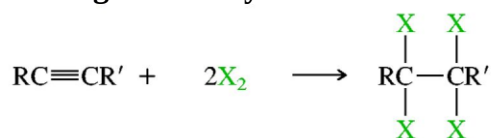
In general, ketones are more stable than their enol precursors and are the products actually isolated when alkynes undergo acid-catalyzed hydration. The standard method for alkyne hydration employs aqueous sulfuric acid as the reaction medium and mercury(II) sulfate or mercury(II) oxide as a catalyst.

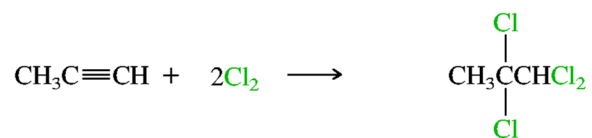


Hydration of alkynes follows Markovnikov's rule; terminal alkynes yield methylsubstituted ketones.



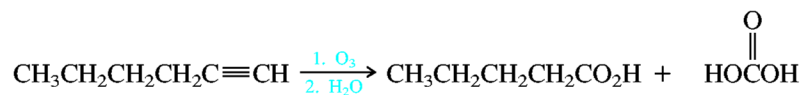
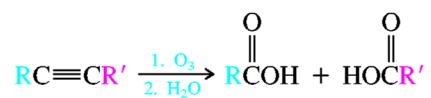
## D. Addition of Halogens to Alkynes





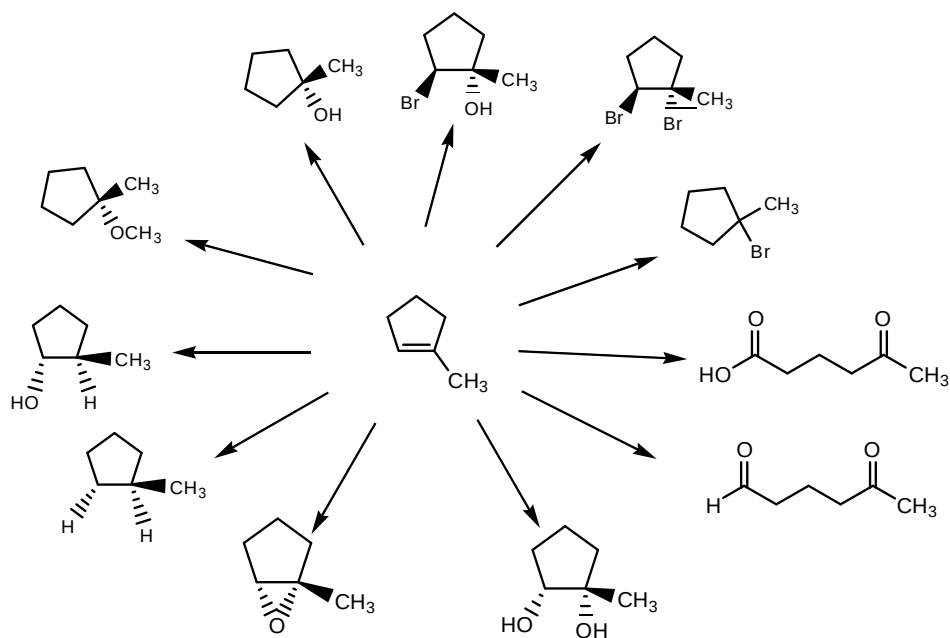
### E. Ozonolysis of Alkynes

Carboxylic acids are produced when alkynes are subjected to *ozonolysis*.

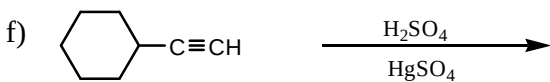
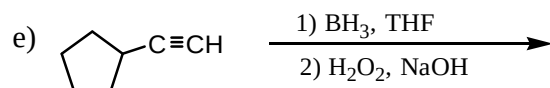
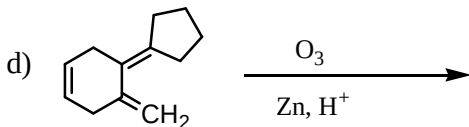
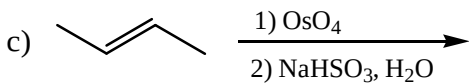
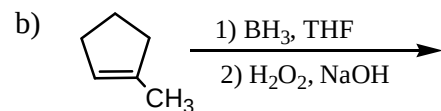
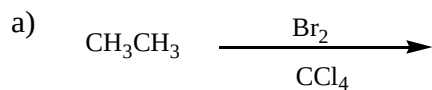


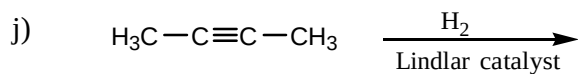
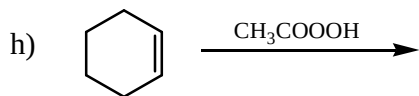
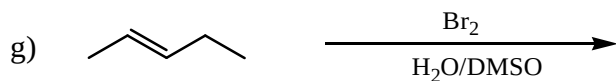
## SELF -TEST EXERCISES 2.2

1 Provide the necessary organic or inorganic reagents for each reaction shown below



2 Complete the following reactions.





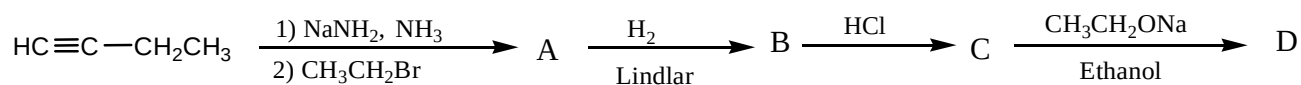
3 Starting from 1-Butyne, any alkyl halide and reagent needed outline the synthesis of

a) Hexane                      b) 5-Methyl-3-hexene

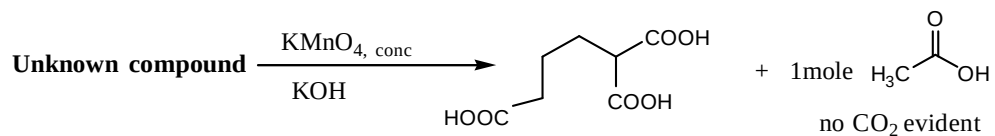
4 Starting from acetylene and any alkyl halide needed, synthesize

a) 2-Chlorobutane    b) 4-Methyl-2-pentanol

5 Suggest the possible structure of A, B, C and D for the following multistep synthesis



6 Propose a structure for the unknown compound given the following data:



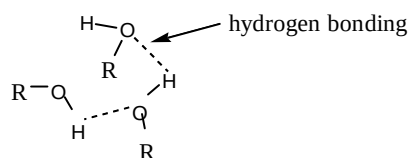
## 2.3.2Alcohols

### Physical properties of alcohols

Most of the common alcohols are liquids at room temperature, the higher alcohols are somewhat viscous, and some of the highly branched isomers are solid at normal temperature.

### Boiling Points

The boiling points of alcohols are in general much higher than comparably sized hydrocarbon, alkylhalide, ethers and carbonyl compounds. This can be explained by the extra intermolecular forces between the alcohol molecules i.e. the hydrogen bonding.



There is no such hydrogen bonding between the other molecules because all the hydrogen atoms are not bonded to oxygen or nitrogen. The presence of hydrogen bonding in alcohol is responsible for its higher boiling point than other hydrocarbons.

$\text{CH}_3\text{CH}_2\text{OH}$   
ethanol  
b.pt=  $78^\circ\text{C}$

$\text{CH}_3-\text{O}-\text{CH}_3$   
diethylether  
b.pt=  $-24^\circ\text{C}$

in diethylether there is no  
hydrogen bonding

Branching will tend to decrease the boiling point since a more symmetrical molecule will have less London Dispersion forces and with more hydrocarbons branching around the carbinol carbon (carbon bonded to the hydroxyl group).

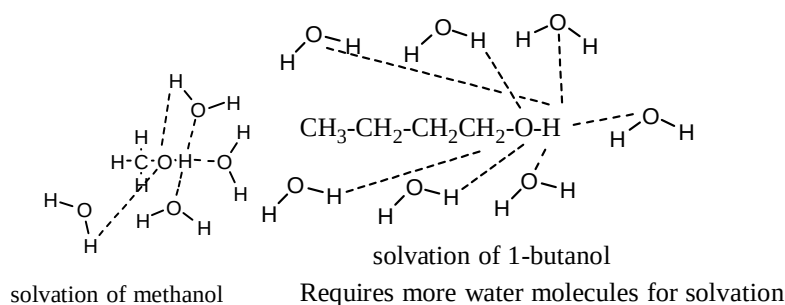
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$   
1-butanol  
b.pt=  $118^\circ\text{C}$

$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$   
2-butanol  
b.pt=  $100^\circ\text{C}$

$\text{CH}_3\text{CH}_2\text{C}(\text{OH})(\text{CH}_3)_2$   
2-methyl-2-propanol  
b.pt=  $83^\circ\text{C}$

### Water Solubility

Low molecular weight alcohols (methyl, ethyl, n-propyl, and isopropyl) are soluble in water in all proportions. This can be explained by considering the way that water molecules can disperse solute molecules into a solution. The polar water molecules are attracted to the Hydroxyl group by hydrogen bonding that occurs between the hydrogens of the water molecules and the Oxygen of the alcohol, with smaller molecules water can more easily surround the alcohol molecules effectively separating them from one another. This is called solvation.



Increased branching will increase water solubility. This can be explained by the fact that increased branching around the carbinol carbon will mean that water molecules will have a smaller molecular surface area to surround in trying to encapsulate the organic molecule.

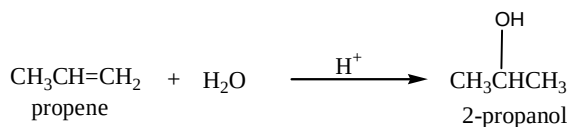
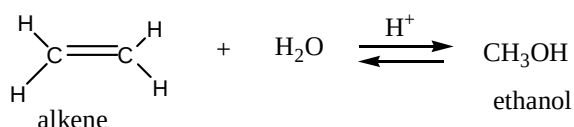
Higher alcohols become more "hydrocarbon-like" and less water-soluble. 1-Octanol, for example, dissolves to the extent of only 1 mL in 2000 mL of water. As the alkyl chain gets longer, the hydrophobic effect (Section 2.17) becomes more important, to the point that it, more than hydrogen bonding, governs the solubility of alcohols.

## Preparation of alcohols

### By Hydration of Alkenes

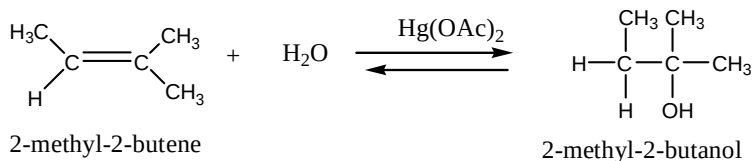
#### a. Acid Catalyzed Hydration of Alkenes

In an aqueous acid solution alkenes will add water across the Pi bond thereby saturating the molecule and producing an alcohol. The reaction is regioselective in that only the more highly substituted alcohol is produced. It follows the Markovnikov rule.

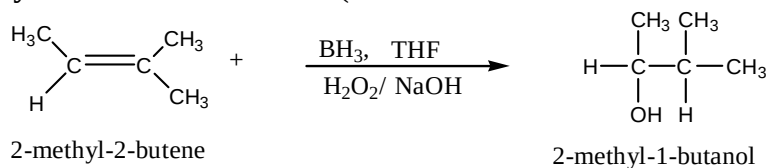


#### b. Oxymercuration-demercuration.

Alkenes add water in the presence mercury acetate, the reaction follows Markovnikov's rule

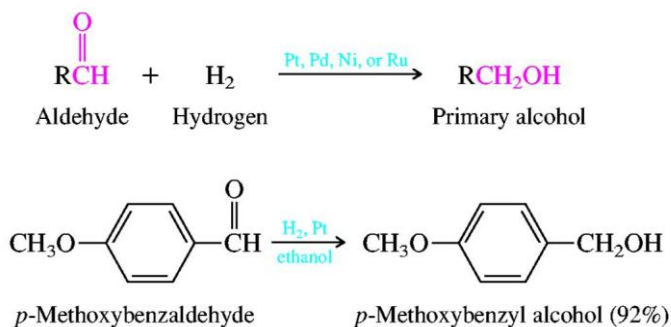


c. **Hydroboration-oxidation** (the reaction follow anti-Markovnikove rule)

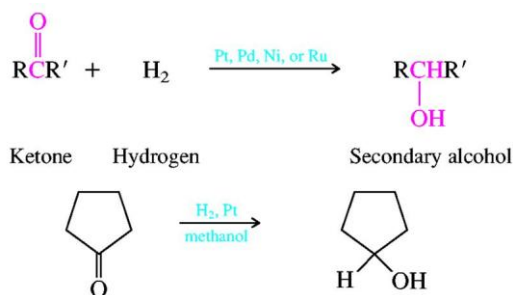


## By Reduction of Aldehydes and ketones

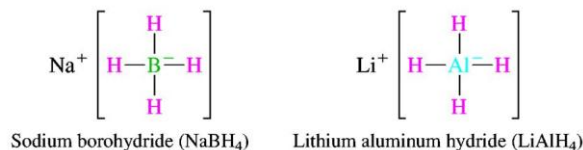
The most obvious way to reduce an aldehyde or a ketone to an alcohol is by hydrogenation of the carbon-oxygen double bond. Like the hydrogenation of alkenes, the reaction is exothermic but exceedingly slow in the absence of a catalyst. Finely divided metals such as platinum, palladium, nickel, and ruthenium are effective catalysts for the hydrogenation of aldehydes and ketones. Aldehydes yield primary alcohols:



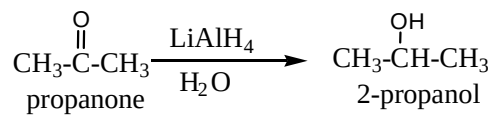
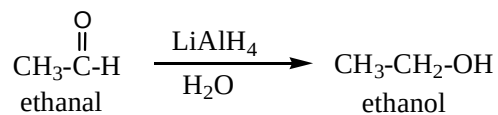
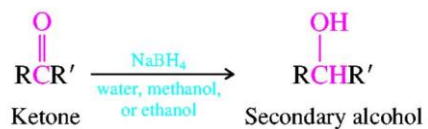
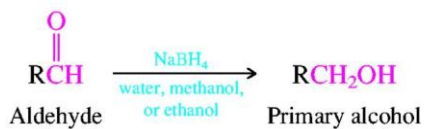
Ketones yield secondary alcohols:



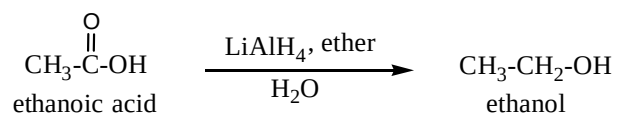
For most laboratory-scale reductions of aldehydes and ketones, catalytic hydrogenation has been replaced by methods based on metal hydride reducing agents. The two most common reagents are sodium borohydride and lithium aluminum hydride.



Sodium borohydride is especially easy to use, needing only to be added to an aqueous or alcoholic solution of an aldehyde or a ketone:



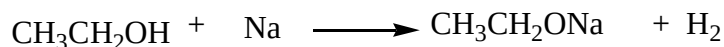
**By Reduction of carboxylic acid and ester using lithium aluminum hydride (LiAlH<sub>4</sub>)**



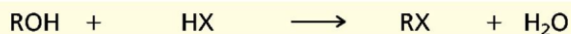
## Reaction of Alcohols

Alcohols are versatile starting materials for the preparation of a variety of functional groups.

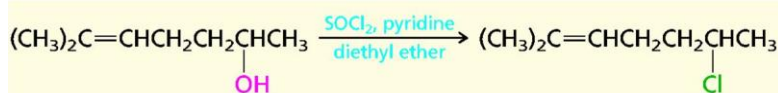
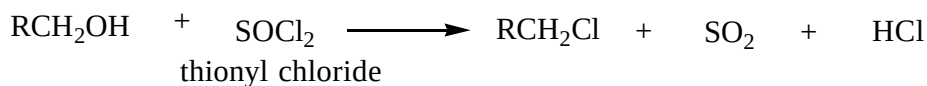
Alcohols are slightly less acidic than water and react readily with group I metals to liberate hydrogen and to form metal alkoxide



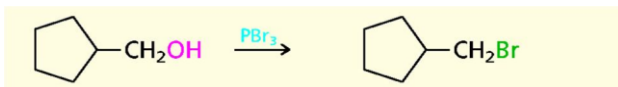
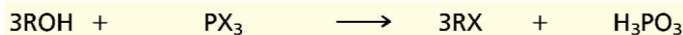
### A. Reaction with hydrogen halides



### B. Reaction with thionyl chloride

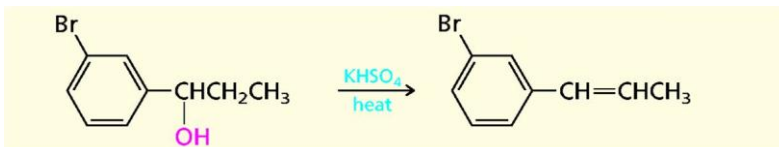


### C. Reaction with phosphorous trihalide to form alkylhalide



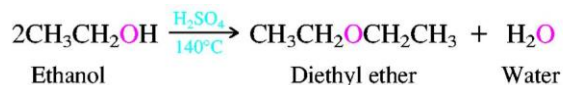
#### D. Acid catalyzed dehydration of alcohol

In the presence of acid catalyst water is eliminated from the adjacent carbon to form alkene



### E. Conversion of alcohols to ether

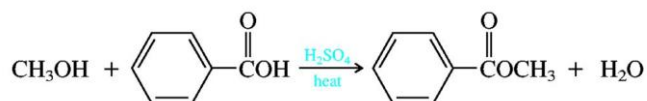
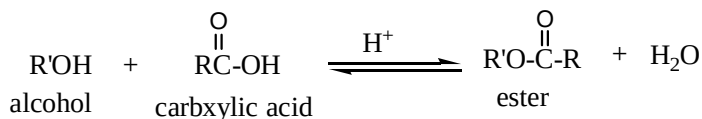
On heating in the presence of acid catalyst primary alcohols are converted to ether



## F. Esterification

Acid-catalyzed condensation of an alcohol and a carboxylic acid yields an ester and water and is known as the Fischer esterification.

In the presence of acid catalyst alcohols react with carboxylic acid to form ester



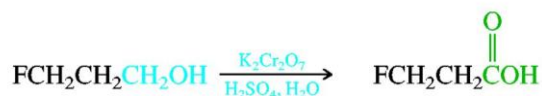
## G. Oxidation of alcohol

Oxidation of alcohols give carbonyl compounds (aldehydes, ketones, and carboxylic acid) the type carbonyl compounds formed depend on the alcohol and oxidizing agent.

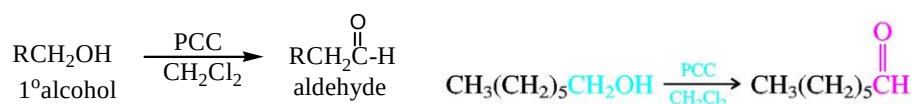
Primary alcohols may be oxidized either to an aldehyde or to a carboxylic acid:



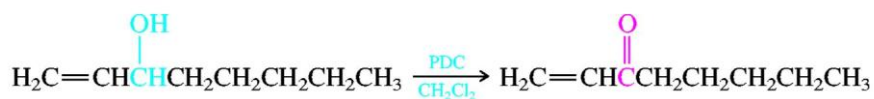
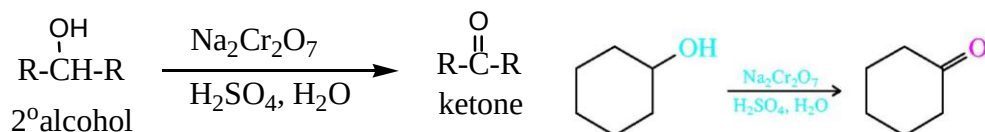
With vigorous oxidizing agent such as chromic acid it gives carboxylic acid



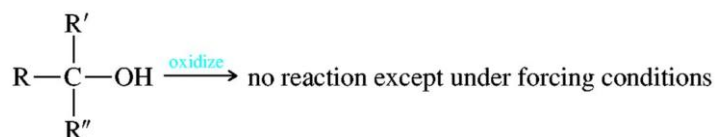
With mild oxidizing agent such as pyridinium chlorochromate (pcc) primary alcohol is oxidized to an aldehyde



Secondary alcohol is oxidized to ketone

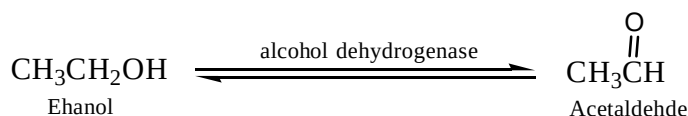


Tertiary alcohols have no hydrogen on their hydroxyl-bearing carbon and do not undergo oxidation readily:



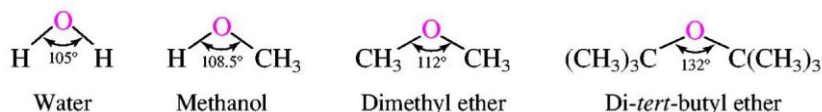
## BIOLOGICAL OXIDATION OF ALCOHOLS

Many biological processes involve oxidation of alcohols to carbonyl compounds or the reverse process, reduction of carbonyl compounds to alcohols. Ethanol, for example, is metabolized in the liver to acetaldehyde. Such processes are catalyzed by enzymes; the enzyme that catalyzes the oxidation of ethanol is called alcohol dehydrogenase.



## 2.3.3 Ethers and Epoxides

Bonding in ethers is readily understood by comparing ethers with water and alcohols. Van der Waals strain involving alkyl groups causes the bond angle at oxygen to be larger in ethers than in alcohols, and larger in alcohols than in water. An extreme example is di-*tert*-butyl ether, where steric hindrance between the *tert*-butyl groups is responsible for a dramatic increase in the C-O-C bond angle.



### A. Ethers

It is instructive to compare the physical properties of ethers with alkanes and alcohols. With respect to boiling point, ethers resemble alkanes more than alcohols. With respect to solubility in water the reverse is true; ethers resemble alcohols more than alkanes. Why?

|                             |                                                 |                                                           |                                                         |
|-----------------------------|-------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|
|                             | $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ | $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ | $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ |
|                             | Diethyl ether                                   | Pentane                                                   | 1-Butanol                                               |
| <b>Boiling point:</b>       | 35°C                                            | 36°C                                                      | 117°C                                                   |
| <b>Solubility in water:</b> | 7.5 g/100 mL                                    | Insoluble                                                 | 9 g/100 mL                                              |

In general, the boiling points of alcohols are unusually high because of hydrogen bonding.

### Preparation of Ethers

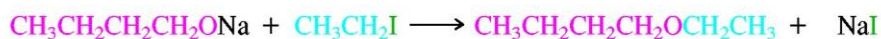
### A. By acid catalyzed condensation of alcohols



In general, this method is limited to the preparation of symmetrical ethers in which both alkyl groups are primary. Isopropyl alcohol, however, is readily available at low cost and gives high enough yields of diisopropyl ether to justify making  $(\text{CH}_3)_2\text{CHOCH}(\text{CH}_3)_2$  by this method on an industrial scale.

### B. Williamson ether synthesis.

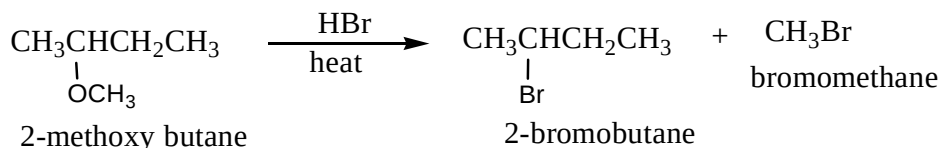
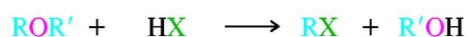
It is an important method of synthesizing ether, it involve the reaction of sodium alkoxide with alkylhalide, sodium alkoxide is prepared by the reaction of alcohol with sodium metal.



## Reactions of ethers

Ethers are one of the unreactive functional group compounds; it is this low reactivity that makes ethers most often to use as a solvent in many organic reactions.

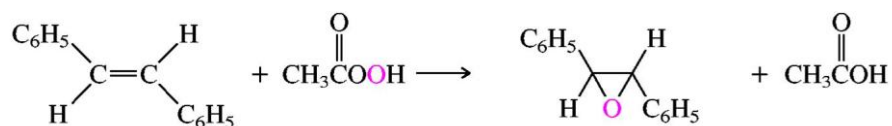
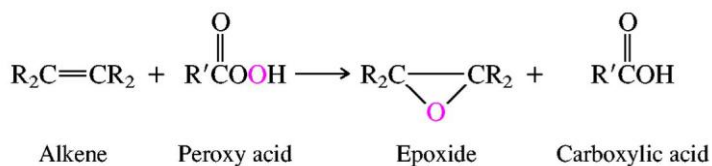
Ethers react with strong acids such as hydrogen halides to form alkylhalide.



## B. Epoxides

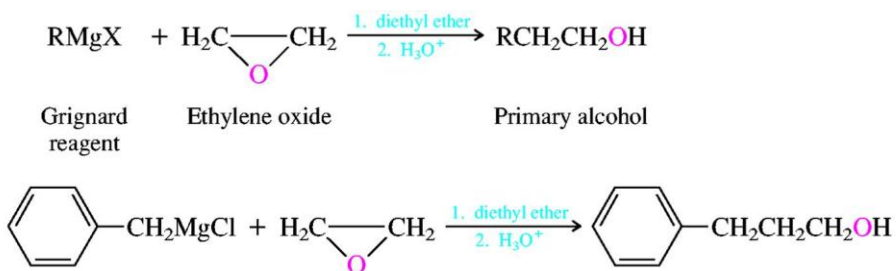
### Preparation of Epoxides:

Epoxidation of alkenes by reaction with peroxy acids

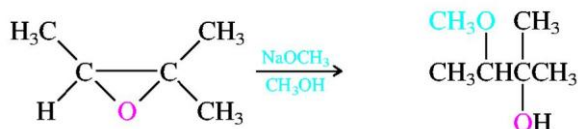


### Reactions of Epoxides

Ring opening of epoxides using Grignard reagents

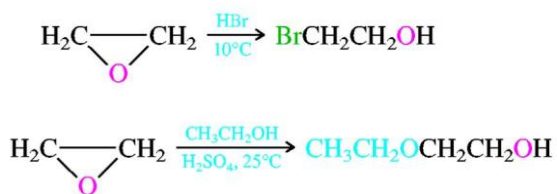


### Nucleophilic Ring-Opening of Epoxides

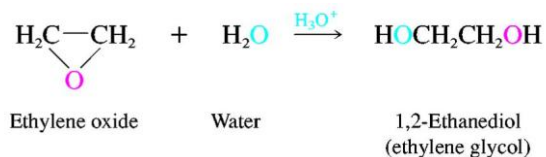


Unsymmetrical epoxides are attacked at the less substituted, less sterically hindered carbon of the ring:

### Acid-Catalyzed Ring-Opening of Epoxides



The industrial preparation of ethylene glycol ( $\text{HOCH}_2\text{CH}_2\text{OH}$ ) by hydrolysis of ethylene oxide in dilute sulfuric acid.

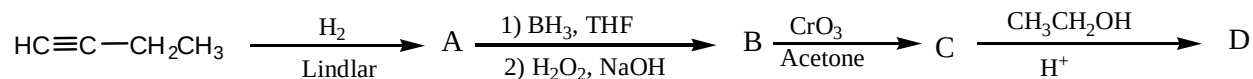


Dear Students, you will study the following functional groups in Organic chemistry II

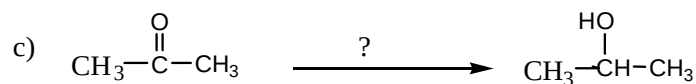
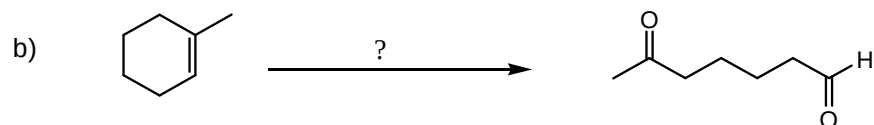
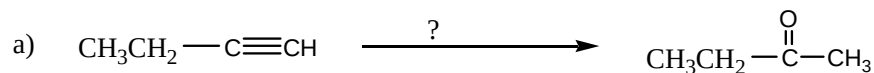
- ✓ Aromatic Hydrocarbons,
- ✓ Amines
- ✓ Aldehydes and Ketones
- ✓ Carboxylic Acids and their Derivatives.

## Self Test Exercise 2.2

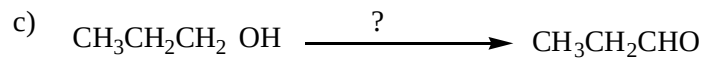
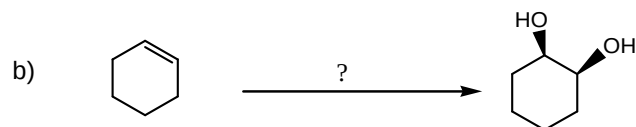
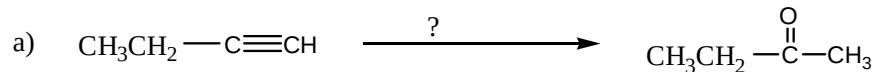
1. Suggest the possible structure of A,B,C,and D for the following reaction. (4pts)



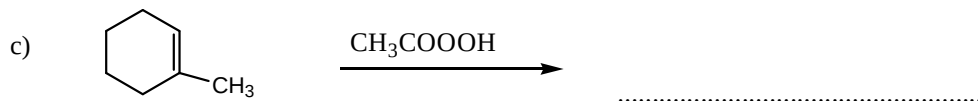
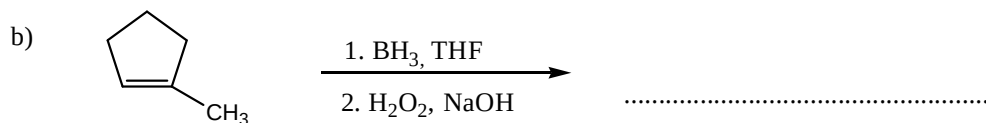
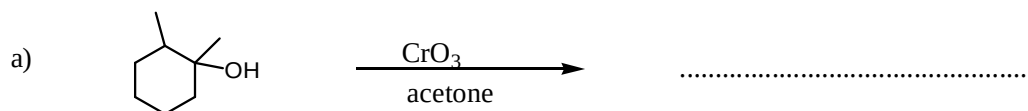
2. Give suitable reagents to accomplish the transformations below (3 pts)



3. Give suitable reagents to accomplish the transformations below (3 pts)



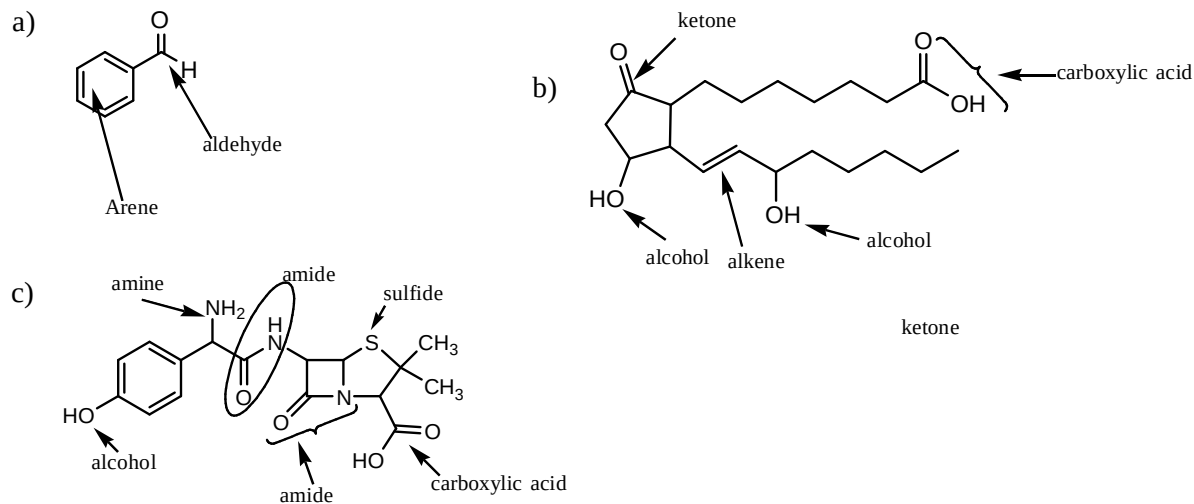
4. Complete the following reactions. If there is no reaction, write "NO REACTION."



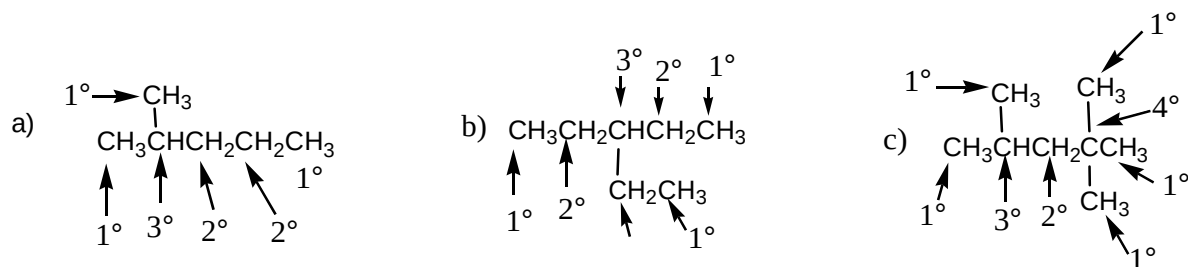
5. An organic compound **A** ( $C_3H_6O$ ) is quite resistant to oxidation. Reduction of **A** forms another compound **B** ( $C_3H_8O$ ). **B** reacts with HBr to give alkyl bromide **C**. The Grignard reagent derived from **C** reacts with **A** to give **D** ( $C_6H_{14}O$ ). What are the structures of **A**, **B**, **C**, and **D**? Write equations for all the reactions.
6. What alkene of molecular formula  $C_7H_{14}$ , when treated with ozone followed by zinc and acetic acid will give propanal and 2-methylpropanal.
7. How would you synthesis ethyl butanoate starting with propene and using any necessary organic or inorganic reagents with no more than two carbon atoms.
8. Starting from 1-Butyne, any alkyl halide and reagent needed outline the synthesis of
  - a) Hexane
  - b) 5-Methyl-3-hexene
9. Starting from acetylene and any alkyl halide needed, show stepwise synthesize of
  - a) 2-Chlorobutane
  - b) 4-Methyl-2-pentanol
10. Using any necessary reagents show reaction sequence to accomplish the following synthesis.
  - a) acetylene as your only source of carbon  $\xrightarrow{???}$  3-methyl-3-pentanol
  - b) 1-butene  $\xrightarrow{???}$  1-butanol
  - c) acetylene  $\xrightarrow{???}$  2-butanone
  - d) 1-butanol  $\xrightarrow{???}$  butanal
  - e) 1-butanol  $\xrightarrow{???}$  butanoic acid
  - f) cyclohexane  $\xrightarrow{???}$  cyclohexanone

## ANSWER FOR EXERCISES

### Exercise 2.1



### Exercise 2.2



### Exercise 2.3

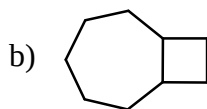
- 5-Ethyl-2,4,6-trimethyldecane
- 4-Ethyl-2-methylhexane
- 8-ethyl-4-isopropyl-6-methyldecane
- 3,5-dimethylheptane
- 3-ethyl-4,5,6-trimethylheptane

### Exercise 2.4

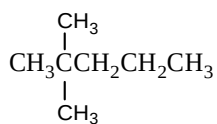
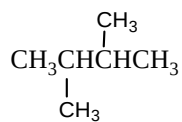
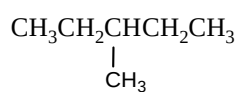
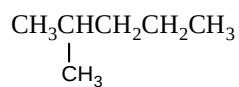
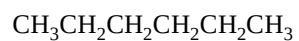
- tert-butylcyclononane or could be named systematically as (1,1-dimethylethyl)cyclononane
- 4-isopropyl-1,1-dimethyldecane
- Clohexylcyclohexane

### Exercise 2.5

1. a) 2,10-dimethylbicycl[4,4,0]decane  
b) 1,7-dichlorobicyclo[3,2,1]octane  
c) Bicyclo[2,2,1]heptane  
d) Spiro[2,2]pentane
- 2.



### Exercise 2.6



### Exercise 2.7

- a) (Z)                      b) (Z)                      c) (Z)                      d) (E)

### Exercise 2.8

- a) 2-chloro-2-methylbutane
- b) 2-chloro-2-methylbutane
- c) 2-chlorobutane
- d) 1-chloro-1-ethylcyclohexane

## UNIT 3

### 3. STEREOCHEMISTRY

#### Learning outcomes:

At the end of this unit, you will be able to:

- understand the concept of stereoisomerism
- identify the type of stereoisomerism
- distinguish enantiomers and diastereomers
- differentiate between optically active and inactive substances
- calculate specific optical rotation
- draw fisher projection formulas of a stereoisomers
- identify meso compounds

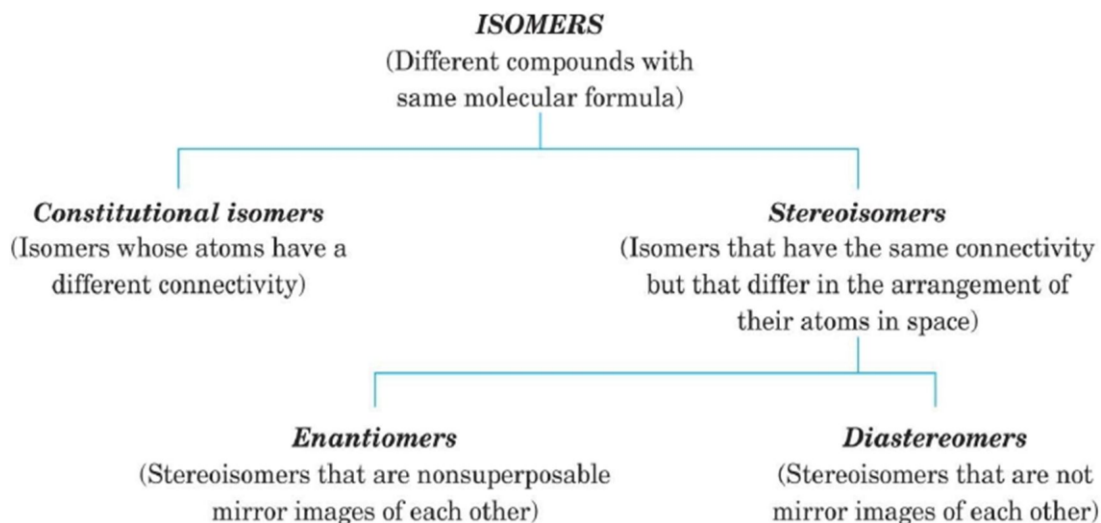
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### 3.0 Stereoisomerism

**Stereochemistry** refers to chemistry in three dimensions. Isomers that have the same constitution but differ in the spatial arrangement of their atoms are called **stereoisomers**. We have already had considerable experience with certain types of stereoisomers-those involving cis and trans substitution patterns in alkenes and in cycloalkanes.

Our major objectives in this chapter are to develop a feeling for molecules as three-dimensional objects and to become familiar with stereochemical principles, terms, and notation. A full understanding of organic and biological chemistry requires an awareness of the spatial requirements for interactions between molecules; this chapter provides the basis for that understanding.

#### SUBDIVISION OF ISOMERS



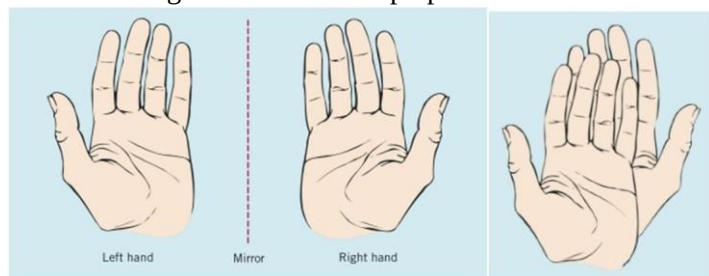
### 3.1 Molecular Chirality: Enantiomers

Everything has a mirror image, but not all things are superimposable on their mirror images.

Superposable: every identical atoms lie on one another like left and right hands

Mirror-image superimposability characterizes many objects we use every day. Cups and saucers, forks and spoons, chairs and beds are all identical with their mirror images. Many other objects though--and this is the more interesting case--are not. Your left hand and your right hand, for example, are mirror images of each other but can't be made to coin-cide point for point, palm to palm, knuckle to knuckle, in three dimensions.

- Left and right hand are not superposable



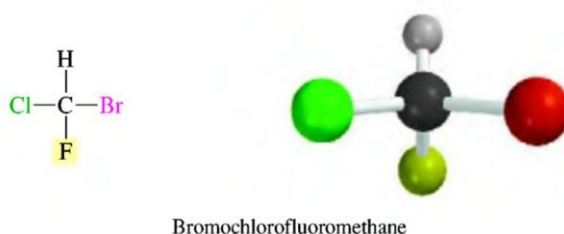
An object is chiral if it is not superimposable on its mirror image.

A molecule is **chiral** if its two mirror-image forms are not superimposable in three dimensions. The word chiral is derived from the Greek word cheir, meaning "hand," and it is entirely appropriate to speak of the "handedness" of molecules. The opposite of chiral is achiral.

A molecule that is superimposable on its mirror image is **achiral**.



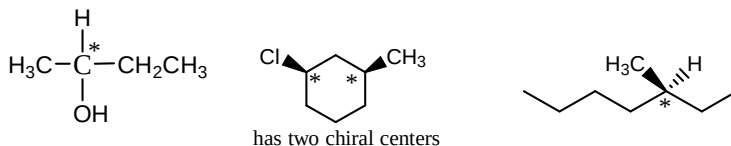
In organic chemistry, chirality most often occurs in molecules that contain a carbon that is attached to four different groups. An example is bromochlorofluoromethane ( $\text{BrClFCH}$ ).



### 3.2 The Chirality Center (\*)

(Also called asymmetric center, asymmetric carbon, chiral center, stereogenic center, and stereocenter)

A tetrahedral carbon atom that bears four different atoms or groups be called a **chirality center**



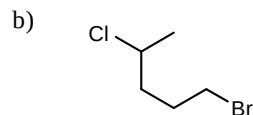
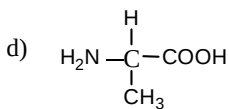
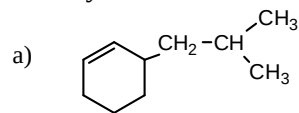
has two chiral centers

a chiral center cannot be:

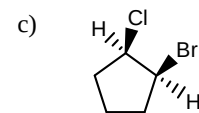
- sp- or sp<sup>2</sup>-hybridized (must be sp<sup>3</sup>)
- an atom with 2 identical substituents (e.g., any -CH<sub>2</sub>- group)

#### Exercise 3.1

Identify each chiral center and put asterisk at each chiral center.



e) 1-chloro-2-methyl-butane



f) 2-bromopentane

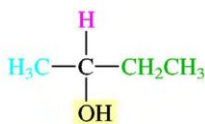
g) 2-methyl-1-butanol

#### Test for chirality

1. a molecule with one chiral center is **chiral**

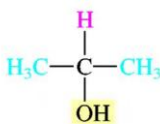
a molecule that has no chiral center is **achiral**

a molecule with more than one chiral center may or may not be chiral. This case use the second test for chirality



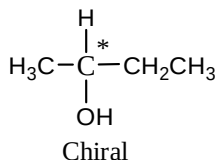
2-Butanol

Chiral; four different substituents at C-2

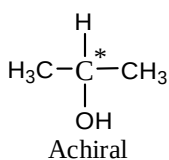


2-Propanol

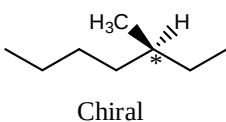
Achiral; two of the substituents at C-2 are the same



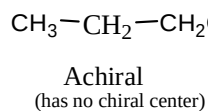
Chiral



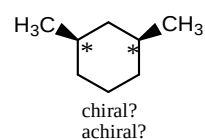
Achiral



Chiral



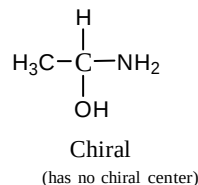
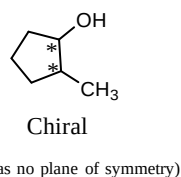
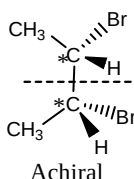
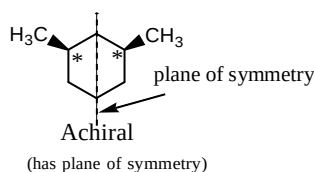
Achiral  
(has no chiral center)



chiral?  
achiral?

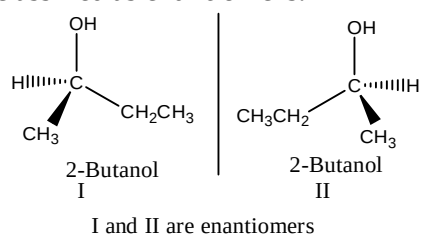
2. All molecules with planes of symmetry are **achiral**

(A plane of symmetry divides an entire molecule into two pieces that one half of the molecule is the mirror image of the other half)



## Enantiomers

Chiral molecule and its mirror are classified as **enantiomers**.



a chiral molecule can have one, and only one, enantiomer.

## 3.3 Configuration of Chiral Compounds (R-S Notation System)

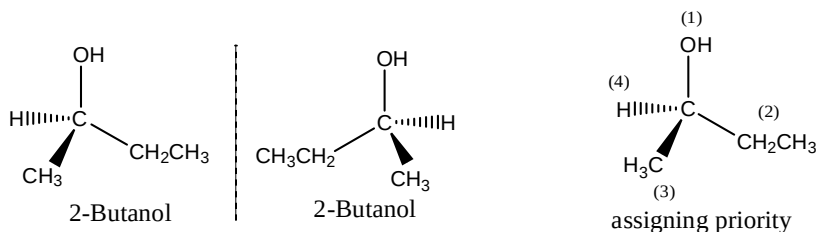
Also called the Cahn-Ingold-Prelog system

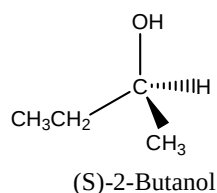
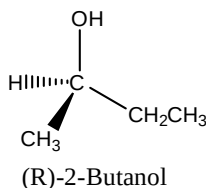
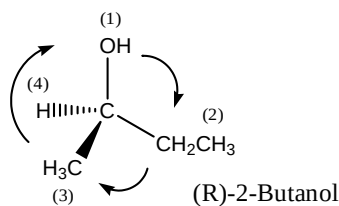
- nomenclature method for designating the specific arrangement of groups about a stereogenic center
- differentiates between enantiomers

The four groups attached to the stereogenic carbon are assigned priorities from highest (1) to lowest (4)

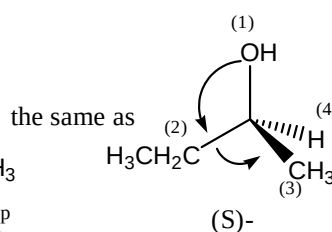
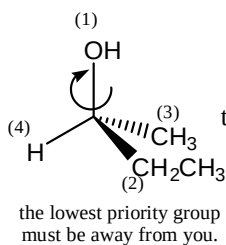
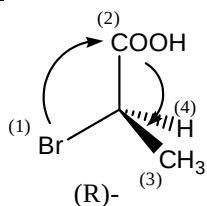
Priorities are assigned as follows

- Atoms directly attached to the stereogenic center are compared.
- Atoms with higher atomic number are given higher priority(1)
- align the lowest priority group (4 ) behind the chiral carbon.
- if the direction of 1→2→3 is **clockwise**, the enantiomer is **R**
- if 1→2→3 is **counterclockwise**, the enantiomer is **S**

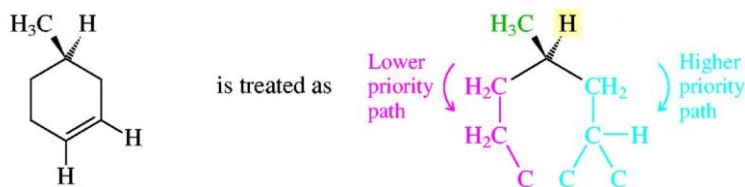




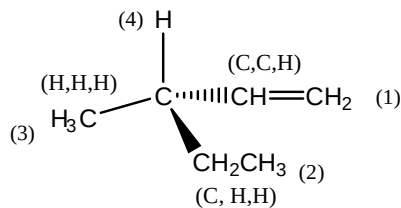
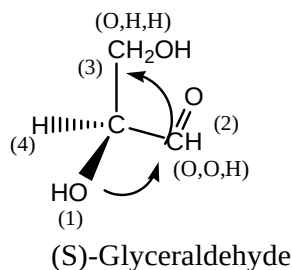
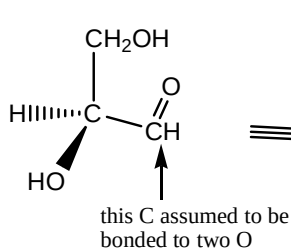
### R or S?



Compounds in which a chirality center is part of a ring are handled in an analogous fashion. To determine, for example, whether the configuration of (+)-4-methylcyclohexene is R or S, treat the right- and left-hand paths around the ring as if they were independent substituents.

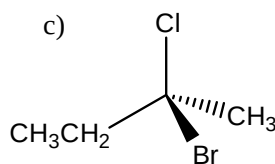
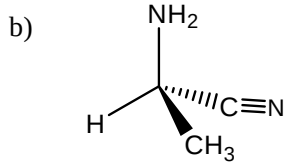
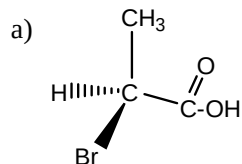


Groups with double or triple bonds are assigned priorities as if their atoms were duplicated or triplicated.



### Exercise 3.2

#### 1. Assign R/S



#### 2. Draw the structure of (S)-2-pentanol

## Properties of Enantiomers

- Enantiomers have almost all identical physical properties (melting point, boiling point, density)

| Physical Properties | (R)-2-Butanol | (S)-2-Butanol |
|---------------------|---------------|---------------|
| Boiling point       | 99.5 °C       | 99.5 °C       |
| Density(g/mL)       | 0.808         | 0.808         |
| Refractive index    | 1.397         | 1.397         |

- However enantiomers rotate the plane of plane-polarized light in equal but opposite directions (Optical Activity)

## 3.4 Optical Activity and Plane Polarized Light

Optical activity is the ability of a chiral substance to rotate the plane of plane-polarized light and is measured using an instrument called a polarimeter.

The light used to measure optical activity has two properties: it consists of a single wavelength and it is plane-polarized. The wavelength used most often is 589 nm (called the D line), which corresponds to the yellow light produced by a sodium lamp.

Chiral compounds rotate the plane of plane-polarized light. Rotation measured in degrees

*clockwise (dextrorotatory or +) or*

*counterclockwise (levorotatory or -)*

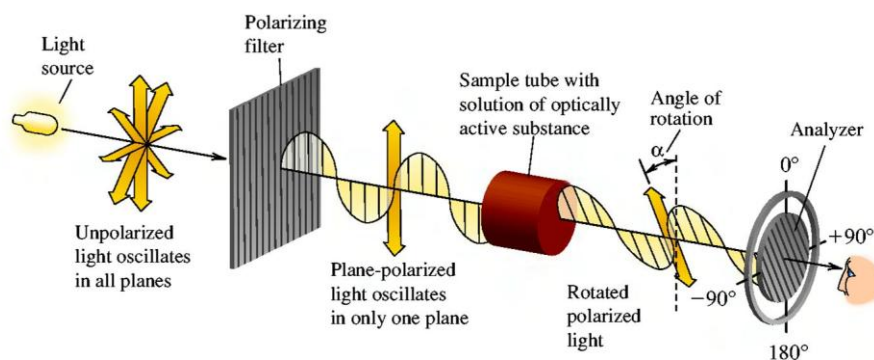


Fig 3.1 Polarimeter

The sample is "optically active" if it rotates the plane of polarized light. The direction and magnitude of rotation are measured using a second polarizing filter (the "analyzer") and cited as  $\alpha$ , the observed rotation. To be optically active, the sample must contain a chiral substance and one enantiomer must be present in excess of the other. A substance that does not rotate the plane of polarized light is said to be optically inactive. All achiral substances are *optically inactive*.

Mixtures containing equal quantities of enantiomers are called **racemic mixtures**. Racemic mixtures are optically inactive. Conversely, when one enantiomer is present in excess, a net rotation of the plane of polarization is observed.

---

### 3.5 Measurement of Optical Rotation

---

Rotation of the plane of polarized light in the clockwise sense is taken as positive (+), and rotation in the counterclockwise sense is taken as a negative (-) rotation. Older terms for positive and negative rotations were *dextrorotatory* and *levorotatory*, from the Latin prefixes dextro- ("to the right") and levo- ("to the left"), respectively.

The observed rotation  $\alpha$  of an optically pure substance depends on how many molecules the light beam encounters. A filled polarimeter tube twice the length of another produces twice the observed rotation, as does a solution twice as concentrated. To account for the effects of path length and concentration, chemists have defined the term specific rotation, given the symbol  $[\alpha]$ .

**Specific Rotation**  $[\alpha]$ : standard amount of optical rotation by 1 g/mL of compound in a standard 1dm cell.

$$[\alpha] = \frac{\alpha}{c \cdot l}$$

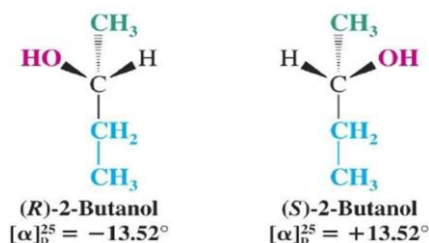
where  $[\alpha]$  = the specific rotation

$\alpha$  = the observed rotation

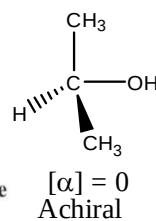
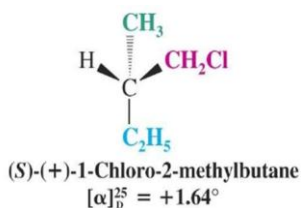
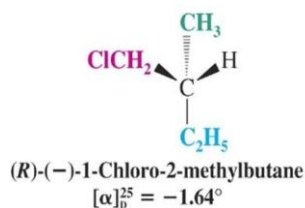
$c$  = the concentration of the solution in grams per milliliter of solution (or density in  $\text{g mL}^{-1}$  for neat liquids)

$l$  = the length of the tube in decimeters (1 dm = 10 cm)

It is convenient to distinguish between enantiomers by prefixing the sign of rotation to the name of the substance. For example, we refer to one of the enantiomers of 2-butanol as (+)-2-butanol and the other as (-)-2-butanol. Optically pure (+)-2-butanol has a specific rotation  $[\alpha]$  of + 13.5°; optically pure (-)-2-butanol has an exactly opposite specific rotation  $[\alpha]$  of - 13.5°



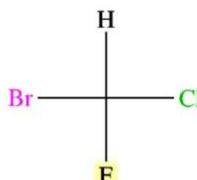
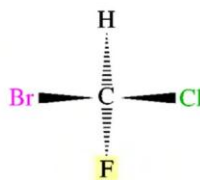
- There is no straightforward correlation between the *R,S* designation of an enantiomer and the direction [(+) or (-)] in which it rotates plane polarized light
- Achiral molecules have zero degree of rotation



### 3.6 Fischer Projections

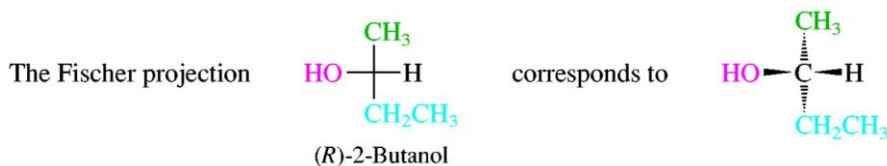
It is a 2-dimensional representation of chiral molecules.

Fischer projections are always generated the same way: the molecule is oriented so that the vertical bonds at the chirality center are directed away from you and the horizontal bonds point toward you. A projection of the bonds onto the page is a cross. The chirality center lies at the center of the cross but is not explicitly shown.



(*R*)-Bromochlorofluoromethane

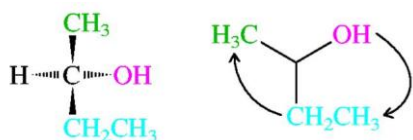
It is customary to orient the molecule so that the carbon chain is vertical with the lowest numbered carbon at the top as shown for the Fischer projection of (*R*)-2-butanol.



To verify that the Fischer projection has the *R* configuration at its chirality center, rotate the three-dimensional representation so that the lowest-ranked atom (H) points away from you. Be careful to maintain the proper stereochemical relationships during the operation.



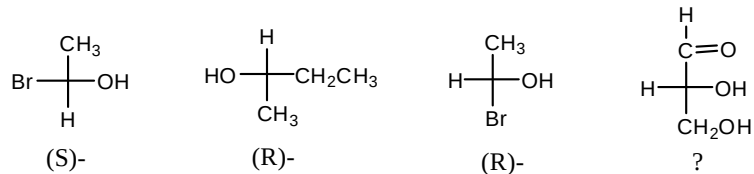
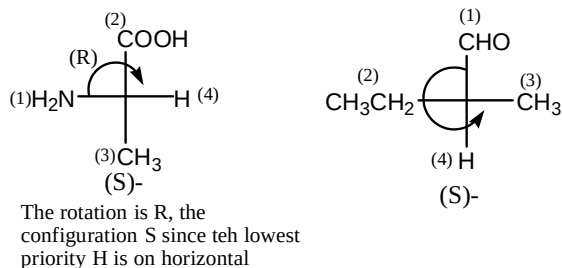
With H pointing away from us, we can see that the order of decreasing precedence  $\text{OH} > \text{CH}_2\text{CH}_3 > \text{CH}_3$  traces a clockwise path, verifying the configuration as *R*.



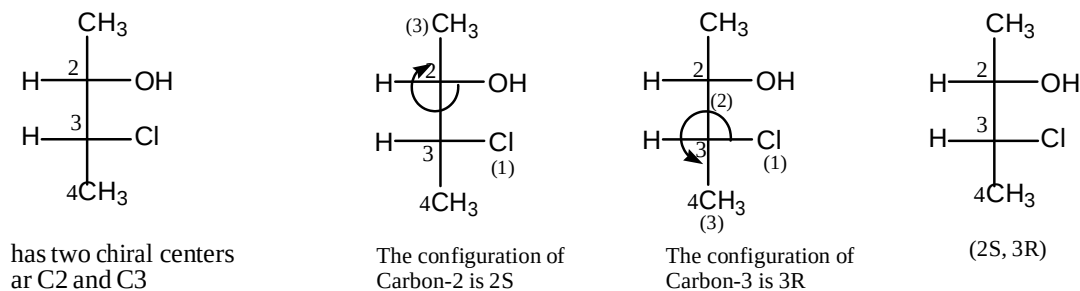
### R/S in Fischer

When the lowest priority is on the top or bottom, we can assign R/S directly as the usual R/S rule.

When the lowest priority is on horizontal line, we can assign R/S by taking the opposite direction of R/S rule.

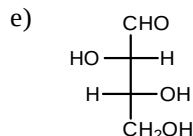
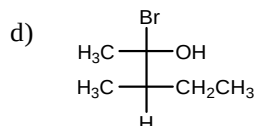
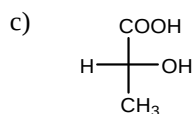
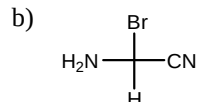
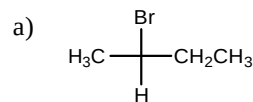


Assigning R/S for molecules with more than one chiral center.



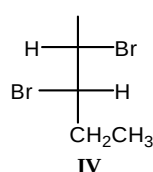
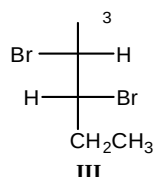
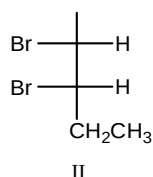
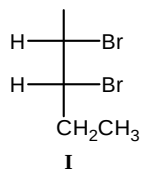
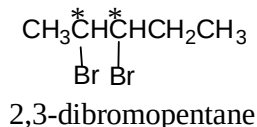
### Exercise 3.3

Assign R/S



## 3.7 Multiple Stereogenic Centers: Diastereomers

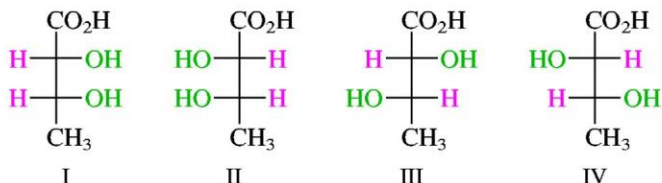
Compounds with more than 2 stereocenters have more than 2 stereoisomers. How many stereoisomers are possible for 2,3-dibromopentane?



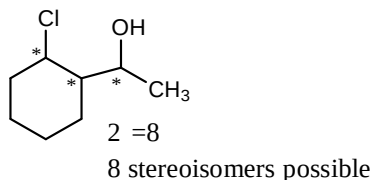
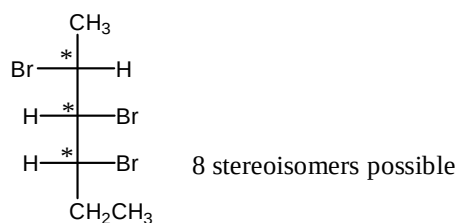
Four stereoisomers possible.

Stereoisomers I and II are enantiomers of each other; the enantiomer of (S,R) is (R,S). Likewise stereoisomers III and IV are enantiomers of each other, the enantiomer of (R,R) being (S,S).

Stereoisomer I is not a mirror image of III or IV, so it is not an enantiomer of either one. Stereoisomers that are not related as an object and its mirror image are called **diastereomers**; diastereomers are stereoisomers that are not enantiomers. Thus, stereoisomer I is a diastereomer of III and a diastereomer of IV. Similarly, II is a diastereomer of III and IV.



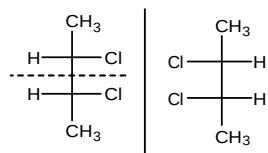
In general, n stereocenters give  $2^n$  stereoisomers



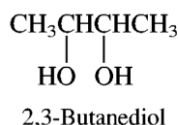
### 3.8 Meso Compounds

Compounds with chiral centers but which are not chiral are called **Meso**.

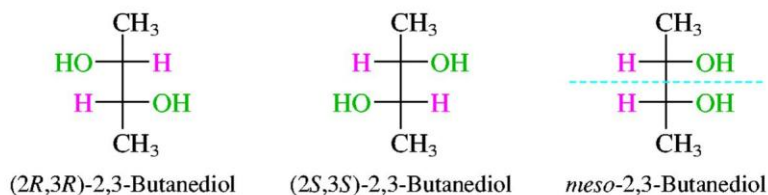
- e.g., (2R,3S)-2,3-dibromobutane (same as its mirror image)



Now consider a molecule, such as 2,3-butanediol, which has two chirality centers that are equivalently substituted.



Only three, not four, stereoisomeric 2,3-butanediols are possible.

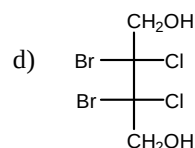
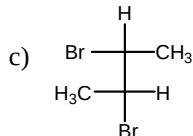
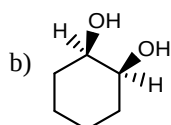
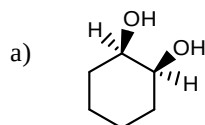


The (2R,3R) and (2S,3S) forms are enantiomers of each other and have equal and opposite optical rotations. A third combination of chirality centers, (2R,3S), however, gives an achiral structure that is superimposable on its (2S,3R) mirror image. Because it is achiral, this third stereoisomer is optically inactive (*meso*).

The *meso* form is known as *meso*-2,3-butanediol.

#### Exercise 3.4

1. Which of the following structure represent a *meso* compound?



2. Which of the following has a *meso* form?

a) 2,3-dibromobutane

b) 2,3-dibromopentane

c) 2,4-dibromopentane

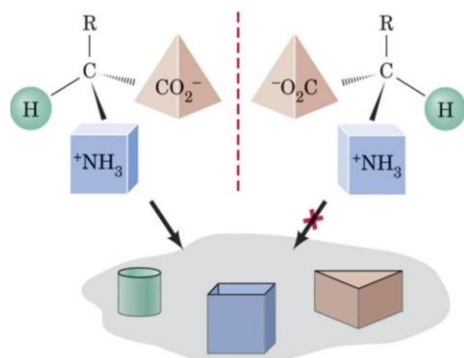
### 3.9 The Biological importance of Chirality

- Most of the molecules that make up plants and animals are chiral.

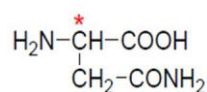
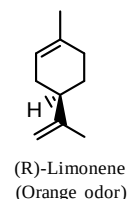
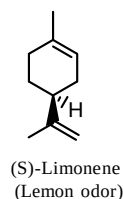
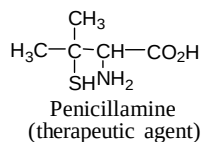
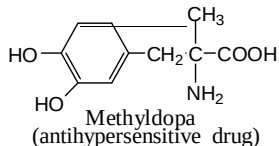
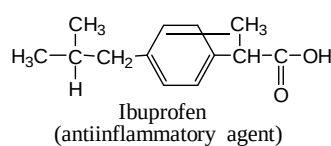
Eg. The aminoacids that make up proteins, natural sugars are chiral

- Chiral molecules affect the human body in different way.

The binding specificity of a chiral receptor site for a chiral molecule is usually only favorable in one way

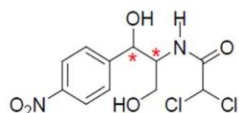


Enantiomers of natural molecules show different biological activities.



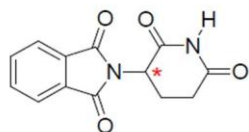
**Asparagine**

**R-** sweet  
**S-** bitter



**Chloramfenikol**

**R,R-** antibiotic  
**S,S-** inactive

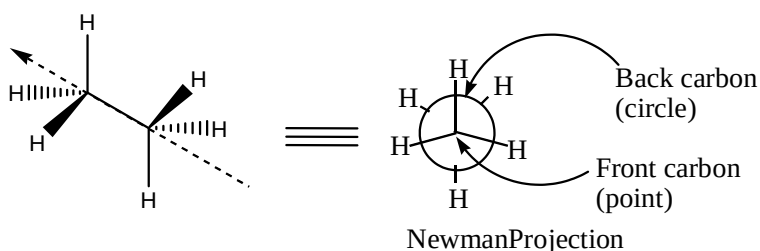
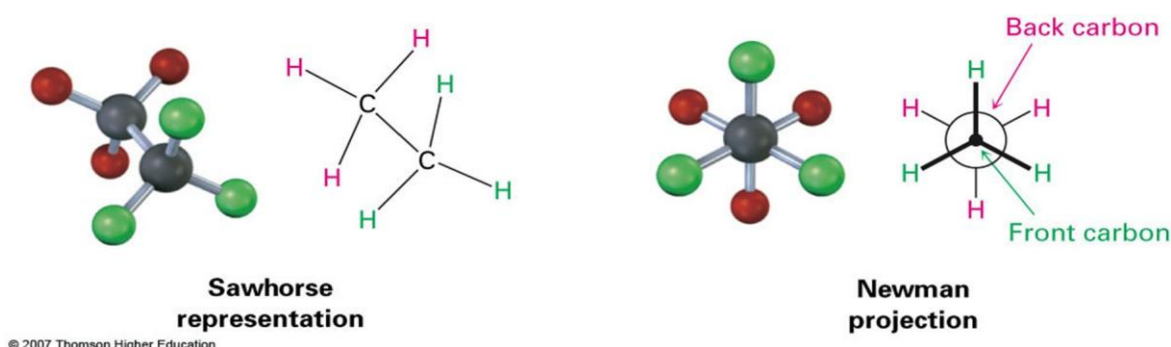


**Thalidomide**

**R-** inactive  
**S-** extremely  
 teratogenic

## 3.10 Conformational Isomerism

- **Conformations:** the **temporary molecular shapes** that result from **rotations of groups** about **single bonds**; structures related by bond rotations.
- **Conformational analysis:** the analysis of the **energy changes** that a molecule undergoes as **groups** rotate about **single bonds**. It is the study of how conformational factors affect the structure of a molecule and its physical, chemical, and biological properties.
- **Conformer:** conformation at potential energy minima. Conformers usually interconvertible at room temperature.
- Conformations can be represented in 2 ways: **Newman and Sawhorse formula**



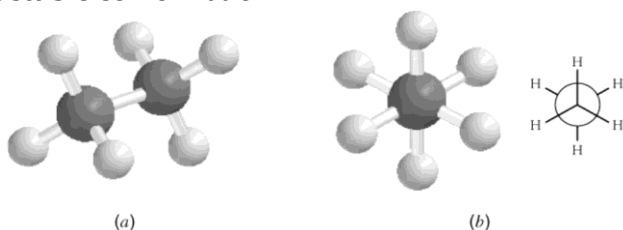
### 3.10.1 Conformational Analysis in Alkanes:

#### Conformation of Ethane

There are two conformation of ethane: Staggered and eclipsed.

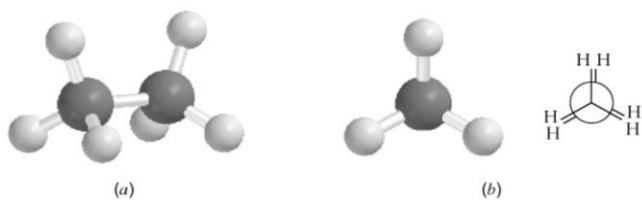
**Staggered** conformation of ethane: all 6 C-H bonds are as far away as possible.

⇒ **most stable** conformation



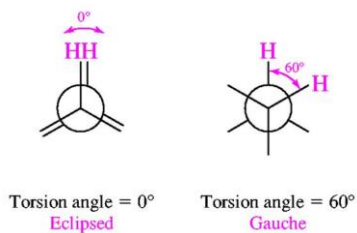
**Eclipsed conformation:** all 6 C-H bonds are as close as possible to each other.

⇒ **least stable** conformation.

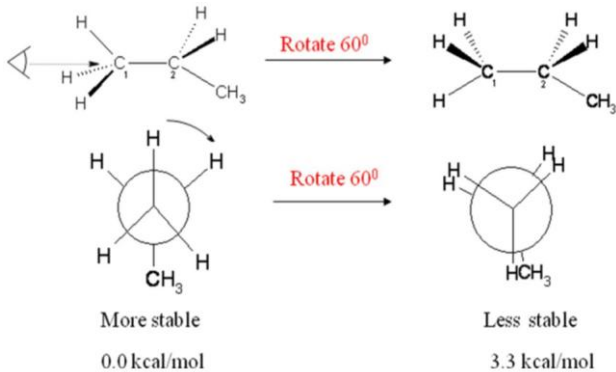


Two main explanations have been offered for the difference in stability between the two conformations. One explanation holds that repulsions between bonds on adjacent atoms destabilize the eclipsed conformation. The other suggests that better electron delocalization stabilizes the staggered conformation. The latter of these two explanations is now believed to be the correct one.

Conformations in which the torsion angles between adjacent bonds are other than  $60^\circ$  are said to have **torsional strain**.

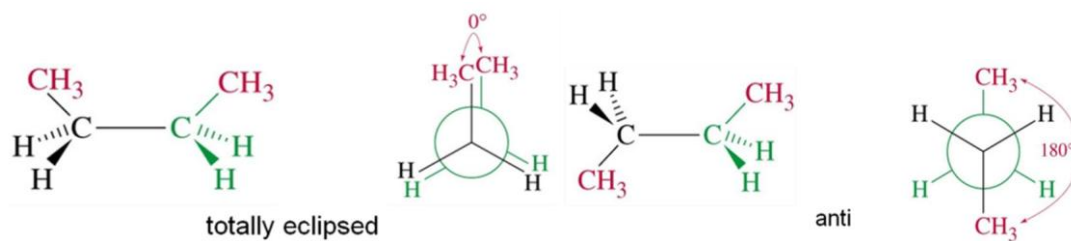


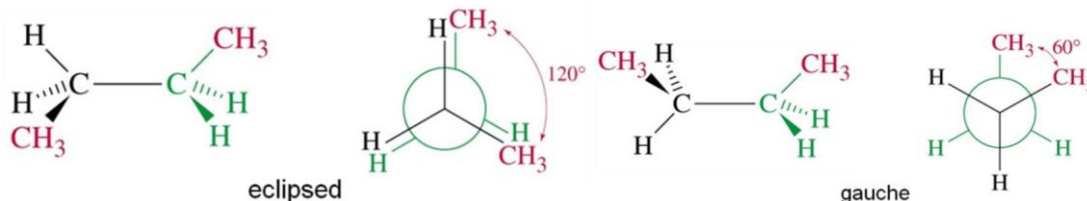
### Conformations of Propane



The additional 0.3 kcal/mol energy is due to **steric strain**.

### Conformations of Butane



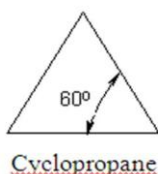


- Gauche conformations are more stable than eclipsed conformations.
- Anti conformation is the most stable conformation of butane.

### 3.10.2 Conformations and Stabilities of Cycloalkanes

- Cycloalkanes possess types of strain that do not exist in noncyclic alkanes i.e angle strain.

**Angle Strain** -is the strain a molecule has because one or more of its bond angles deviate from the ideal value; in the case of alkanes the ideal value is  $109.5^\circ$ .



What is the normal bond angle for an  $sp^3$  carbon?

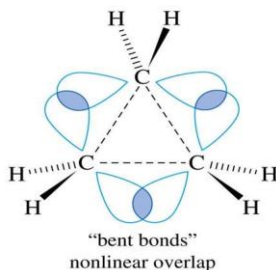
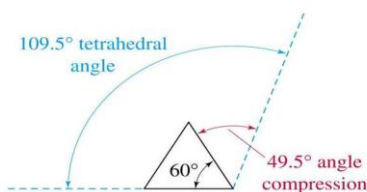
$109.5^\circ$

Deviation  $109.5 - 60 = 49.9^\circ$

**Ring Strain** – total strain (sum of torsional, steric and angle strain) in a cycloalkane compared to an open, noncyclic reference compound.

#### A. Cyclopropane

- Large ring strain due to angle compression
- Very reactive, weak bonds



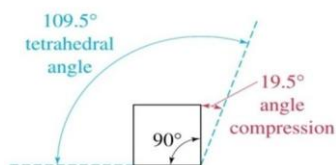
Strong  $sp^3-sp^3\sigma$  bonds are not possible for cyclopropane, because the  $60^\circ$  bond angles of the ring do not permit the orbitals to be properly aligned for effective overlap. As a result cyclopropane is unstable compound.

## Cyclobutane

Angle strain due to compression

Torsional strain partially relieved by ring-puckering

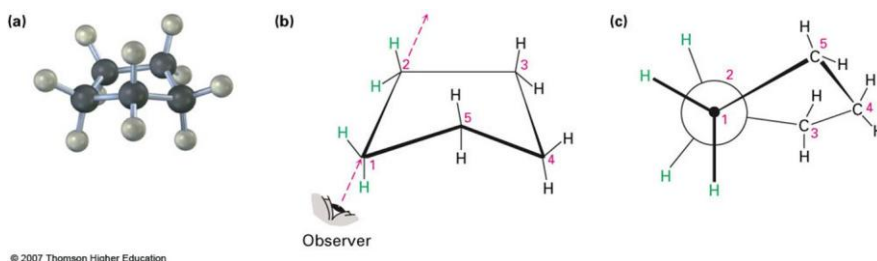
Unstable molecule



## Cyclopentane

Angle strain in the planar conformation of cyclopentane is relatively small because the  $108^\circ$  angles of a regular pentagon are not much different from the normal  $109.5^\circ$  bond angles of  $sp^3$ -hybridized carbon.

- Small angle strain. Stable molecule



## Cyclohexane

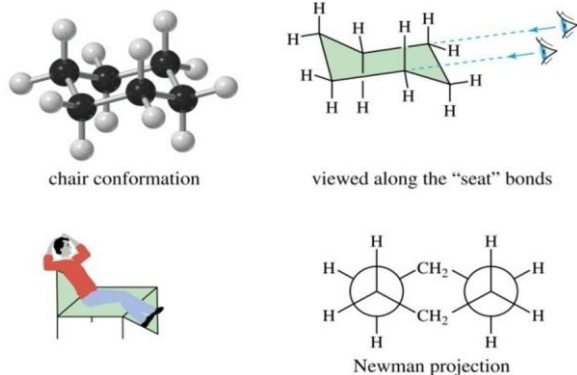
The bond angles is  $109.5^\circ$  and all hydrogens are staggered.

No angle strain and no torsional strain.

Two conformations: Chair and boat

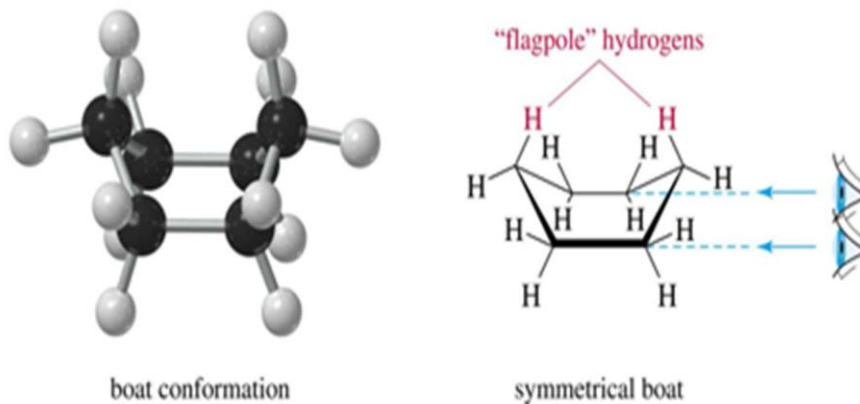
### Chair Conformer

- C—C bond angles are all  $109.5^\circ$
- free of angle strain.
- free of torsional strain



### Boat Conformer

- free of angle strain.
- C—C bond angles are all  $109.5^\circ$
- has torsional strain and **flagpole interaction**



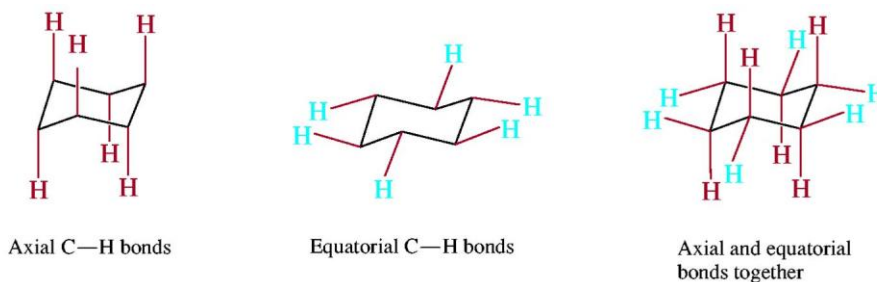
- The most stable form is chair conformer. Boat cyclohexane is less stable because of the repulsion between the two flagpole hydrogen atoms.

### Equatorial and Axial Hydrogens of Cyclohexane

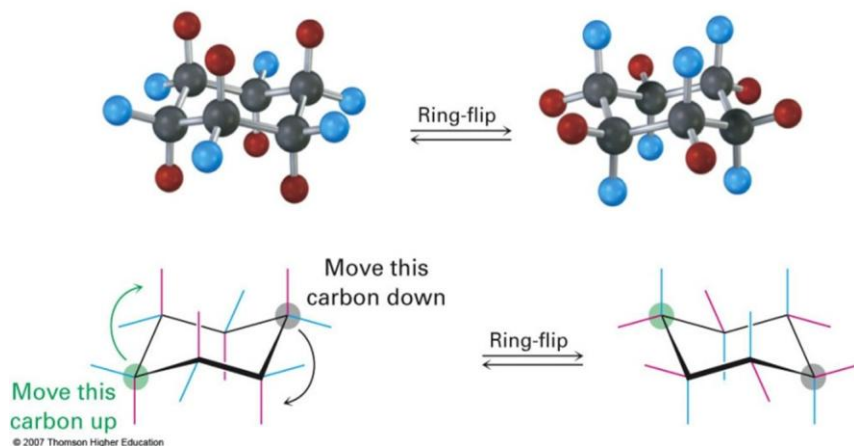
The chair conformation has two kinds of positions for substituents on the ring: *axial* positions and *equatorial* positions

Six of the hydrogens, called *axial* hydrogens, have their bonds parallel to a vertical axis that passes through the ring's center. These axial bonds alternately are directed up and down on adjacent carbons. The second set of six hydrogens, called *equatorial* hydrogens, are located approximately along the equator of the molecule.

Each carbon atom in cyclohexane has one axial and one equatorial hydrogen.

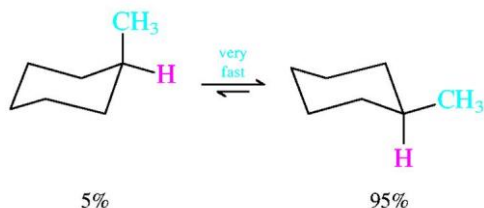


- The axial and equatorial hydrogens are exchangeable by a process called “**ring flipping**”.

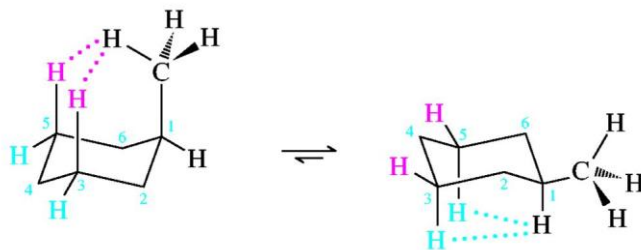


## Conformational Analysis of Monosubstituted Cyclohexanes

Ring inversion in methylcyclohexane differs from that of cyclohexane in that the two chair conformations are not equivalent. In one chair the methyl group is axial; in the other it is equatorial. At room temperature approximately 95% of the molecules of methylcyclohexane are in the chair conformation that has an equatorial methyl group, whereas only 5% of the molecules have an axial methyl group.



A methyl group is less crowded when it is equatorial than when it is axial. One of the hydrogens of an axial methyl group is within 190-200 pm of the axial hydrogens at C-3 and C-5. This distance is less than the sum of the van der Waals radii of two hydrogens (240 pm) and causes van der Waals strain in the axial conformation. When the methyl group is equatorial, it experiences no significant crowding.

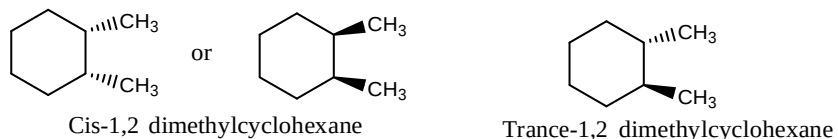


The greater stability of an equatorial methyl group, compared with an axial one, is another example of a steric effect. An axial substituent is said to be crowded because of **1,3-diaxial repulsions** between itself and the other two axial substituents located on the same side of the ring.

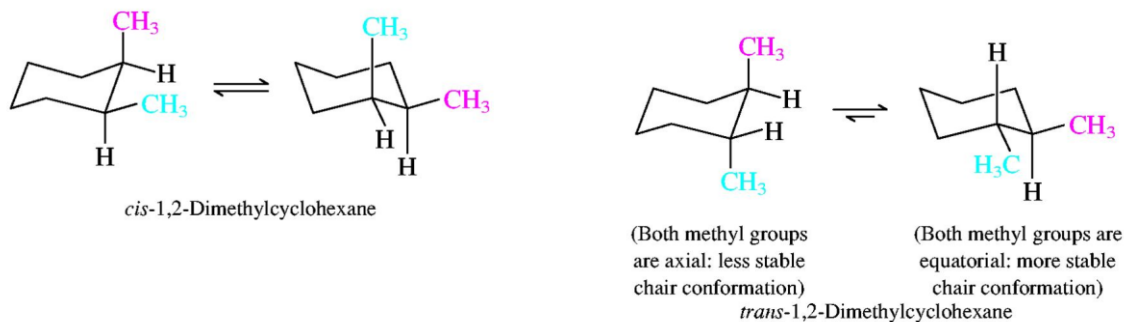
Generally a branched alkyl group such as isopropyl exhibits a greater preference for the equatorial orientation than does methyl.

## Disubstituted Cyclohexanes

In a **trans-disubstituted** cyclohexane, one group is attached by an **upper** bond and one by the **lower** bond; in a **cis-disubstituted** cyclohexane, both groups are attached by an **upper** bond or both by the **lower** bond. There are two isomers of 1,2-dimethylcyclohexane. *cis* and *trans*

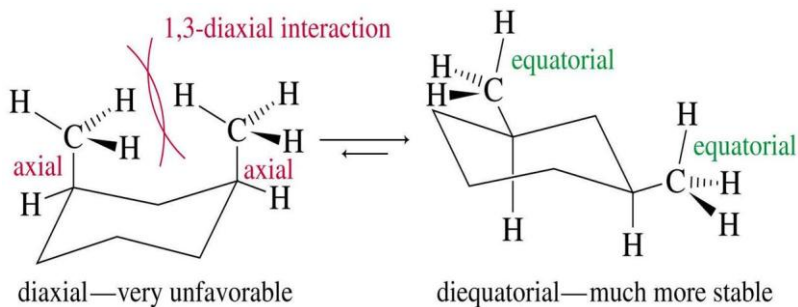


The *cis* stereoisomer has two chair conformations of equal energy, each containing one axial and one equatorial methyl group.

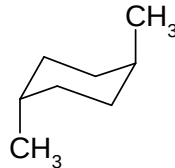
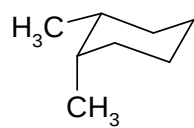


Both methyl groups are equatorial in the most stable conformation of *trans*-1,2-dimethylcyclohexane.

*Trans*-1,3-Disubstituted cyclohexane with two **different alkyl groups**, the conformation of **lower energy** is the one having the **larger group** in the **equatorial** position.



Can you name these dimethylcyclohexanes?



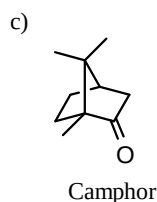
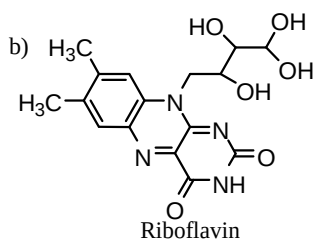
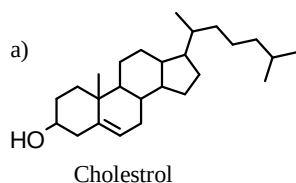
Which of these represent the most stable conformation possible?

## Self-TestExercise 3.0

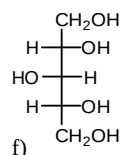
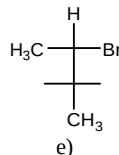
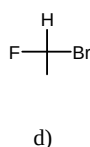
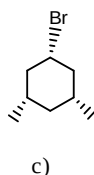
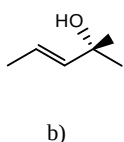
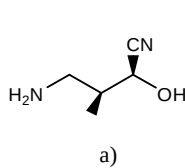
1. Define the following terms

Superposable molecule, stereogenic center, chiral molecule, achiral molecule, enantiomers, diastereomers, plane of symmetry, meso compound.

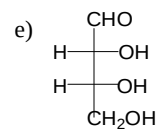
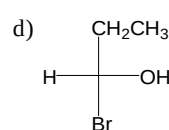
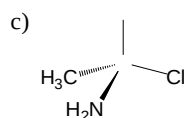
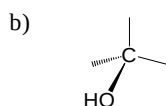
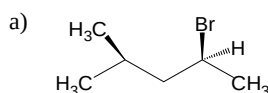
2. Which of the following molecules have chiral center? Place an asterisk to each chiral center.



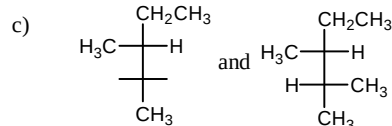
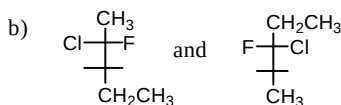
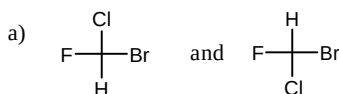
3. Label each molecule as chiral, achiral, or achiral /meso. Designate each chirality center as R or S



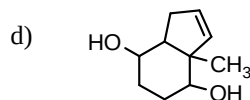
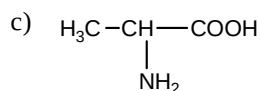
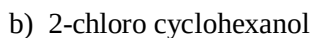
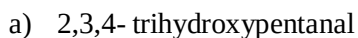
4. Assign **R** or **S** configuration for each chiral carbon.



5. What is the relationship between each or the following pairs of compounds?(identical, enantiomers, diastereomers, none)



6. How many stereoisomers possible from each of the following molecules?



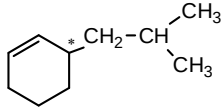
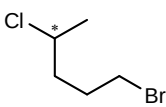
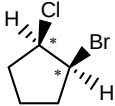
7. What is the difference between a) enantiomers and diastereomers,

b) meso compounds and achiral compounds

8. Identify the most stable conformation of a) ethane b) Butane c) Cyclohexane
9. Draw chair cyclohexane ring, and label all position of axial and equatorial.
10. What is torsional strain, ring strain?
11. What is 1,3 diaxial interaction?
12. Why is a 1,3- cis disubstituted cyclohexane more stable than its trans-isomer?
13. Why is a 1,2- trans disubstituted cyclohexane more stable than its cis- isomer.?

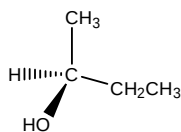
## ANSWER FOR EXERCISES

### Exercise 3.1

- a) 
- b) 
- c) 
- d)  $\text{H}_2\text{N}-\underset{\text{CH}_3}{\overset{\text{H}}{\text{C}}}-\text{COOH}$  no chiral center
- e) 1-chloro-2-methyl-butane  
 $\text{CH}_2-\overset{*}{\text{CH}}(\text{Cl})\text{CH}_2\text{CH}_3$
- f) 2-bromopentane  
 $\text{CH}_3-\overset{*}{\text{CH}}(\text{Br})\text{CH}_2\text{CH}_2\text{CH}_3$
- g) 2-methyl-1-butanol  
 $\text{CH}_2-\overset{*}{\text{CH}}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

### Exercise 3.2

1. a) (S)- b) (R) - c) (S)-
- 2.

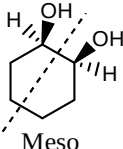


### Exercise 3.3

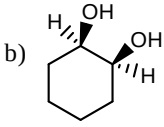
- a) (R) b) (S) c) (R) d) (2S, 3R) e) (2S, 3R)

### Exercise 3.4

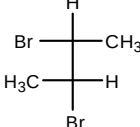
- 1)
 



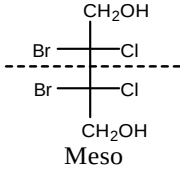
Meso



not meso



not meso



Meso
- 2)
  - a) has meso form
  - b) has no meso form
  - c) has meso form

## UNIT 4

### 4. MAJOR ORGANIC REACTIONS

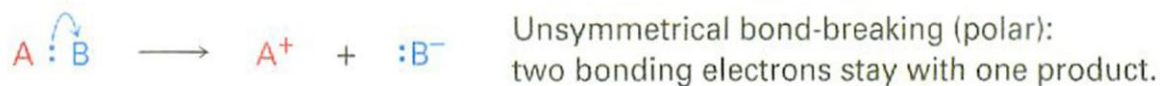
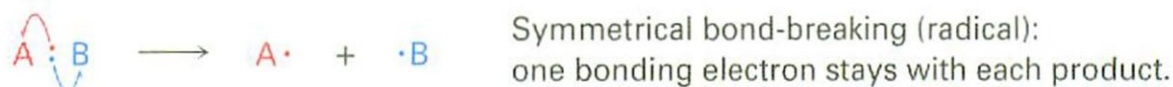
#### Learning outcomes:

At the end of this unit, you will be able to:

- understand the concept of reaction mechanism
- show electron movement in organic reactions
- identify types of organic reaction
- predict organic reaction products with mechanism.

#### Introduction

All chemical reaction involve bond breaking and bond formation. There are two ways in which a covalent bond can break:

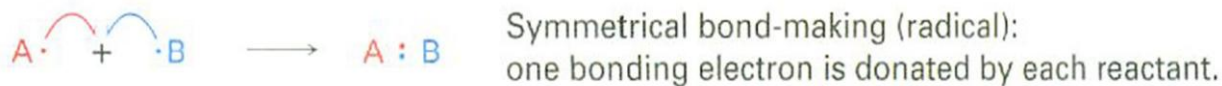


Curved arrows indicate breaking and forming of bonds

Arrowheads with a “half” head (“fish-hook”) indicate homolytic and homogenic steps 

Arrowheads with a complete head indicate heterolytic and heterogenic steps 

Similarly there are two ways bond formation



Kind of organic reaction : Substitution , Addition, Elimination, Rearrangement

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#### 4.1 Substitution Reaction

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## Introduction

**Nucleophile (Nu):**-is a substance that is nucleus-loving.

has a -vely polarized, electron rich atom and can form a bond by donating electron

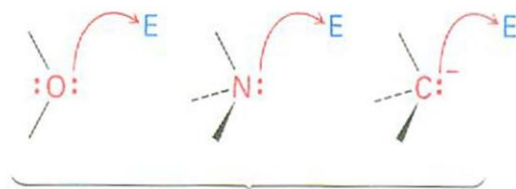
**Electrophile (E):**-is a substance that is electron-loving.

has a +vely charge, electron poor atom and can form a bond by accepting a pair of electrons from nucleophiles.

### Using curved arrows in polar reaction

Electrons move from a nucleophilic source (Nu) to an electrophilic sink(E)

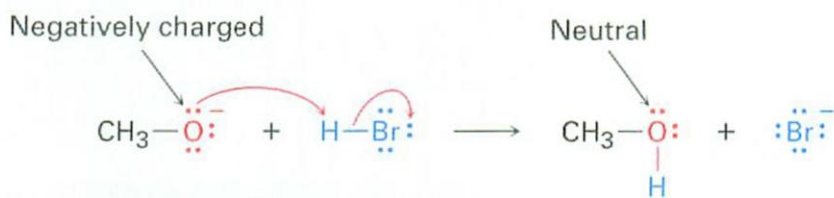
Electrons usually flow  
from one of these  
nucleophiles.



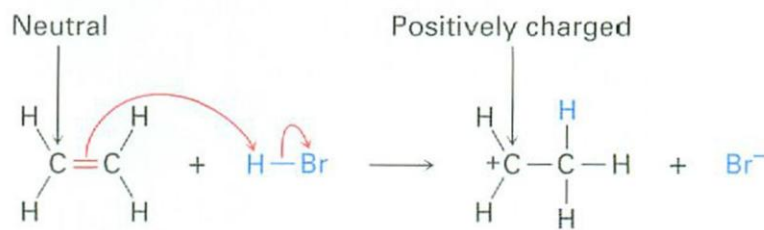
Electrons usually flow  
to one of these  
electrophiles.

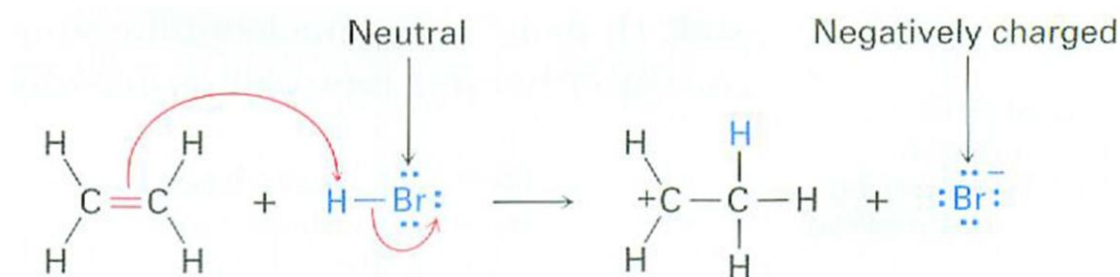


**The nucleophile can be either negatively charged or neutral.** If the nucleophile is negatively charged, the atom that gives away an electron pair becomes neutral. For example:

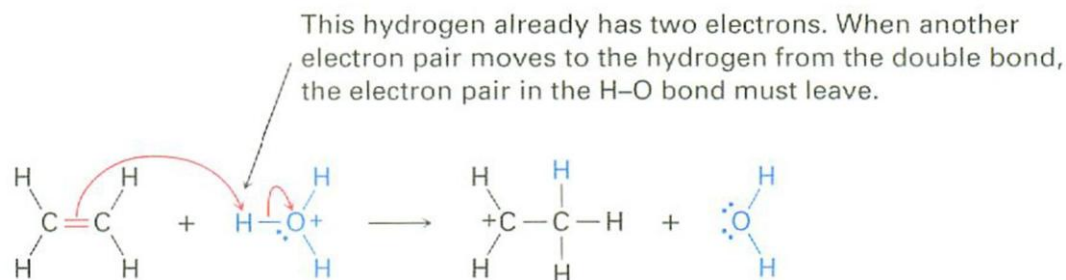


If the nucleophile is neutral, the atom that gives away an electron pair acquires a positive charge. For example:





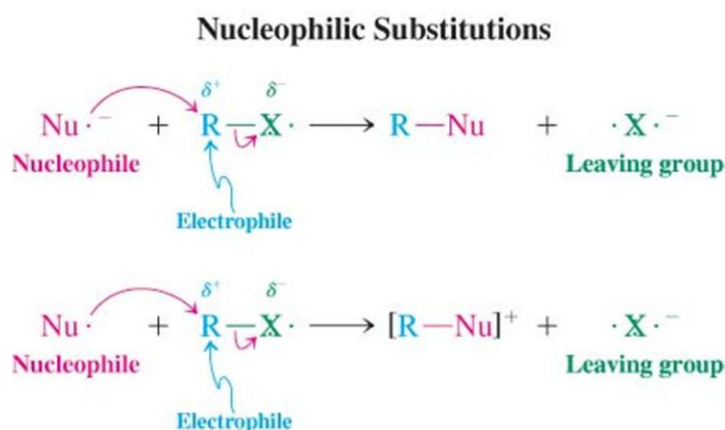
The octet rule must be followed during electron movement



### 4.1.1S<sub>N</sub>1and S<sub>N</sub>2 Mechanism

#### Nucleophilic Substitution

Nucleophiles attack electrophilic centers. Nucleophilic substitution of a haloalkane can be described by two general equations:



In both cases, the leaving group is the halide anion, X<sup>-</sup>. In describing reactions, the organic starting material is called the substrate of the reaction. Here, the substrate is being attacked by a nucleophile.

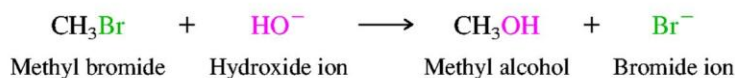
There are two mechanisms of substitution: S<sub>N</sub>1 & S<sub>N</sub>2

#### S<sub>N</sub>2 Mechanism

The mechanisms by which nucleophilic substitution takes place have been the subject of much study.

Extensive research by Sir Christopher Ingold and Edward D. Hughes and their associates at University

College, London, during the 1930s emphasized kinetic and stereochemical measurements to probe the mechanisms of these reactions. Recall that the term kinetics refers to how the rate of a reaction varies with changes in concentration. Consider the nucleophilic substitution in which sodium hydroxide reacts with methyl bromide to form methyl alcohol and sodium bromide:

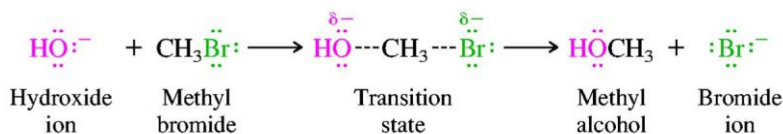


The rate of this reaction is observed to be directly proportional to the concentration of both methyl bromide and sodium hydroxide. It is first-order in each reactant, or second order overall.

$$\text{Rate} = k[\text{CH}_3\text{Br}][\text{HO}^-]$$

Hughes and Ingold interpreted second-order kinetic behavior to mean that the rate determining step is bimolecular, that is, that both hydroxide ion and methyl bromide are involved at the transition state. The symbol given to the detailed description of the mechanism that they developed is  $\text{S}_{\text{N}}2$ , standing for substitution **nucleophilic bimolecular**.

The  $\text{S}_{\text{N}}2$  mechanism is a single-step process in which both the alkyl halide and the nucleophile are involved at the transition state. Cleavage of the bond between carbon and the leaving group is assisted by formation of a bond between carbon and the nucleophile. In effect, the nucleophile "pushes off" the leaving group from its point of attachment to carbon. The  $\text{S}_{\text{N}}2$  mechanism for the hydrolysis of methyl bromide may be represented by a single elementary step:

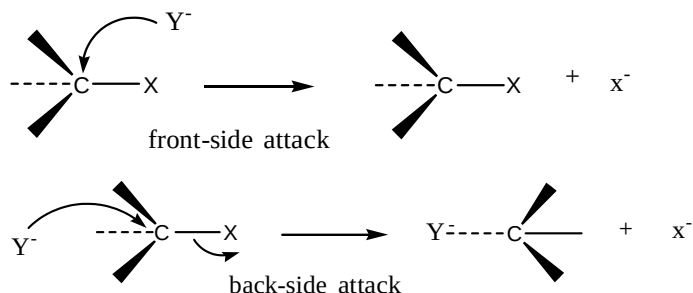


## STEREOCHEMISTRY OF $\text{S}_{\text{N}}2$ REACTIONS

What is the structure of the transition state in an  $\text{S}_{\text{N}}2$  reaction? In particular, what is the spatial arrangement of the nucleophile in relation to the leaving group as reactants pass through the transition state on their way to products?

Two stereochemical possibilities present themselves. The nucleophile simply assumes the position occupied by the leaving group. It attacks the substrate at the same face from which the leaving group departs. This is called **front-side displacement**, or substitution with retention of configuration.

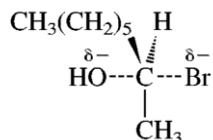
In a second possibility the nucleophile attacks the substrate from the side opposite the bond to the leaving group. This is called back-side displacement, or substitution with **inversion of configuration**.



Which of these two opposite stereochemical possibilities operates was determined in experiments with optically active alkyl halides. In one such experiment, Hughes and Ingold determined that the reaction of 2-bromooctane with hydroxide ion gave 2-octanol having a configuration opposite that of the starting alkyl halide.

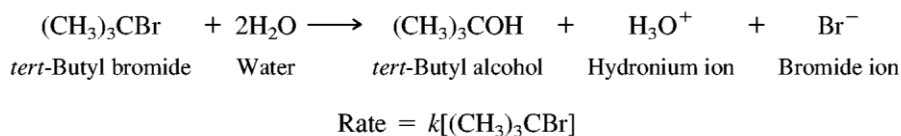


SN2 had occurred with inversion of configuration, consistent with the following transition state:



## S<sub>N</sub>1 Mechanism

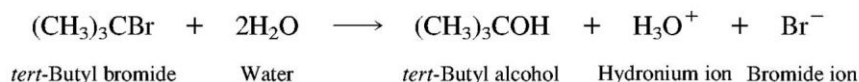
Hughes and Ingold observed that the hydrolysis of tert-butyl bromide, which occurs readily, is characterized by a first-order rate law:



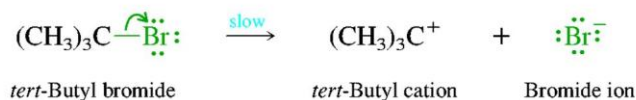
They found that the rate of hydrolysis depends only on the concentration of tert-butyl bromide. Adding the stronger nucleophile hydroxide ion, moreover, causes no change in the rate of substitution, nor does this rate depend on the concentration of hydroxide. Just as second-order kinetics was interpreted as indicating a bimolecular rate-determining step, first-order kinetics was interpreted as evidence for a unimolecular rate-determining step—a step that involves only the alkyl halide.

The proposed mechanism is outlined below and is called S<sub>N</sub>1, standing for **substitution nucleophilic unimolecular**. The first step, a unimolecular dissociation of the alkyl halide to form a carbocation as the key intermediate, is rate-determining.

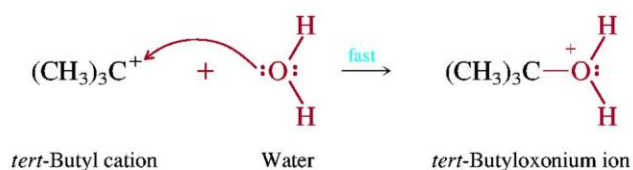
### The Overall Reaction:



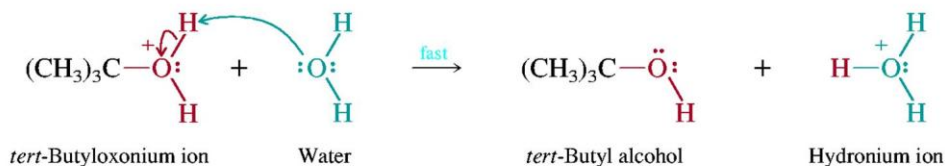
**Step 1:** The alkyl halide dissociates to a carbocation and a halide ion.



**Step 2:** The carbocation formed in step 1 reacts rapidly with a water molecule. Water is a nucleophile. This step completes the nucleophilic substitution stage of the mechanism and yields an alkyloxonium ion.



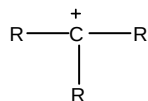
**Step 3:** This step is a fast acid-base reaction that follows the nucleophilic substitution. Water acts as a base to remove a proton from the alkyloxonium ion to give the observed product of the reaction, *tert*-butyl alcohol.



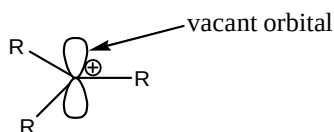
Carbocations are formed in  $\text{S}_{\text{N}}1$  reaction. Thus the rate depends on the stability of carbocations

## Carbocation

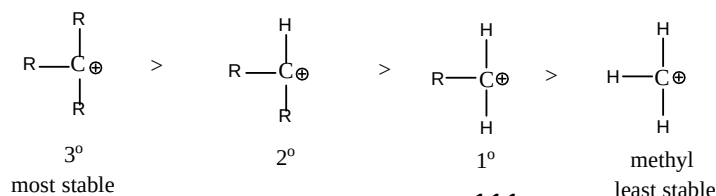
- Is a carbon cation
- Substance that contains a trivalent positively charged carbon atom



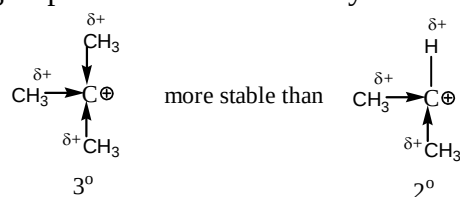
- the structure of carbocation is trigonal planar



- The order of carbocation stability is:  $3^\circ > 2^\circ > 1^\circ > \text{methyl}$

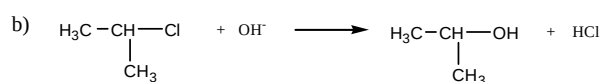
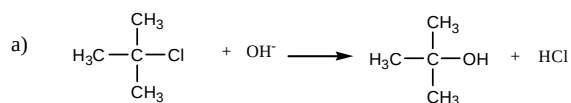


- Alkyl group stabilize carbocation by inductive effect.

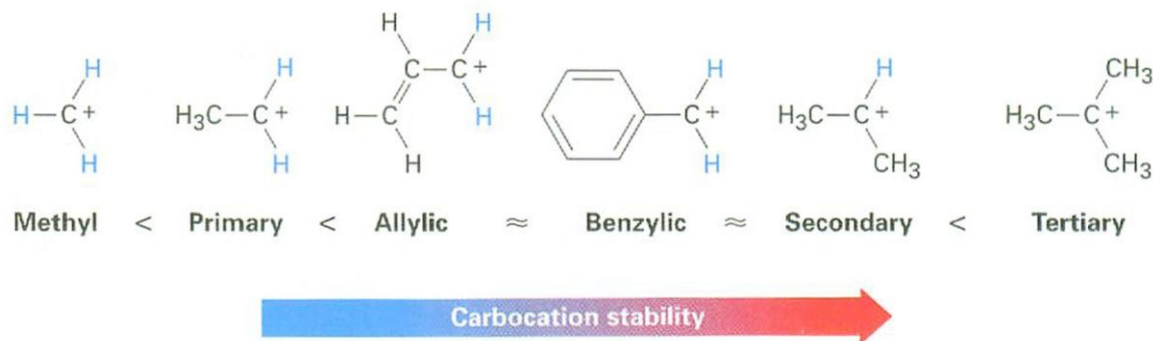


The more stable the carbocation, the rapid the reaction by  $\text{S}_\text{N}1$

The relative reactivity of alkyl halides in  $\text{S}_\text{N}1$  reactions is: methyl < primary < secondary < tertiary

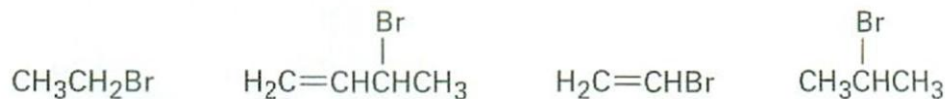


Reaction (a) takes place more rapidly than (b) since it passes through more stable  $3^\circ$  carbocation.  
Carbocation stability



### Exercise

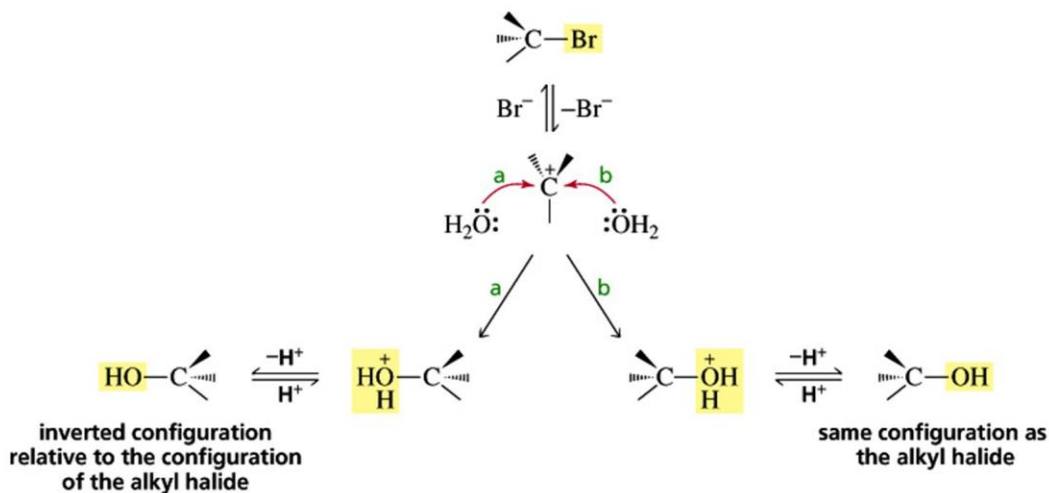
Rank the following substances in order of their expected  $\text{S}_\text{N}1$  reactivity:



## STEREOCHEMISTRY OF SN1 REACTIONS

SN1 is not stereospecific

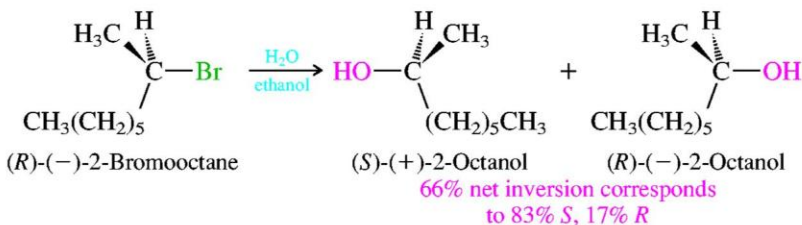
Since carbocation has a trigonal planar structure, the Nucleophile can attack from front side or back side.



A mixture of (R) and (S) products are possible= Racemization

The leaving group partly block one face of the molecule, thus giving a major product of inversion

For example, the hydrolysis of optically active 2-bromooctane in the absence of added base follows a first-order rate law, but the resulting 2-octanol is formed with 66% inversion of configuration.



### 4.1.2 Factors Affecting $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ Reactions

- The structure of substrate
- The concentration and reactivity of nucleophile
- The nature of leaving group
- The effect of solvent

#### Factors Affecting $\text{S}_{\text{N}}1$

##### A. The Structure of Substrate

The more stable the carbocation, the faster the reaction.

In general more highly substituted alkyl halides are more reactive toward SN1 reaction than less substituted ones.

SN1 reactivity rates follow the trend:  $3^\circ > 2^\circ > 1^\circ > \text{methyl}$

We can illustrate this order in the reaction of alkyl halides with acetic acid

Table 5.1 Relative reactivities

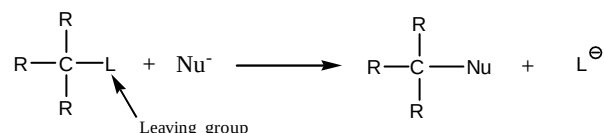
| Substrate                                   | Relative rate of $S_N1$ |
|---------------------------------------------|-------------------------|
| $\text{CH}_3\text{CH}_2\text{Br}$ (primary) | Too slow to measure     |
| $(\text{CH}_3)_2\text{CHBr}$ (secondary)    | 1                       |
| $(\text{CH}_3)_3\text{CBr}$ (tertiary)      | $10^5$                  |

### B. The Concentration and Reactivity of Nucleophile

Since the rate law for an  $S_N1$  reaction is first order in respect to the Organic substrate we should not expect a change in the concentration of the nucleophile to have any effect on the reaction rate of a reaction occurring via the  $S_N1$  path. Nucleophilic strength is the measure of the ability of the nucleophile to more rapidly attach or bond itself to the carbon releasing the leaving group. Since the rate determining step in the mechanism for  $S_N1$  does not involve the nucleophile, we would not expect the nucleophilicity to affect the rate and it does not.

### C. The Nature of Leaving Group

The leaving group ability (LGA) is defined as the ability of the leaving group to be displaced and remain stable after depart.



Leaving groups that are weak bases (coming from strong acids) are more stable after leaving the molecule and do not have any desire to return. However, the more basic the leaving group is (coming from weak acids), the poorer will be its ability to depart without wanting to return to the molecule.

Good leaving groups are essential for  $S_N1$  reaction. Some common leaving groups are:

Anions:  $\text{Cl}^-$ ,  $\text{Br}^-$ ,  $\text{I}^-$ ,  $\text{RSO}_3^-$  (Sulphonate)

Neutral:  $\text{H}_2\text{O}$ ,  $\text{ROH}$ ,  $\text{R}_3\text{N}$ ,  $\text{R}_3\text{P}$

Chloride, Bromide, and Iodide all come from strong acids,  $\text{HCl}$ ,  $\text{HBr}$ , and  $\text{HI}$ . Strong acids produce weak conjugate bases, which make fantastic leaving groups with high LGA. Fluoride comes from  $\text{HF}$ , which is a weak acid so its conjugate base, Fluoride ion, would make a very poor leaving group.

### D. The effect of solvent

The solvent used in a substitution reaction will influence which pathway will be chosen. There are three major categories of solvent.

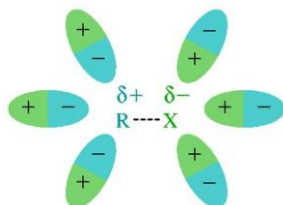
1. Polar Protic Solvents
2. Polar Aprotic Solvents
3. Non-Polar Aprotic Solvents

**Polar Protic solvents** are solvents in which at least one Hydrogen atom is bonded to either an Oxygen or a Nitrogen. This creates a polar molecule that will attract other molecules or ions using Hydrogen bonding. Water ( $\text{H}_2\text{O}$ ) or an alcohol ( $\text{R-O-H}$ ), amine ( $\text{R-NH}_2$ ), Carboxylic Acids like formic acid,  $\text{H-COOH}$ , or acetic acid,  $\text{CH}_3\text{COOH}$  are described as Polar Protic solvents.

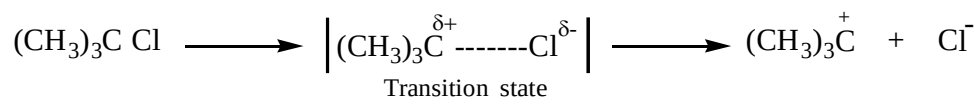
**Polar aprotic solvents** are solvents whose molecules exhibit a molecular dipole moment but whose hydrogen atoms are not bonded to an oxygen or nitrogen atom. Therefore no hydrogen bonding interactions can take place between particles. Some examples of such solvents would include aldehydes, ketones, Dimethyl Sulfoxide (DMSO),  $\text{CH}_3\text{-SO-CH}_3$ , Dimethyl Formamide (DMF),  $\text{H-CO-N(CH}_3)_2$ .

Non-polar aprotic solvents are solvents whose molecules have a zero molecular dipole and whose hydrogen atoms are not bonded to an oxygen or nitrogen. Examples include all the hydrocarbons (Alkanes, Alkenes, and Alkynes).

Polar protic solvents are useful for  $\text{S}_\text{N}1$  reaction. According to the  $\text{S}_\text{N}1$  mechanism, a molecule of an alkyl halide ionizes to a positively charged carbocation and a negatively charged halide ion in the rate-determining step. As the alkyl halide approaches the transition state for this step, a partial positive charge develops on carbon and a partial negative charge on the halogen.



Polar protic solvents surround carbocations by hydrogen bonding to the charged species, which has the effect of stabilizing the intermediate.

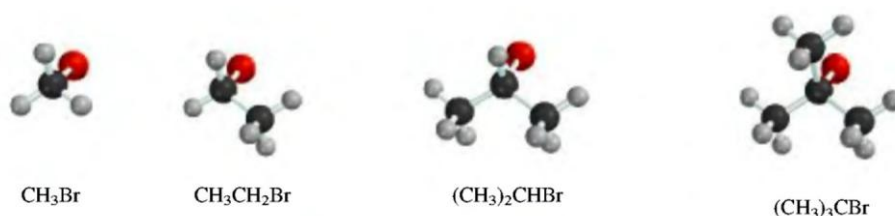


$(\text{CH}_3)_3\text{C}^+$  become more stable with the help of protic solvent.

## Factors Affecting $\text{S}_\text{N}2$

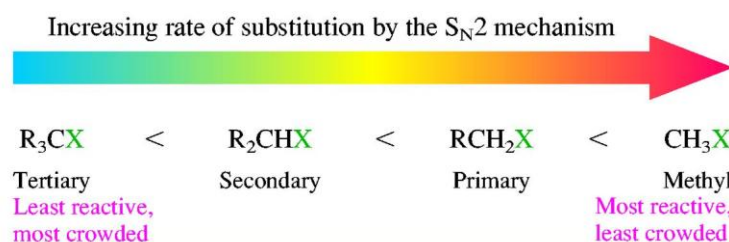
### A. The Structure of Substrate

Large group hinders the approach of a nucleophile in  $\text{S}_\text{N}2$  reaction. The nucleophile must approach the alkyl halide from the side opposite the bond to the leaving group.



The three hydrogens of methyl bromide offer little resistance to approach of the nucleophile, and a rapid reaction occurs. Replacing one of the hydrogens by a methyl group somewhat shields the carbon from attack by the nucleophile and causes ethyl bromide to be less reactive than methyl bromide. Replacing all three hydrogens by methyl groups almost completely blocks back-side approach to the tertiary carbon of  $(\text{CH}_3)_3\text{CBr}$  and shuts down  $\text{S}_{\text{N}}2$ .

In general,  $\text{S}_{\text{N}}2$  reactions of alkyl halides show the following dependence of rate on structure:



## B. The Concentration and Reactivity of Nucleophile

The rate of  $\text{S}_{\text{N}}2$  depends on the nature of nucleophile. Strong nucleophiles will react more rapidly than nucleophiles that are described as poor or weak. Negatively charged nucleophiles tend to be stronger than the neutral nucleophiles with the same central atom.

For example,  $\text{OH}^-$  is stronger than  $\text{H}_2\text{O}$  and  $\text{RO}^-$  are better nucleophiles than  $\text{R-O-H}$ .

Stronger nucleophiles will enhance the reaction rate of a reaction operating under  $\text{S}_{\text{N}}2$  conditions. Increasing the concentration of the nucleophile will enhance the rate of an  $\text{S}_{\text{N}}2$  reaction since kinetically the reaction is first order in respect to the nucleophile concentration. This is in contrast to  $\text{S}_{\text{N}}1$  where the concentration and nucleophilicity have no effect on the rate of an  $\text{S}_{\text{N}}1$  reaction.

Table 5.2 Relative nucleophilicities

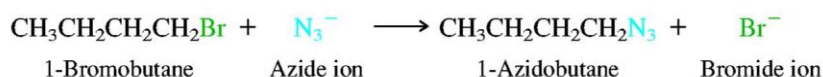
|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| Very Good | $\text{I}^-$ , $\text{HS}^-$ , $\text{RS}^-$                                   |
| Good      | $\text{Br}^-$ , $\text{HO}^-$ , $\text{RO}^-$ , $\text{NC}^-$ , $\text{N}_3^-$ |
| Fair      | $\text{NH}_3$ , $\text{Cl}^-$ , $\text{F}^-$ , $\text{RCO}_2^-$                |
| Weak      | $\text{H}_2\text{O}$ , $\text{ROH}$                                            |
| Very Weak | $\text{RCO}_2\text{H}$                                                         |

### C. The Nature of Leaving Group

The reaction rate of a reaction occurring via S<sub>N</sub>2 path would increase with higher leaving group affinity (LGA) for the leaving group. This is the same effect as was observed for an S<sub>N</sub>1 reaction.

### D. The Effect of Solvent

Polar Aprotic solvents are important in S<sub>N</sub>2 reaction. They lack OH groups and do not solvate anions very strongly, leaving them much more able to express their nucleophilic character. The Table below compares the second-order rate constants *k* for S<sub>N</sub>2 substitution of 1-bromobutane by azide ion (a good nucleophile) in some common polar aprotic solvents with the corresponding *k*'s for the much slower reactions observed in the polar protic solvents methanol and water.



| Solvent                       | Structural formula                    | Dielectric constant $\epsilon$ | Type of solvent | Relative rate |
|-------------------------------|---------------------------------------|--------------------------------|-----------------|---------------|
| Methanol                      | CH <sub>3</sub> OH                    | 32.6                           | Polar protic    | 1             |
| Water                         | H <sub>2</sub> O                      | 78.5                           | Polar protic    | 7             |
| Dimethyl sulfoxide            | (CH <sub>3</sub> ) <sub>2</sub> S=O   | 48.9                           | Polar aprotic   | 1300          |
| <i>N,N</i> -Dimethylformamide | (CH <sub>3</sub> ) <sub>2</sub> NCH=O | 36.7                           | Polar aprotic   | 2800          |
| Acetonitrile                  | CH <sub>3</sub> C≡N                   | 37.5                           | Polar aprotic   | 5000          |

Polar Aprotic solvents do not solvate or encapsulate the nucleophile so they are free to approach the organic substrate. Naked anions are highly reactive as a nucleophile. This means that the rate of an S<sub>N</sub>2 reaction will be increased when Polar Aprotic solvents are used.

### Comparison of S<sub>N</sub>1 and S<sub>N</sub>2 (Summary)

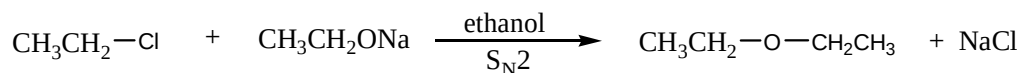
| Factor        | S <sub>N</sub> 1                                                  | S <sub>N</sub> 2                                       |
|---------------|-------------------------------------------------------------------|--------------------------------------------------------|
| Substrate     | Required formation of stable carbocation<br>3° > 2° > 1° > Methyl | Required unhindered substrate<br>Methyl > 1° > 2° > 3° |
| Nucleophile   | Weak nucleophile                                                  | Strong nucleophile                                     |
| Leaving group | Better leaving group                                              | Better leaving group                                   |
| Solvent       | Polar protic solvent                                              | Polar Aprotic solvent                                  |

### 4.1.3 Applications of Substitution Reactions

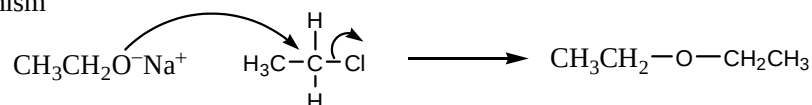
Substitution reactions are of great synthetic value, and they can be used in the preparation of various functional groups

#### 1. Alkyl halides

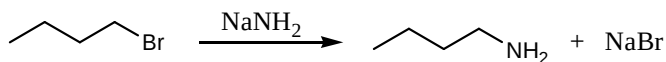
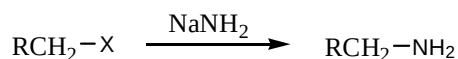
- Williamson ether synthesis



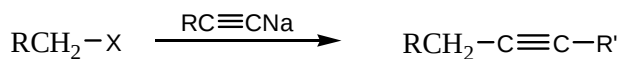
mechanism



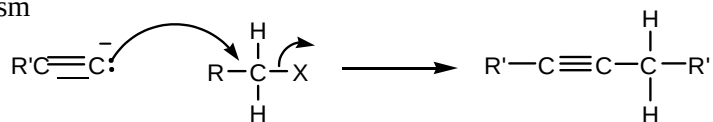
- preparation of primary amine



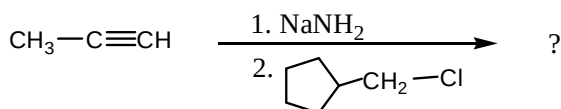
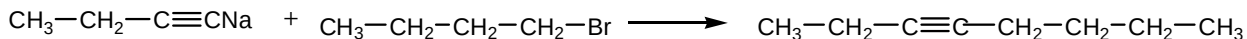
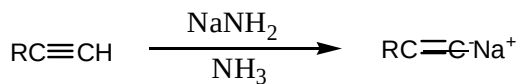
- Preparation of alkynes



mechanism

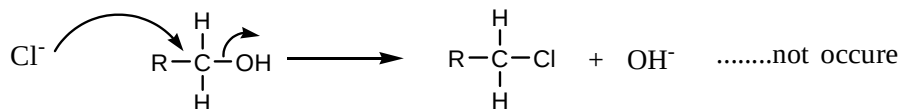


$\text{RC}\equiv\text{CNa}$  can be prepared from a terminal alkyne using strong base like  $\text{NaNH}_2$



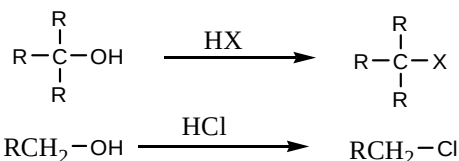
#### 2. Alcohols

-OH is too basic to be displaced by a nucleophile

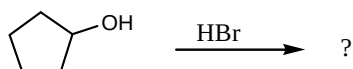
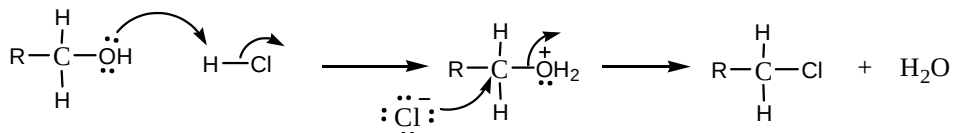


- We need to make OH to be good leaving group.
- Acid protonation change  $\text{-OH}$  to  $\text{-OH}^+$  (good leaving group)

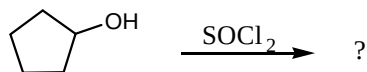
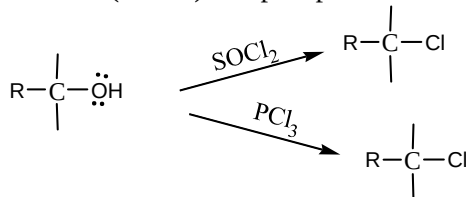
### Preparation of alkyl halide from alcohol



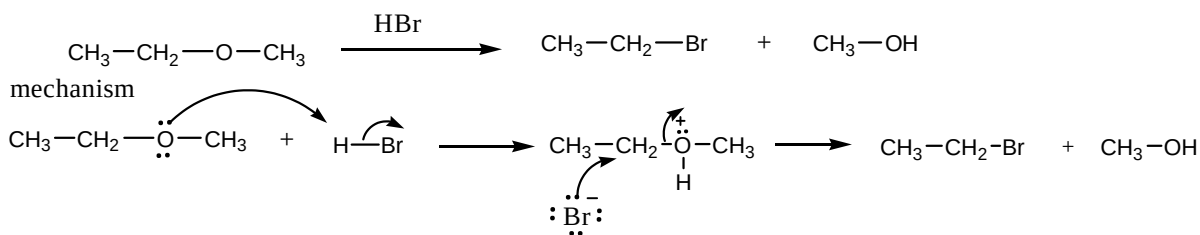
mechanism



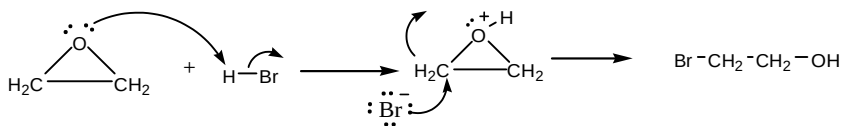
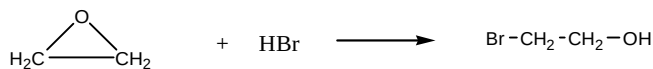
- Thionyl chloride ( $\text{SOCl}_2$ ) and phosphorus halide ( $\text{PX}_3$ ) are also used for this purpose



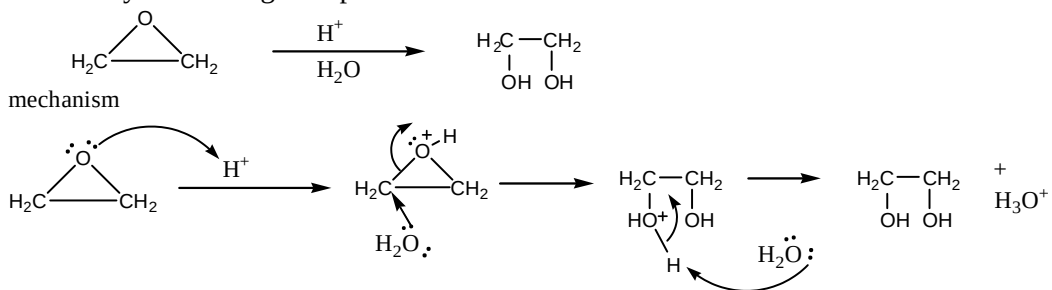
### 3. Ether



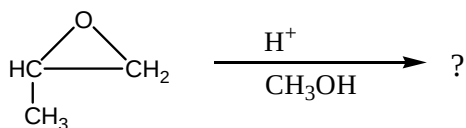
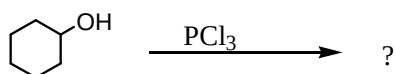
### 4. Epoxides



### Acid catalyzed cleavage of epoxides



### Exercise

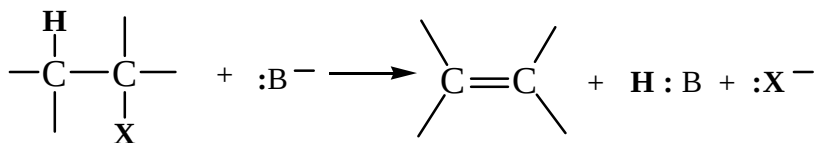


## 4.2 Elimination Reactions

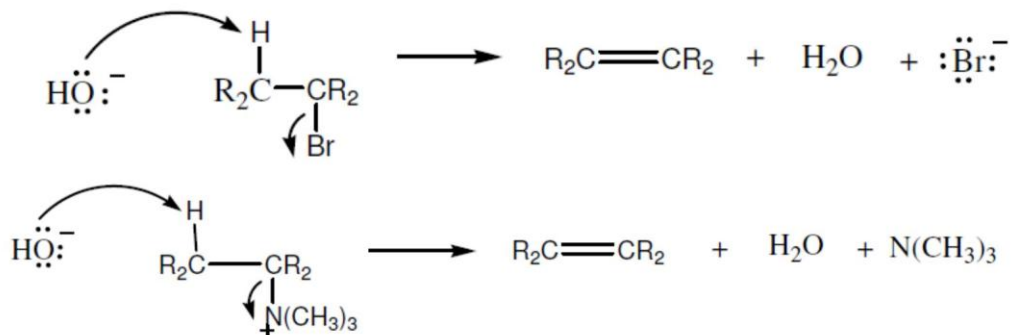
### Introduction

An elimination reaction is a type of organic reaction in which two substituents (two atoms or groups) are removed from a molecule in either a one or two-step mechanism, usually with the introduction of unsaturation (multiple bonding)

- base removes  $\text{H}^+$  as  $\text{X}^-$  leaves
- base attacks H (nucleophile attacks C)



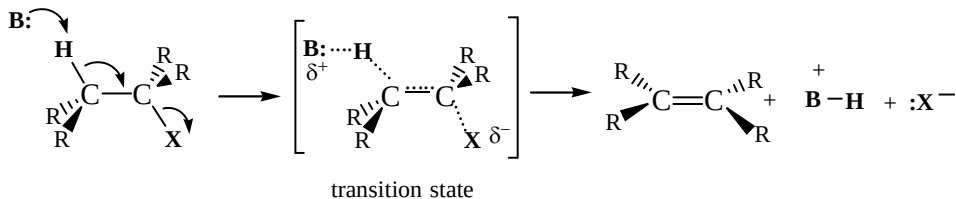
Some elimination reactions are illustrated below:



## 4.2.1 E2 and E1 Mechanism

### E2 Elimination Mechanism

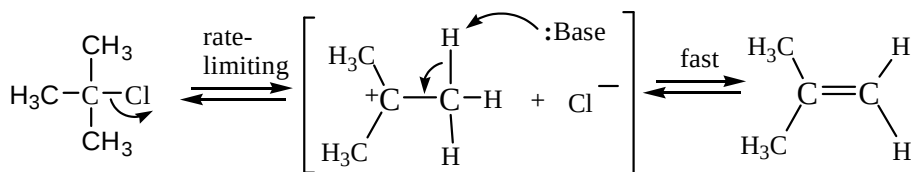
- concerted (single-step)
- the H and X eliminated must be aligned anti to one another



### E1 Elimination Mechanism

- two-step mechanism

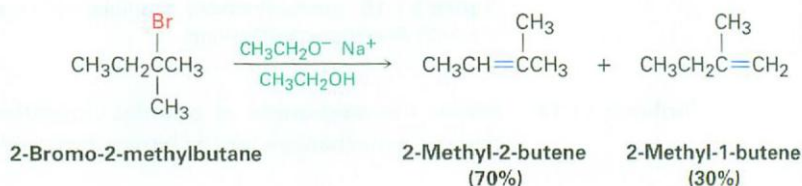
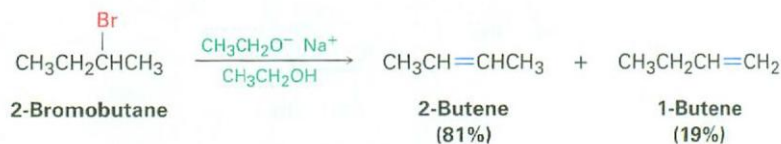
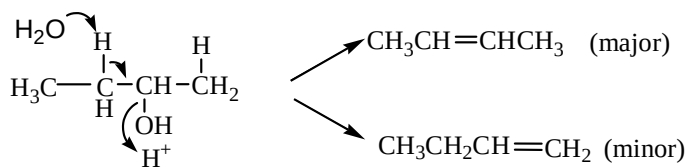
first  $X^-$  leaves, then  $H^+$  leaves (carbocation intermediate) same first step as  $S_N1$



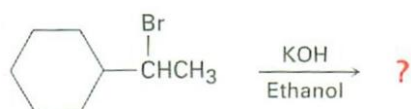
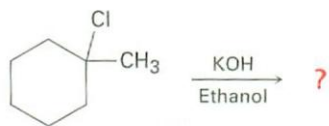
## 4.2.2 Zaitsev's Rule

Elimination reactions almost always give mixtures of alkene products.

Zaitsev's rule (Saytzeff's) rule states that if more than one alkene can be formed by an elimination reaction, the more stable alkene is the major product. In general, the compound that has a more highly substituted  $C=C$  double bond is more stable due to the electron donating properties of the alkyl group.

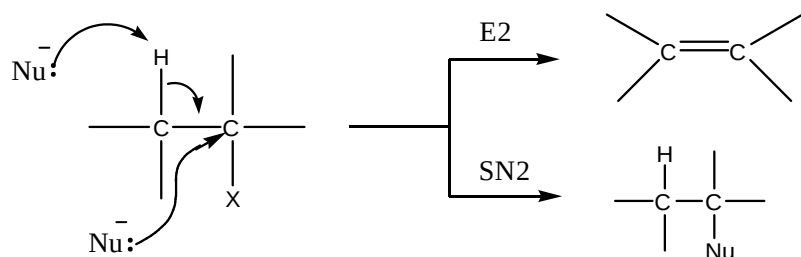


- Ex: predict the major product

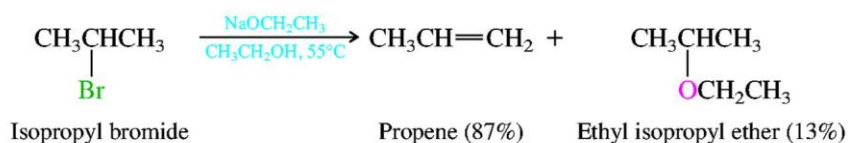


### 4.2.3 Elimination versus Substitution

Nucleophilic substitution and elimination reactions often compete with each other, since bases are nucleophilic, they can undergo substitution



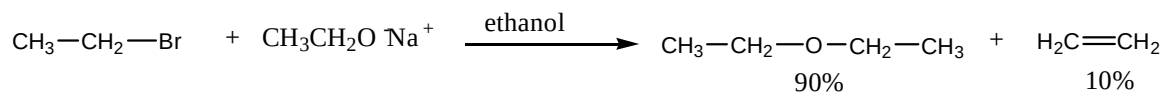
How can we predict whether substitution or elimination will be the principal reaction observed with a particular combination of reactants? The two most important factors are the structure of the alkyl halide and the basicity of the anion. It is useful to approach the question from the premise that the characteristic reaction of alkyl halides with Lewis bases is elimination, and that substitution predominates only under certain special circumstances. In a typical reaction, a typical secondary alkyl halide such as isopropyl bromide reacts with a typical Lewis base such as sodium ethoxide mainly by elimination:



- to favor elimination:**
  - use a strong, hindered base
  - e.g., KOtBu
- to favor substitution:**
  - use a small, unhindered nucleophile

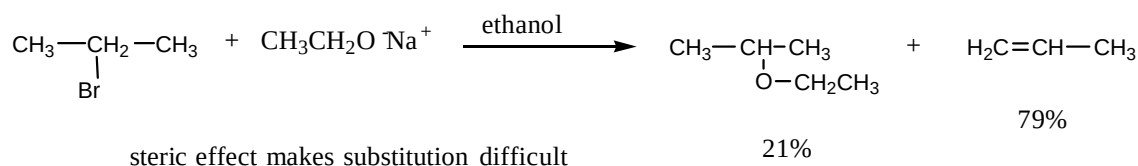
- Reactivity Patterns**

With 1° halide substitution is highly favoured

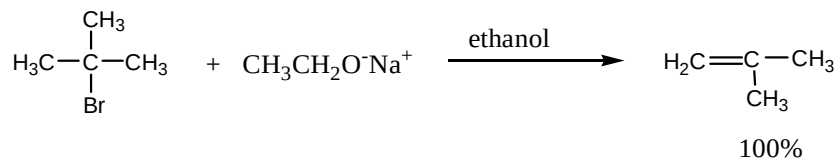


The base can easily approach the carbon bearing the leaving group

With 2° halide elimination favoured



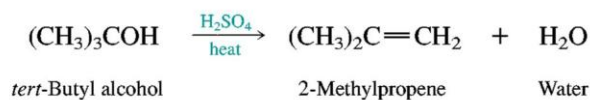
With 3° halide elimination highly favoured



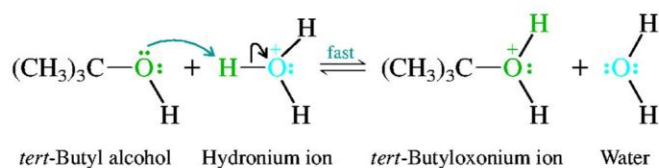
## 4.2.4 Applications of Elimination Reactions

### 1. Dehydration of alcohol

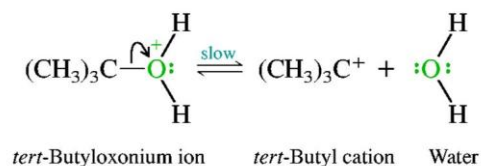
**The overall reaction:**



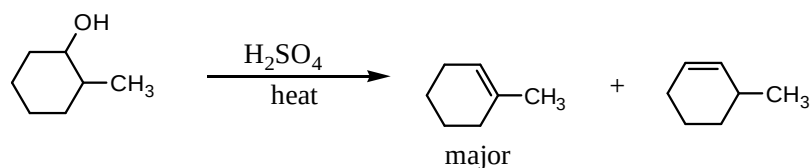
**Step (1):** Protonation of *tert*-butyl alcohol.



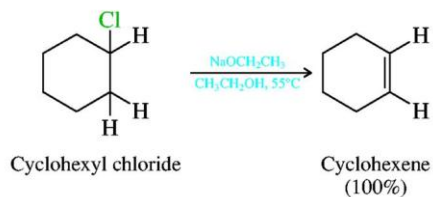
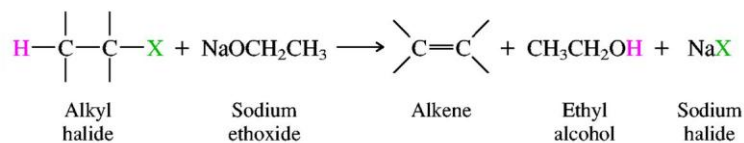
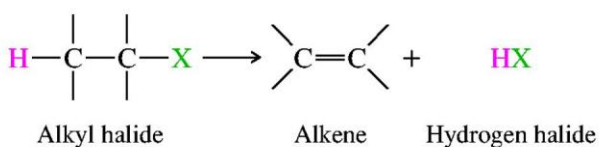
**Step (2):** Dissociation of *tert*-butyloxonium ion.



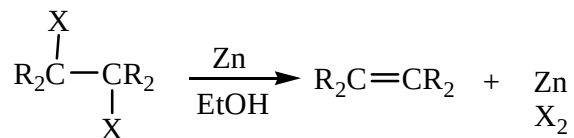
**Step (3): Deprotonation of *tert*-butyl cation**



## 2. Dehydrohalogenation



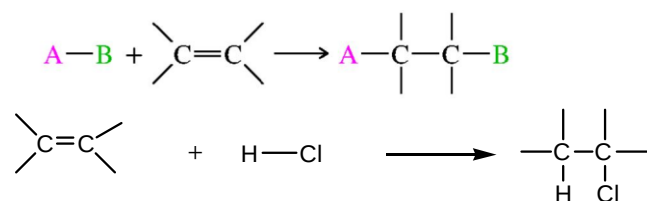
## 3. Elimination of vicinal dihalides – synthesis of alkenes



## 4.3 Addition Reactions

Addition reaction is a reaction in which two molecules combine to yield a single molecule of product.

Addition reactions are limited to compounds that contain multiply-bonded atoms.



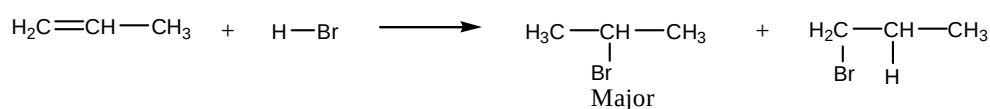
### Orientation of Addition

Regiochemistry: specific orientation of addition (which C gets H, which gets X?)

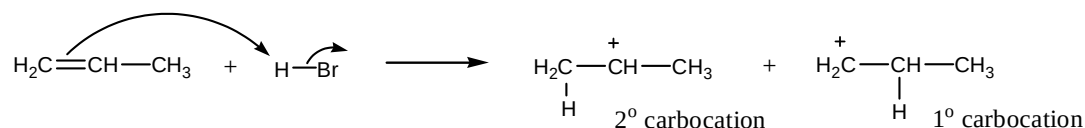
Alkene additions are regioselective: one direction of addition is usually preferred. Governed by Markovnikov's Rule

#### 4.3.1 Markovnikov's Rule

The original: add H to the C with more H's (or to the C with fewer alkyl groups)



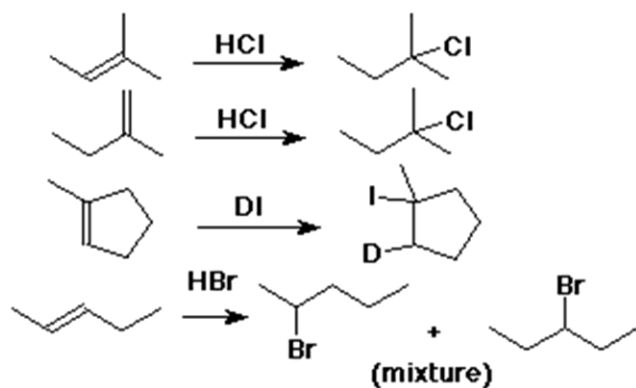
mechanism



The reason: the reaction tends to pass through more stable carbocation. (add H to form the more stable cation)

carbocation stability:  $3^\circ > 2^\circ > 1^\circ$

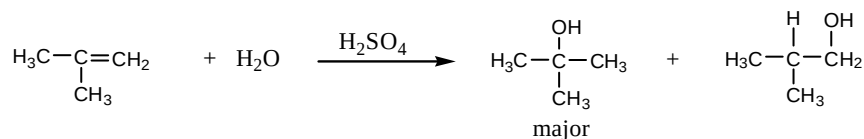
Modern statement of **Markovnikov's Rule**: in ionic addition reaction the reaction passes through the formation of more stable carbocation as an intermediate



## 4.3.2 Application of Addition Reactions

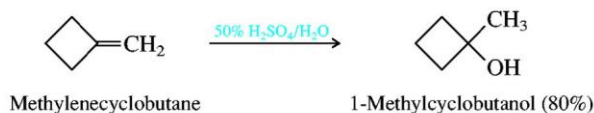
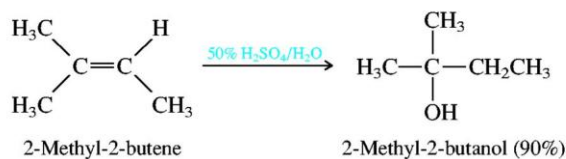
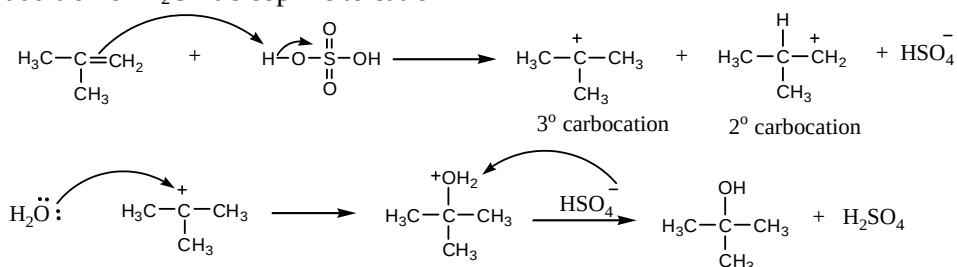
### 1. Hydration of Alkenes

alkene + water  $\rightarrow$  alcohol

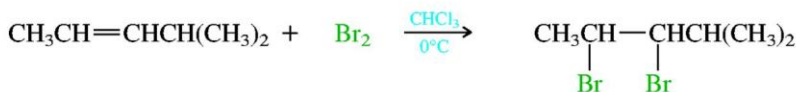
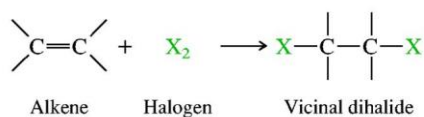


the reaction follows Markovnikov's rule

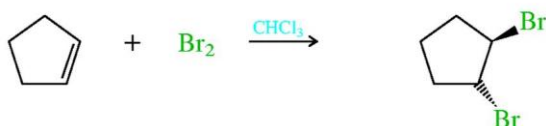
- **mechanism:**
- step 1:  
addition of  $\text{H}^+$  electrophile to pi bond
- step 2:  
addition of  $\text{H}_2\text{O}$  nucleophile to cation



### 2. Halogenation of Alkenes



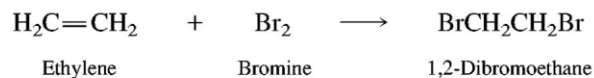
- **stereochemistry of halogen Addition**
- anti stereochemistry:  
two new groups are added to opposite sides of the original pi bond



- **Mechanism of the anti addition**

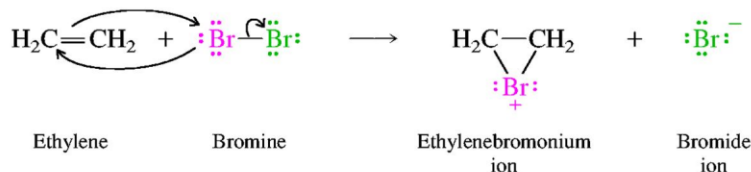
Bromonium Ion : carbocations can be stabilized by bonding to a neighboring Br

**The overall reaction:**

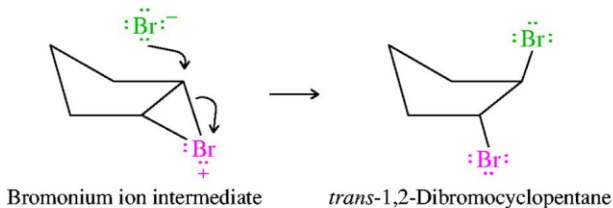
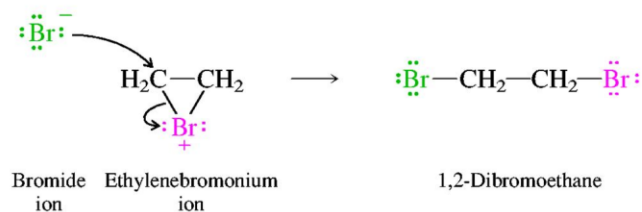


**The mechanism:**

**Step 1:** Reaction of ethylene and bromine to form a bromonium ion intermediate:

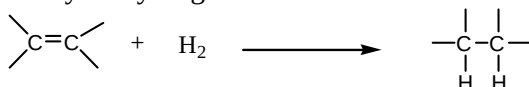


**Step 2:** Nucleophilic attack of bromide anion on the bromonium ion:

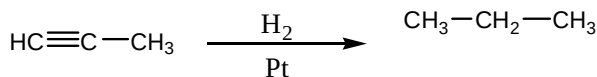
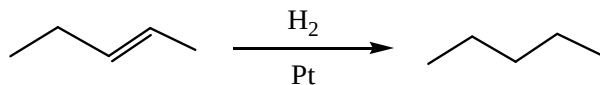
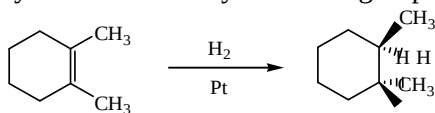


### 3. Reduction of Alkenes

- reduction - addition of H<sub>2</sub> (or removal of O)
- Catalytic Hydrogenation



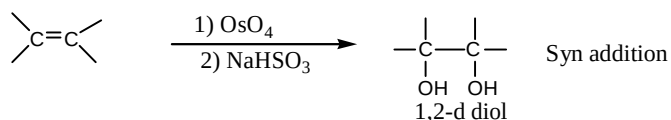
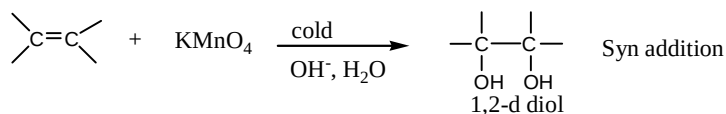
- requires an active catalyst, typically Pt, Pd, Ni, PtO<sub>2</sub>
- reaction occurs on the surface
- both Hs are delivered to the same side of the pi bond=Syn Addition
- syn stereochemistry: two new groups are added to the same side of the original pi bond



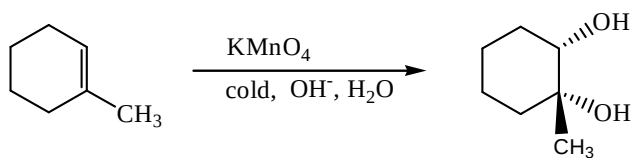
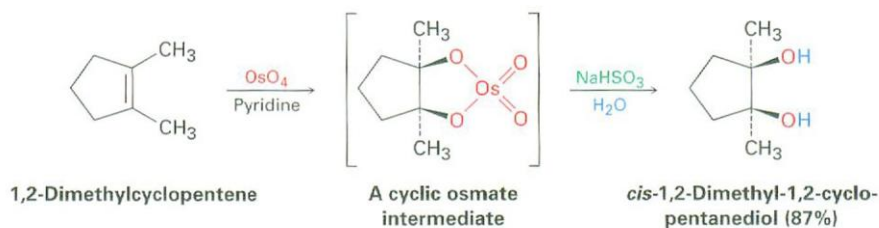
## 4. Oxidation of Alkenes

- oxidation - addition of O (or removal of H<sub>2</sub>)
- there are a wide variety of oxidizing agents: O<sub>2</sub>, O<sub>3</sub>, KMnO<sub>4</sub>, CrO<sub>3</sub>, Na<sub>2</sub>Cr<sub>2</sub>O<sub>7</sub>

**Hydroxylation:** Addition of two OH groups to each of the two double bonded carbon

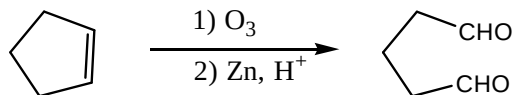
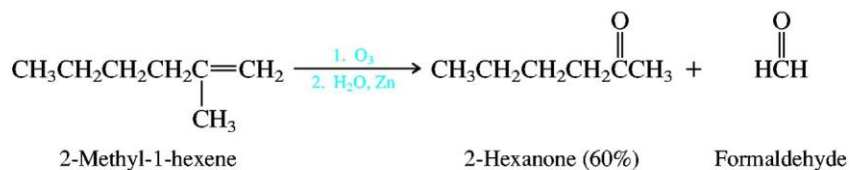
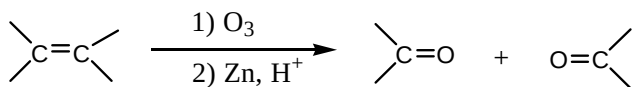


- The two OH adds to the same side of the double bond (syn)



## 5. Ozonization

- ozone (O<sub>3</sub>) causes double bond cleavage
- sometimes useful degradation method to identify unknown compounds



## Self-Test Exercise 4.0

1. Provide the missing organic product for the following reaction. Indicate the major product when more than one product is possible

